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Investigating the Application of a Recommender System for Irish College Courses

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Masters in Computer Science

Supervisor: Owen Conlan

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Declaration

I, the undersigned, declare that this work has not previously been submitted as an exercise for a degree at this, or any other University, and that unless otherwise stated, is my own work.

Brendan McCann

April 10, 2026

Investigating the Application of a Recommender System for Irish College Courses

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University of Dublin, Trinity College, 2026

Supervisor: Owen Conlan

In 2024, over 48,000 Irish secondary school students progressed to college after graduation, whereas in 2018, about 10,000 Irish graduates reported they were not likely to study their college course again, demonstrating the importance and difficulty in choosing a relevant college course in Ireland. This dissertation proposes a Recommender System (RS) that provides users with relevant Irish college course recommendations, assisting students in this difficult decision-making process. The RS is a hybrid RS with Knowledge-Based Filtering and Content-Based Filtering, utilising a student's academic interests, RIASEC model, expected Leaving Certificate (LC) points, NFQ level and college preferences to provide 20 course recommendations. This RS was evaluated in an experiment with $n = 31$ participants, where participants blindly evaluated the relevance of recommendations provided by this RS and a baseline RS, which returns 20 courses sorted by LC points. Large, statistically significant ($p \leq 0.05$) improvements were observed across the precision, serendipity and trust metrics, and insignificant differences for the diversity metric. Moreover, participants liked four out of 20 baseline recommendations on average, all four of which were not on their original college application list. Alternatively, participants liked nine out of 20 actual recommendations, eight of which were not on their original college application list, demonstrating the potential effectiveness of an RS for Irish college courses.

I would like to dedicate this dissertation to my Dad.

I'm grateful for the value you've placed on my education, and I don't take for granted the opportunities I've gotten from it. Thank you for being so opinionated on this topic and inspiring this research, and for all of your advice which led me here.

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Chapter 1

Introduction

1.1 Problem Statement

This section will outline the problem this dissertation seeks to address, and the context surrounding it.

According to Higher Education Authority (2024b), over 48,000 Irish secondary school students progressed to college after graduation in 2024. The individuals who have considered progressing to college have all been posed with the question: *What college course should I study?* This is a daunting question for many reasons. The majority of secondary school students are faced with making this decision at 16-19 years old, an age with limited life experience and knowledge about the world. This decision also comes with many personally confronting questions, which may include: *“What do I want to do with the rest of my life?”* and *“What do I love doing?”*, among many others. Moreover, an individual may choose a college course they will later regret. For instance, in Higher Education Authority (2018), about 10,000 (31%) of recent Irish college graduates reported they were not likely to study their course again, adding more weight to this crucial life decision.

Furthermore, individuals are shown to be influenced by various external factors when choosing their college course. For instance in Mullen (2011); Mullen et al. (2003); Leighton and Speer (2023), college course choice is largely influenced by the parent’s educational background of the student. Additionally, Slattery et al. (2023); I Wish (2025) demonstrate the downsides of gender representation in certain college courses, acting as barriers to college students of under-represented genders. Lastly, Hinrichs (2015); Dickson (2010); Xie and Goyette (2003) highlight the influence of race and ethnicity on an individual’s college course choice. The above points illustrate the fact that this decision is not made in a vacuum, adding complexity to a decision that is already challenging.

To assist students in this difficult decision, it would be appropriate to introduce the role of a Guidance Counsellor (GC). GCs are trained professionals in Irish secondary schools, primarily known for assisting students in their career prospects after graduation. Most notably, the role of a GC extends well beyond vocational support. According to Institute of Guidance Counsellors (2016), GCs are also trained to provide educational, personal and social support. In this context, it is worth introducing the Institute of Guidance Counsellors (IGC)[1], the professional body for GCs in Ireland. In a report by the Institute of Guidance Counsellors (2022), mental health issues were more commonly encountered by GCs compared to career decision-making issues[2], marking the first time that career decision-making issues were not the most commonly encountered issue by GCs since 2019. In the IGC's 2024-2027 strategic plan outlined in Institute of Guidance Counsellors (2024b), they signalled a shift in priorities for GCs across Irish secondary schools: vocational support may not be the most important aspect of a GCs role going forward.

As is evident, choosing a college course is a difficult, multi-faceted decision, and the decision is made more challenging by the shifting priorities of Irish GCs. This leads us to the motivation behind this research.

1.2 Research Motivation

In light of the identified problems, we can shift to a more technological focus. A Recommender System (RS) is a technology that recommends items to a user. Well-known examples of RSs include Netflix[3] recommending films to their customers, or Amazon[4] suggesting products to their customers. One of the earliest known RSs was *Tapestry* in Goldberg et al. (1992), a system that made tailored mailing list recommendations to its users. Around this time, the field of RSs received considerable interest as e-commerce websites could increase their revenue if they suggested more personalised products to their users. As discussed in Schafer et al. (1999), Amazon and Netflix are the most notable adopters of RSs in order to increase company revenue.

As previously explained, an RS recommends items to a user. However, we can re-

[1] `https://igc.ie/`

[2] In Institute of Guidance Counsellors (2024a), career decision-making issues did surpass mental health issues as the most commonly encountered issue for GCs; however, both are pressing issues.

[3] `https://www.netflix.com/`

[4] `https://www.amazon.com/`

conceptualise an RS in the context of college courses: college courses can take the place of items, and a student can take the place of a user. Given the established difficulty and influences of an individual's college course choice, the shifting priorities of GCs towards student's mental health as opposed to career-making decisions, and a technology that could be appropriately applied to this domain, an opportunity presents itself: a Recommender System for Irish college courses.

A technology, such as an RS, could assist an individual in navigating a difficult decision-making process. For instance, there are over 45 higher education institutes in Ireland offering a total of over 1700 courses[5], representing a huge set of choice to the individual. An advantage of a personalised RS is to provide a user with a much smaller list of relevant recommendations. Long et al. (2025) demonstrated that smaller recommendation sets are often equally as effective as larger recommendations sets, suggesting that a much narrower set of personalised college course recommendations could be effective in this space.

For the 2026/2027 academic year, 52[6] new higher education courses were introduced in Ireland. Moreover, as discussed in section 2.4.1, the requirements for Irish higher education courses change every year as a reflection of demand and changing priorities. This introduces administrative strain on GCs, making it difficult to have up-to-date awareness on the newest courses and current course entry requirements. A technology, such as an RS, can be automated to maintain the latest college course information and utilise this information in recommending college courses that a human GC may not even be aware of, posing a potential advantage of an RS in this domain.

As we have argued, an RS is a technology that could be appropriately applied to the domain of Irish college courses, potentially alleviating the difficulty of the college course decision-making process. With the motivation behind our research described, we can now define our research question.

1.3 Research Question

In light of the motivation behind this research, the research question for this dissertation can be formulated: *"To what extent can a Recommender System assist an Irish student in exploring college courses they would consider studying?"*

[5] Higher education colleges and courses can be found here: `https://careersportal.ie/`

[6] See `https://careersportal.ie/courses/coursefinder?types_in=2` for new courses

In this research question, the meaning of ‘assist’ is intentionally broad, and we can further understand the meaning of ‘assist’ by exploring the three qualities of an effective RS, as outlined in Aggarwal et al. (2016), a prominent and popular textbook on RSs:

According to Aggarwal, the recommendations should be **relevant** to the user’s preferences. In this context, an effective RS would recommend college courses the user would consider studying based off their gathered preferences. For example, it would generally not make sense to recommend healthcare courses to a student that has demonstrated no interest in healthcare. In context to our research question, an RS with relevant recommendations would assist a student by recommending college courses they would consider studying.

Aggarwal also argues that the recommendations should be **diverse** and avoid being uniform in a particular category. For example, an effective RS would not suggest 20 different computer science courses to a user, even if the user has demonstrated a strong interest in computers. If we revisit the research question, we desire an RS that assists a student’s college course exploration, and by suggesting college courses across a diverse range of the user’s preferences, the RS would assist the user in discovering college courses they would consider studying.

A user may receive recommendations that are relevant to their preferences and diverse across their interests, however the user may already have an awareness of these recommendations. For example, a student may receive recommendations in biology and engineering, but they may have already considered studying these college courses previously. As a result, the user does not actually gain any new information after interacting with the RS. Aggarwal claims that the quality that separates a good RS from a great one is **serendipity:** these are recommendations that are genuinely unexpected but incredibly relevant. The RS demonstrates a deep knowledge of the user, providing the user with relevant recommendations that the user was genuinely unaware of. Continuing our example, a serendipitous recommendation to this user might be biomedical engineering, merging both of the user’s interests in an unexpected but useful way, and the user may now be considering studying the college course after interacting with the RS. This would be incredibly effective in assisting the user with their college course exploration, as outlined in the research question.

Serendipity is the gold standard for RSs. However, it is very challenging to consistently produce serendipitous recommendations, because by their nature the recommendations are ‘high-risk, high-reward’. RSs are often faced with the dilemma between producing ‘safe’ recommendations that are likely relevant but uninteresting, or more ‘adventurous’ recommendations that could be serendipitous or unsatisfactory. This is a well-known tradeoff in the field of RSs, and it is the price paid for attempting to achieve the gold standard in RSs: serendipity.

In summary, according to Aggarwal et al. (2016), an effective RS provides recommendations that are:

- Relevant
- Diverse
- Serendipitous

It is worth mentioning that the above three qualities will be referenced frequently throughout this dissertation, and thus we recommend that the reader grows familiar with these concepts. In summary, the research question for this dissertation is: *"To what extent can a Recommender System assist an Irish student in exploring college courses they would consider studying?"*, and we hypothesise that an RS that provides relevant, diverse, and serendipitous recommendations would assist an Irish student in exploring college courses they would consider studying. Therefore, this dissertation's ability to answer this research question will be measured by its performance on the relevance, diversity and serendipity metrics.

1.4 Research Objectives

With our research question formulated, we can define our research objectives below:

1. Provide an answer to the question: *"What college course should people study?"*.
2. Identify the relevant information an RS should question a user to recommend suitable college courses.
3. Design an objective, easily reproducible baseline RS appropriate to the Irish college course context.
4. In comparison to a baseline college course RS, design an RS with statistically significant improvements for the user-perceived trust metric.
5. Analyse the qualitative insights and findings from a conducted experiment.

1.5 Overview

In this section, an overview of this dissertation's chapters is provided.

- Chapter 2 - Background Reading. This chapter will describe the relevant background knowledge to understand the Irish college admission system.
- Chapter 3 - Literature Review. This chapter will review the literature surrounding college course choice, and closely-related works.
- Chapter 4 - Design. This chapter will describe the high-level design of the college course RS, with a justification of different design decisions.
- Chapter 5 - Implementation. This chapter will explore the implementation details behind the design ideas exposed in chapter 4.
- Chapter 6 - Evaluation. This chapter will explain the conducted experiment methodology, and evaluate the RS's performance on various metrics.
- Chapter 7 - Conclusions & Future Work. This chapter will summarise the dissertation, and explore potential areas for future development.

Chapter 2

Background Reading

As mentioned in chapter 1, this dissertation specifically focuses on a college course RS in the context of the Irish secondary school and college system. Grading systems and college admission systems often differ between countries, so this chapter seeks to provide essential background knowledge behind the Irish secondary school and the college admission system.

2.1 The Leaving Certificate

The Leaving Certificate (LC) is the final examination taken by the majority of secondary school students in Ireland. Instead of taking the LC, students can take the Leaving Certificate Applied (LCA) which is a more practical alternative to the LC. Students who take the LCA cannot directly enter higher education institutes; they commonly go for direct employment, apprenticeships or enter further education institutes. For the sake of narrowing scope, apprenticeships and further education courses are considered beyond the scope of this RS. The RS proposed in this dissertation will only recommend higher education college courses in Ireland.

A student's overall performance in the LC is measured in terms of points, a number that ranges between 0 and 625, where 625 is the maximum achievable points. A student chooses to be examined in each subject at increasing levels of difficulty, known as Foundation Level (FL)[1], Ordinary Level (OL) or Higher Level (HL). A student receives a grade depending on the level and mark they achieve in the subject. For example, a student that gets a mark of 91% in the HL LC English exam receives a H1 grade, for which they

[1] Foundation Level is only offered for the LC Irish and mathematics subjects.

receive 100 points, and a student that gets a mark of 64% in their OL LC Geography exam receives an O4 grade, for which they receive 28 points. For a full list of LC exam grades and their translated LC points, see Figure 2.1 below. A student is examined in a minimum of six subjects, and their overall LC points is an accumulation of the points received in each examined subject.

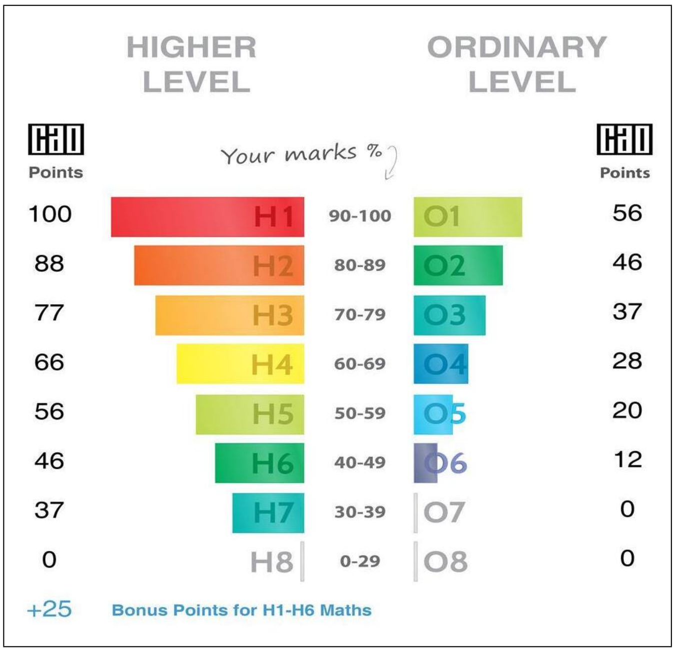


Figure 2.1: A full list of LC exam grades and their translated LC points.

2.2 NFQ Levels

The Irish National Framework of Qualifications (NFQ)[2] is a system that describes the majority of education qualifications in Ireland. It is a 10-level system, where 1 is the lowest and 10 is the highest (doctorate degree). For instance, the LC is classed as NFQ level 5. The NFQ levels that are relevant to this dissertation are NFQ level 6 (Advanced or Higher Certificate), NFQ level 7 (Ordinary Bachelors degree) and NFQ level 8 (Honours Bachelor degree). A full graphic of the 10 NFQ levels are shown below in figure 2.2.

[2]https://www.qqi.ie/what-we-do/the-qualifications-system/national-framework-of-qualifications

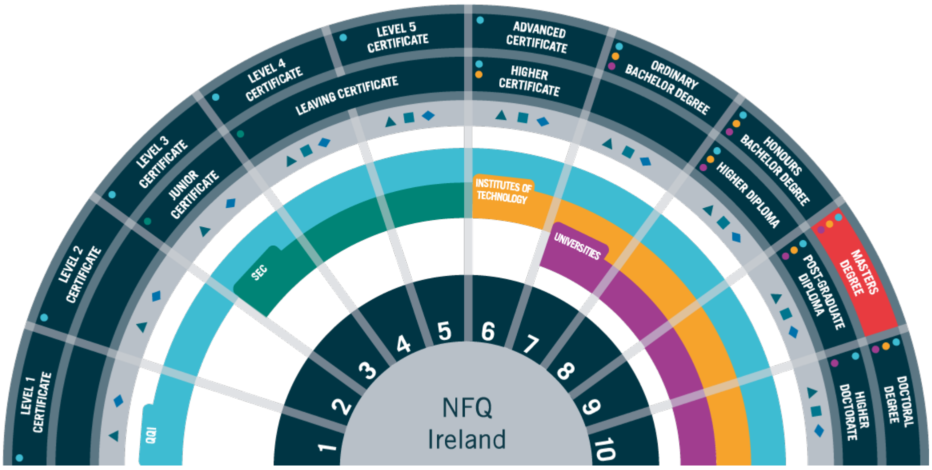
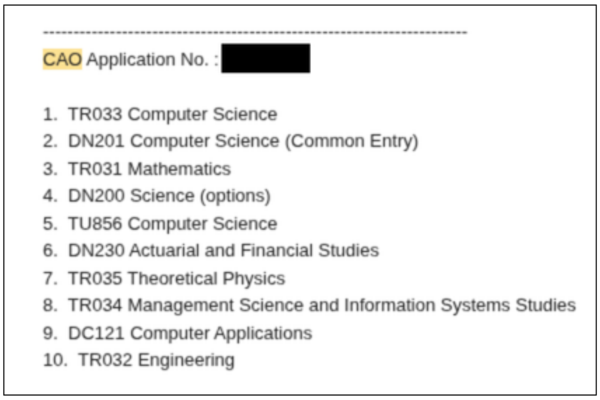


Figure 2.2: The 10-level NFQ system. Source.

2.3 The Central Applications Office

Rather than going to the colleges directly to apply for a specific college course, in Ireland all college applications are done through the Central Applications Office (CAO)[3], the centralised college admissions system in Ireland. For reference, there were over 83,000 applications to the CAO in 2025 according to Central Applications Office (2025). As part of a CAO application, students are encouraged to create two numbered lists, each with up to ten college courses. One of these numbered lists is for NFQ level 8 college courses, and the other list is for NFQ level 6 and 7 college courses. These college courses are ordered by the student's preference of study - for instance, the student's number one choice would be the course they want to study the most, and their number ten choice is the course they want to study the least out of this list. An example of this list is given below in figure 2.3



CAO Application No. : [redacted]

1. TR033 Computer Science
2. DN201 Computer Science (Common Entry)
3. TR031 Mathematics
4. DN200 Science (options)
5. TU856 Computer Science
6. DN230 Actuarial and Financial Studies
7. TR035 Theoretical Physics
8. TR034 Management Science and Information Systems Studies
9. DC121 Computer Applications
10. TR032 Engineering

Figure 2.3: An example of an NFQ level 8 CAO college course list.

[3] https://www.cao.ie/

The preference ranking of a student's CAO application courses is relevant here. For example, if a student receives an offer for a course that is ranked first in their list, they will not receive offers for courses below this preference, even if they satisfy the course's entry requirements. Entry requirements will be explained in section 2.4.

2.4 Entry Requirements

Entry requirements for colleges differ by country, and in this section the entry requirements for Irish college courses are described.

2.4.1 LC Points

The primary entry requirement for most college courses in Ireland is LC points. For example, the computer science course in Trinity College Dublin (TCD) has a minimum entry requirement of 533 points[4], meaning that a student must achieve 533 points or higher in their LC to be considered admission into the course. A course's LC points is the points of the student who was last admitted to the course in the previous academic year. For example, let's suppose that in 2025, 150 people were admitted into computer science in TCD. Out of all these 150 students, the student with the lowest LC points was 533, meaning the course has a minimum entry requirement of 533 LC points. As a result, the LC points associated with a course are not fixed and fluctuate every year. LC points are commonly misunderstood as reflecting the difficulty of a course; however, it would be more accurate to say that the LC points reflect the demand of a course.

There are a handful of ways in which a student can achieve additional LC points. For example, a student will receive an extra 25 points if they achieve a passing ($\geq$40%) grade in the HL mathematics exam. As well as this, if a student takes their LC exams in the Irish language, they can receive up to 10% in bonus marks in each of their LC exams[5].

[4] 533 points as of 2025, see: `https://www.tcd.ie/courses/undergraduate/courses/computer-science/`

[5] For exact bonus marks in answering the LC in the Irish language, see: `https://www.citizensinformation.ie/en/education/state-examinations/leaving-certificate/`

2.4.2 LC Subject Requirements

Many Irish college courses have grade requirements for specific subjects to allow admission into the course. In most cases, the subjects are relevant to the course and the college seeks knowledge in that subject. For example, computer science in TCD has a minimum entry requirement of a H4 (60-69% in HL) in mathematics to be considered for admission into the course.

In Ireland, students are obligated to take the LC English, mathematics and Irish[6] exams, and they choose a minimum of four additional subjects to take exams in. The student makes their subject choices in fifth year, which is the penultimate year in Irish secondary schools. Due to the differing subject entry requirements by college courses, some students may be unable to take certain courses simply because they have not taken the relevant subject. For example, if a student has not taken two science[7] subjects, they are unable to be admitted into medicine[8], meaning that a student's college course options are already limited by the subject choices they made in fifth year, almost two years before they finalise their CAO college course list.

In addition to specific college courses requiring a minimum grade in particular subjects, some subjects are required to be admitted into the college itself. For example, all Royal College of Surgeons in Ireland courses[9] require a third language[10].

2.4.3 Portfolios, Tests and Interviews

In a majority of cases, if a student satisfies the minimum entry requirements for a course with their LC points and subject grades, the student is admitted into the course. However, some Irish college courses have additional entry requirements. For instance, some courses may require the student to be examined in a test outside of the LC. For example, all medicine courses in Ireland require their applicants to sit the Health Professions Admissions Test (HPAT)[11]. Some college courses may also require a portfolio submission, or for the student to undertake an interview. For example, the architecture course in Technological University Dublin[12] requires the student to submit an additional portfolio

[6] Some students may qualify for an exemption from Irish if they satisfy certain requirements.
[7] LC Physics, Chemistry, Physics/Chemistry, Biology and Agricultural Science.
[8] Subject requirements for TCD medicine as of 2026: `https://www.tcd.ie/courses/undergraduate/courses/medicine/`
[9] As of 2026, see: `https://www.rcsi.com/dublin/undergraduate/medicine/entry-requirements`
[10] Languages other than English and Irish, for example: German, Spanish, Italian, etc.
[11] `https://hpat-ireland.acer.org/`
[12] `https://www.tudublin.ie/study/undergraduate/courses/architecture-tu832/`

showcasing the student's competence in sketching, painting, etc., as well as an interview demonstrating their interest in the course.

In the case of portfolios and tests, the student's scores in these are added to their total LC points. For example, if a student achieves 550 LC points and a score of 200 in the HPAT, their combined LC points is 750 only when the student is applying for courses that require the HPAT. This explains why some courses may have a minimum requirement of 732 LC points, despite 625 being the maximum achievable LC points.

2.5 Guidance Counsellors

The role of a Guidance Counsellor (GC) has already been described in section 1.2; however, this section provides additional background information regarding the role of a GC in Irish secondary schools. As legally mandated in Irish Statute Book (1998), every Irish secondary school student has access to a GC to assist them in their educational and career choices. That being said, all GCs do not spend all of their time on guidance counselling. According to Institute of Guidance Counsellors (2024a), 43.6% of GCs are exclusively involved in guidance counselling as part of their role in the school, whereas the other 56.4% combine guidance counselling with other duties, such as subject teaching and school administrative roles.

2.6 Terminology Clarification

In Ireland, a 'college course' defines the collection of college classes taken over the duration of the student's degree. An example of a college course in Ireland is computer science at TCD, encompassing 48 classes taken over the four year degree[13]. In the United States of America (USA), a 'college major' is equivalent to the Irish definition of a college course. As a result, the terms 'college course' and 'college major' are seen as equivalents and will be used interchangeably throughout this dissertation.

[13] https://www.tcd.ie/courses/undergraduate/courses/computer-science/

2.7 Common Types of Recommender Systems

As mentioned in chapter 1, an RS is a technology that recommends items to users. There are many different types of RSs, and in this section we will describe three types of RSs that are relevant to this dissertation. It is worth noting we will not describe their advantages or limitations, instead a straightforward definition will be provided. We will follow the definitions as described in Aggarwal et al. (2016).

A Content-Based Filtering (CBF) RS uses information explicitly provided by the user in order to make recommendations. For example, a CBF RS would recommend biology to a user because they explicitly expressed interest in the biological sciences.

A Collaborative Filtering (CF) RS requires the information about other users who have also interacted with the RS. A CF RS recommends items to a user that other similar users also liked. For example, a CF RS would recommend computer science to a user, because other users with similar interests also liked computer science.

A Knowledge-Based Filtering (KBF) RS is much more direct and asks the user for explicit specifications, recommending items to users that fit these specifications. For example, a user may explicitly specify they want business courses in Dublin, and then the user may be recommended commerce in University College Dublin (UCD).

A Hybrid RS combines multiple types of RSs into one RS. For example, a Hybrid RS may utilise CBF and KBF approaches in generating recommendations. For example, a user may be asked to explicitly specify the colleges they want to study at (KBF), and then be recommended college courses based on their gathered preferences (CBF).

2.8 Chapter Summary

In summary, this chapter described the workings of the Irish secondary school and college admissions system, which is relevant information as our research is focused on the Irish context. As well as this, we described fundamental concepts of RSs. With this chapter complete, we are better equipped to move on to our literature review.

Chapter 3

Literature Review

In this chapter, there will be a review of the literature regarding the fundamentals of college course choice. In light of this, we will examine closely-related works, which will allow us to clearly define our research gap.

3.1 Fundamentals of College Course Choice

In order to design a college course RS, we must first gather some information about the user, and this information will then be used to recommend suitable college courses to the user. Should the RS ask the user about their hobbies, grades, values or other information we are not aware of? Which of these factors are relevant or irrelevant to their college course choice? These difficult questions are what research objective 2 seeks to answer: "Identify the relevant information an RS should question a user to recommend suitable college courses".

However, in order to answer research objective 2, we must first understand the fundamentals of college course choice, which is the intention of this section. This will include a review of why people chose their college course and college, and the socio-demographic influences on their decision. As well as this, we will explore the theories of career choice and job satisfaction. Ultimately, this will assist us answer research objective 1: "Provide an answer to the question: *What college course should people study?*". In turn, this will assist us answer research objective 2, which would ultimately enlighten the final design of our RS.

3.1.1 College Course Choice Justifications

As established in chapter 1, choosing a college course is a major life decision. As a result, we can assume that students choose a particular college course for a set of reasons. According to Matthews et al. (2024); Gillis and Ryberg (2021); Mullen (2011), the most common and prominent justifications students cited when choosing their college course can be broadly categorised as:

- Interest in the subject area.
- A sense of meaning derived from the course.
- Satisfying parent's desires.
- Work/life balance.
- Salary expectations and job prospects after graduation.

This is by no means an exhaustive list; however, it does cover the most prominent justifications provided by students when justifying their college course choice. As well as this, these justifications were all commonly cited across genders and race/ethnicity. However, there were different priorities across gender and race/ethnicity. For instance, Gillis and Ryberg (2021) found that on average, men are more likely to be career-oriented when choosing their college course compared to women, whereas in Matthews et al. (2024), men were found to be more likely to cite salary expectations and job prospects as their justifications for choosing their college course. Despite these differences, the papers still conclude that all the above justifications are relevant across the differing genders and races/ethnicities.

3.1.2 College Choice Justifications

Choosing a college course is one decision, but a separate and closely related decision is choosing the college itself. For example, a student may settle on studying computer science, but the next question to ask is: *"Where do I study computer science?"* Aydin (2015); College MatchPoint (2018); Baharun (2010) have found that students provide a variety of justifications for their college choice, the most prominent justifications include:

- Prestige or reputation.

- Practicality with college location.
- Friends attending the college.
- Departments or academic staff.
- Family members being college alumni.
- Sports clubs or societies.

Once again, this is by no means an exhaustive list; however, it does cover the most prominent justifications of students' college choice. That being said, there seem to be cultural differences in priorities of college choice justifications. For instance, in Baharun (2010), a study of Malaysian college students, the college's reputation was the most important justification in college choice, whereas in Telli Yamamoto (2006), a study of Turkish college students, departments and academic staff were the most important justification in college choice. While we can clearly observe cultural differences in the most important justifications for college choice, we can make a very similar conclusion from the previous section: despite these differences, the papers still conclude that all the above justifications are still relevant across different cultures.

3.1.3 Socio-demographic Influences

In the previous subsections, we examined the justifications students provided in relation to their college and college course choice. However, we will discover that a student does not make this decision in a vacuum, and there are a variety of socio-demographic influences on their college course choice. Understanding these influences will assist us in answering the question posed by research objective 1: *"What college course should people study?"*.

Parent's Educational Background

In this discussion, we will explore the influence of a parent's educational background on a student's college and college course choice. According to Leighton and Speer (2023), there is a strong correlation between a student's college course choice and their parent's educational background. According to their findings, students with more educated parents will lean towards college courses with lower early-career earnings, but with faster growth towards higher earnings later in their career. On the other hand, according to Leighton and Speer (2023), students with less educated parents will lean towards college courses with more defined career paths and higher early-career earnings. As a result, students with less educated parents prioritise 'safer' college courses with more defined career paths that yield higher early-career earnings. There are analogous findings in Mullen (2011): working-class parents generally encourage their children to attend college courses that directly prepare them for careers, whereas middle class parents generally encourage college courses where their children can explore their interests. Moreover, Mullen et al. (2003) has shown that the parental educational background of a student has a strong correlation with the student's attendance in a postgraduate degree programme. The findings above all give credence to the idea that an individual's college course choice is strongly influenced by their parent's educational backgrounds.

Gender

In this discussion we will explore the influence of gender on college course choice. Figure 3.1 visualises the proportional gender distribution across different fields of study in Irish colleges. As is clearly observable, many fields of study are not equal in gender. To give some extreme examples of college course participation in Ireland, according to Higher Education Authority (2024b), fisheries has 100% male participation, and pre-school education has 97% female participation, demonstrating the large gaps in gender proportions across different fields of study in Ireland.

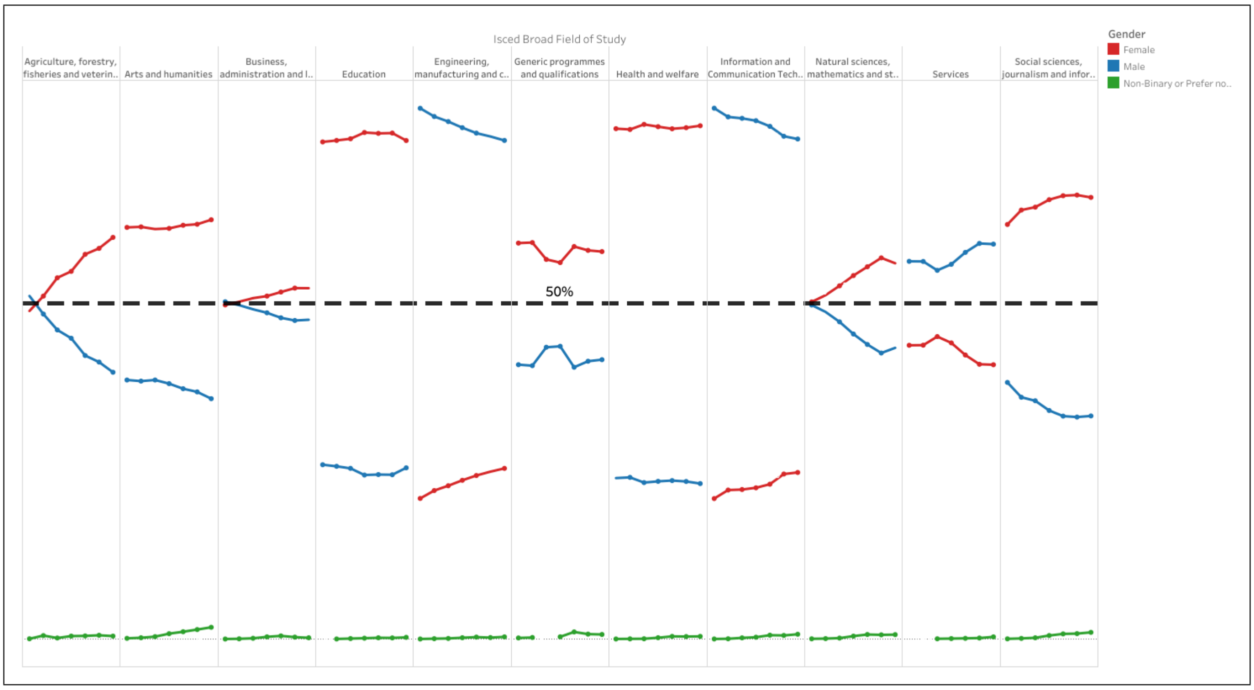


Figure 3.1: Mean gender differences across different fields of study in Irish colleges. Red=Female, Blue=Male, Green=Non-binary or Prefer not to say. Source: Higher Education Authority (2024b).

The above figure 3.1 and aforementioned statistics clearly demonstrate gender differences across different fields of study. These inequalities result in negative experiences for the less-represented gender. For instance, Slattery et al. (2023) conducted a qualitative study on 21 Irish female students studying either mathematics (66% male), physics (68% male), computer science (74% male) or engineering (71% male) in Ireland - parenthesised data is from Higher Education Authority (2024b). Slattery et al. found that during their studies, females felt under-valued by their male peers, which negatively impacted their self-beliefs and interest in the course subject matter. In addition to this, they found that females suffered more pressure to perform, and were less likely to engage in the learning environment, showcasing an overall negative experience for females in male-dominated fields of study.

Experiences of isolation and lack of belonging due to gender differences, such as the example given above, are well-known. To provide an example, in I Wish (2025), a survey of Irish secondary school girl's attitudes towards Science, Technology, Engineering and Mathematics (STEM), 65% of girls reported that "Poor gender equality in STEM careers" is a barrier to continuing their studies in STEM. This is quite striking considering that girls express these viewpoints before they even begin studying in these STEM-related fields. These feelings act as deterrents for girls considering studying in STEM, influencing their college course choice solely on their gender.

Race and Ethnicity

In this discussion, we will explore the influence of race and ethnicity on college course choice. We were unable to find statistics for racial or ethnic differences by college course study in Ireland, so statistics from Hinrichs (2015) are used, which was collected from the National Center for Education Statistics in the US Department of Education.

According to analysis from Hinrichs, there are a handful of striking racial or ethnic differences by college course choice. For example, as shown in Figure 3.2, 31% of Asian students study a STEM-related college course, a much higher proportion compared to 11% black, 13% hispanic and 15% white students. An additional example of racial or ethnic differences by college course choice can be found in Figure 3.3, which examines the proportion of race and ethnicity in Public Administration and Social Service: nearly 4% of black students study in this field, a relatively large difference compared to under 1% for Asian, 2.5% hispanic and 1.5% white students studying in this field.

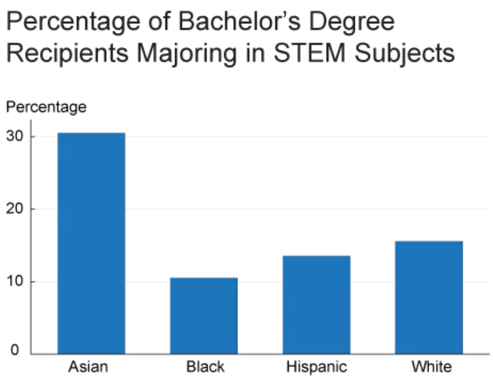


Figure 3.2: Source: Hinrichs (2015).

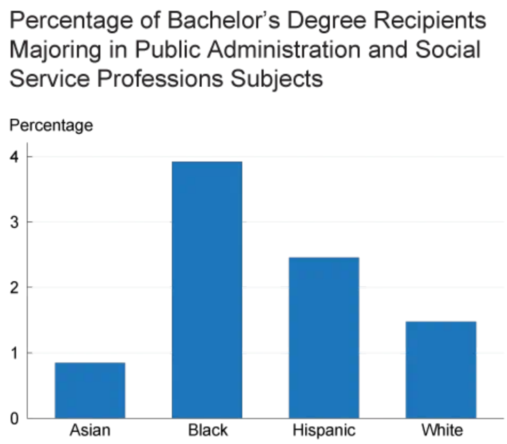


Figure 3.3: Source: Hinrichs (2015).

Further to this analysis, we can examine the work conducted by Dickson (2010), a study performed on administrative data from three Texas Universities (n=127,330). Similar proportions of findings were reported, for example Dickson found that 37% of science-related college course students were Asian, with 26% white, 24% black and 23% hispanic students being the other prominent racial/ethnic groups. As to why this may be the case, Xie and Goyette (2003) argues that Asian-Americans prioritise objective credentials and 'hard skills', which are more difficult to discriminate against in job interviews that are more focused on technical ability, as opposed to job interviews focusing more on 'soft skills'. This poses a potential explanation for the relatively high proportion of Asians in science-related college courses compared to other racial/ethnic groups.

Overall, the findings of Hinrichs (2015) and Dickson (2010) demonstrate the differing proportions of race or ethnicity across college course choices in the USA. In this discussion, there was particular focus on the high proportion of Asians in science-related college courses, for which Xie and Goyette (2003) provided a potential explanation. More generally, this demonstrates the undeniable influence of race and ethnicity on a student's college course choice.

Socio-demographic Influences Summary

As is clearly evident from the review in this subsection, there are a variety of socio-demographic factors that influence a student's college course choice, in particular, their gender, race or ethnicity, and parent's educational background.

3.1.4 Theories and Hypotheses of Career Choice

In this subsection, we will turn our attention to academic literature and popular sentiment to assist us answer the question posed in research objective 1: *"What college course should people study?"*

Holland's Theory of Career Choice

Holland's theory of career choice, first presented in Holland (1959), posits that individual's experience satisfaction when their personality traits match the traits of the role in their work. Holland proposes his own six categories of personality traits present in roles of work, and are defined as:

1. **Realistic:** The doers. They typically enjoy work that involves using their hands, which include making or tinkering with physical objects. Examples of Realistic jobs include construction workers and bakers.
2. **Investigative:** The thinkers. They are intellectually curious and enjoy work that involves observation, research or analysis. Examples of Investigative jobs include scientists and philosophers.

3. **Artistic:** The creators. They enjoy expressing themselves creatively and appreciate a lack of structure in solving problems. Examples of Artistic jobs include graphic designers and actors.

4. **Social:** The helpers. They typically enjoy working with and helping people, and have strong interpersonal skills. Examples of Social jobs include teachers and nurses.

5. **Enterprising:** The persuaders. They typically enjoy managing people or influencing others, and are often confident and assertive. Examples of Enterprising jobs include lawyers and managers.

6. **Conventional:** The organisers. They typically enjoy work that involves order and organisation, with strong attention to detail and adherence to rules. Examples of Conventional jobs include accountants and librarians.

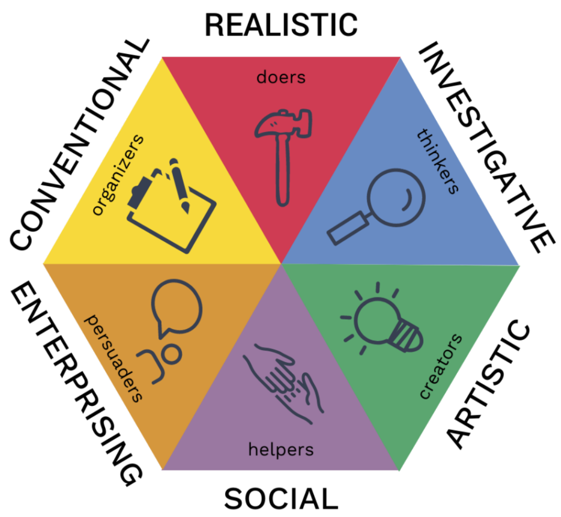


Figure 3.4: Holland's RIASEC model. Source.

These personality traits in work are alternatively defined as the RIASEC model. A visual graphic of Holland's RIASEC model can be seen in figure 3.4. Holland remarks that individuals may exhibit many of the above personality traits simultaneously, and do not necessarily exhibit just one trait. As well as this, the traits are not simply binary and have gradations of strength - in other words, a person may demonstrate strong Investigative qualities, moderate Social qualities and weak Realistic qualities. Holland developed an identification system called "Holland codes" that involves the first letter of each of his six traits. Holland codes are essentially used to classify people or jobs under Holland's proposed categories. To give some examples, a digital marketer would generally be classed as an “AE” Holland code, meaning the role exhibits Artistic and Enterprising qualities, as the role involves designing campaigns to influence public perception (Enterprising) and creating visually-appealing displays (Artistic). A surgeon would generally be classed as an “RIS” Holland code, meaning the role exhibits Realistic, Investigative and Social qualities, as the role involves strong dexterity with hands (Realistic), deep understanding of human biology (Investigative) and strong patient communication skills (Social). Generally, a Holland code is no longer than 3 letters.

It is worth mentioning that Holland’s proposed personality traits are nuanced and not simplistic. For instance, a biologist may demonstrate Investigative qualities through their fascination with human biology, whereas a sociologist may demonstrate Investigative qualities through their interest in subcultures. This does not imply that a biologist would be interested in subcultures, or that a sociologist would be fascinated with human biology. In other words, Holland’s personality traits manifest themselves differently in different people.

The impact of Holland’s theory on the field of Vocational Psychology cannot be overstated - according to Spokane and Cruza-Guet (2005), Holland’s theory is the most extensively researched career theory. That being said, some aspects of Holland’s theory have been challenged. For instance in Einarsdóttir et al. (2020), they argue the personality model may not appropriately fit the vocational interest structure of Icelanders. In spite of these imperfections, large bodies of research have mostly supported Holland’s personality model among high-school students, college students and working adults, according to Nauta (2010), demonstrating the theory’s robustness over time.

Life-Span, Life-Space Theory

Despite the evident success and utility of career fit theories’ such as Holland’s, there are limitations to these kinds of theories. Most notably, matching an individual’s traits to a job’s traits involves the individual making a decision at a particular point in time. Super (1980) argues that an individual’s career is not one static decision. Rather, Super poses that career development is a lifelong process with a series of developmental stages, in which the individual’s self-concept evolves over their lifespan. In this view, an individual’s career is the cumulative outcome of many decisions taken at different points in time. Super coined this as Life-Span, Life-Space Theory (LSLST). In spite of the prominence of LSLST, it has faced criticisms. For instance, LSLST inadequately accounts for the dynamic careers of women, which are often interrupted by caregiving, according to Fitzgerald and Betz

(1994). It is worth noting that Holland's theory is not at odds with Super's theory, rather Super intended to complement the established utility of Holland's theory, according to Greenhaus and Callanan (2006).

The Passion Hypothesis

One of the most popular responses to the question *"What college course should people study?"* is to *"Follow your passion"*. This sentiment was birthed in Bolles (1972), a self-help book for job-hunters, which sold over ten million copies[1]. The sentiment gained a surge in popularity when Steve Jobs, the co-founder of *Apple*[2], famously claimed in his commencement speech at the 2005 Stanford University Graduation: *"The only way to do great work is to love what you do"*[3]. Jobs' speech garnered a large amount of popularity, amassing over 48 million views[4] on YouTube, demonstrating the sentiment's widespread appreciation.

Newport (2016) coins this as the Passion Hypothesis: the idea that to find great work, you simply find out what you love and follow those interests. As appealing as "Follow your passion" sounds, the advice has many limitations. The underlying implication of this advice is that passions are pre-existing, and neglects the idea that passions can be developed. This then prompts the question: *"Can passion be found or developed at work?"*, which is answered in O'Keefe et al. (2018). Following the definition in Dweck (2006), in O'Keefe et al.'s, experiment, the participants were broadly categorised into "fixed theorists", who believed that personal interests are relatively fixed, and "growth theorists" who believed personal interests are developed. According to their findings, the fixed theorists were more likely to dampen their interests in areas outside of their existing interests, suggesting a level of narrow-mindedness for the fixed theorists. As well as this, when engaging in a new interest became difficult, the interest dropped significantly more for fixed theorists in comparison to growth theorists, suggesting that growth theorists have more endurance when encountering inevitable difficulties when pursuing their interests.

As was argued, the Passion Hypothesis is limited career advice, and we desire a more nuanced answer to the question posed by research objective 1: *"What college course should people study?"*.

[1] As of 2016. Source
[2] `https://www.apple.com/`
[3] `https://www.youtube.com/watch?v=UF8uR6Z6KLc`
[4] As of March 2026.

3.1.5 Job & Career Satisfaction

In this subsection, we will examine the satisfaction of individuals in their jobs and careers. These insights will assist us answer the question posed in research objective 1: *"What college course should people study?"*.

Essential to the discussion of job satisfaction is Herzberg's Two-factor theory as outlined in Herzberg (1959). Herzberg's theory posits that job satisfaction is influenced by two factors, which are coined as 'Motivators' and 'Hygiene Factors'. Motivators lead to job satisfaction, whereas Hygiene Factors prevent job dissatisfaction. Examples of Motivators include interesting work, achievement, autonomy and a sense of meaning, and examples of Hygiene Factors include salary, job security, colleague relationships and status. Essentially, Herzberg's theory claims that job satisfaction has different factors to job dissatisfaction. For example, an individual may be in a high-paying, respectable job with positive colleague relationships, however the individual may not derive a sense of meaning or have interest in their job. According to Herzberg's theory, the individual would not be dissatisfied with their job - after all, the job does satisfy certain criteria - although the individual would not be incredibly satisfied with their job given the absence of Motivators.

Many of Herzberg's claims are supported in the literature - for the sake of brevity, we will focus on the impact of interest or salary on job satisfaction. For instance in Hoff et al. (2018); Nye et al. (2012), they found statistically significant positive correlation between interest-job fit and job satisfaction. According to a meta-analysis conducted by Judge et al. (2010), they found a weak relationship between salary and job satisfaction, giving credence to Herzberg's theory with regards to salary being a Hygiene Factor (prevents job dissatisfaction) as opposed to a Motivator (provides job satisfaction). In spite of the findings of a weak relationship between salary and job satisfaction, we would argue that prioritising salary in work is understandable. According to National Youth Council of Ireland (2025), three of five Irish adults under 25 years old are considering emigrating due to the rising cost of living in Ireland, clearly demonstrating the importance of salary in work.

3.1.6 Summary and Synthesis of Findings

In summary, we first examined the common justifications students provided for their college course choice, which included interest, salary and a sense of meaning. Secondly, we examined the common justifications students provided for their college choice, which included prestige/reputation and location/practicality. As well as this, students' college course choices were found to have socio-demographic influences, such as their parent's educational background, gender and race/ethnicity. We also examined Holland's RIASEC model and dissected the popular advice "Follow your passion". In addition to this, we examined the factors of job satisfaction, including the strong relationship between interest-fit and job satisfaction, and the weak relationship between salary and job satisfaction.

Ultimately, the purpose of this section was to answer the question posed by research objective 1: *"What college course should people study?"*. As is evident from our investigations, the answer to this question is not straightforward. Despite this, we will attempt to answer this question below by synthesising this section's findings:

Given the significant relationship between interest-job fit and job satisfaction, an individual should consider studying a college course that they either **find interesting,** or can see themselves **developing an interest in.** As well as this, given the robustness and ubiquity of Holland's RIASEC model, an individual should consider studying a college course that aligns with their **RIASEC model.** Moreover, given the aforementioned rising cost of living in Ireland, students may also consider college courses for their **salary** or **job prospects.** Many students expressed interest in college courses or careers that are meaningful, so a student may also consider a college course for their sense of **meaning** or **impact.** Beyond the college course, a student should also consider the actual **college** at which to study their chosen course, which may impact their decision depending on the availability of the college course at different colleges.

This is of course, a simplified answer to a very multi-faceted question, and we would like to acknowledge that this answer serves more as a guideline, as opposed to a "one-size-fits-all" answer. In addition to this, the factors provided in this answer were deemed to be the most prominent factors given the reviewed literature. Although as is evident from our prior review, there are many reasons as to why a student may be satisfied with their college course choice, and this reason may be beyond the scope of the provided answer. For example, a student may choose to study a particular course primarily because they are seeking positive work/life balance.

The reader may notice that a handful of the previously discussed topics, such as gender, did not feature in this dissertation's attempt at answering the question: *"what college course should people study?"* We would argue that having an awareness of these factors is still useful in understanding the fundamentals of college course choice, the ultimate goal of this section. We can now consider research objective 1 achieved.

3.2 Closely-Related Work

As explored the previous section, choosing a suitable college course is a multi-faceted, difficult decision for students. Technologies, not limited to an RS, have been used to assist students in finding suitable college courses for them. In this section, we will examine the academic literature and real-world products for technologies like this. We will start with the simplest technologies, and work our way up to more sophisticated solutions in assisting students with their college course choice.

3.2.1 Simple Filters

CareersPortal[5] and Qualifax[6] are the two leading college course databases in Ireland[7]. The simplest technology for assisting Irish students in their college course selection is to use the preset filters on these websites as you search their college course database. CareersPortal allows users to filter by multiple features shown in Figure 3.5, which includes filtering by career sector (see Figure 3.6), and career interests[8] (see Figure 3.7). After applying some of these filters, a user would be presented with a result, an example of which is shown in Figure 3.8.

[5] `https://careersportal.ie/`
[6] `https://www.qualifax.ie/`
[7] Qualifax has over 500,000 users annually. Source.
[8] CareersPortal's career interests are explained in chapter 4

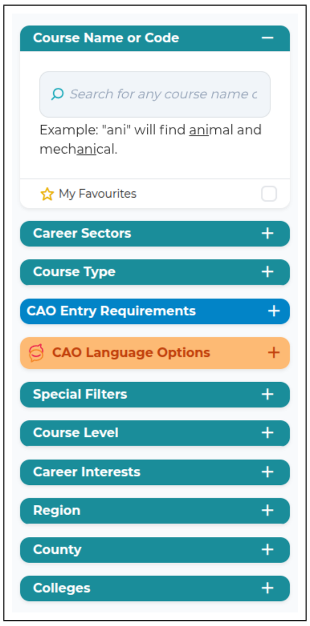


Figure 3.5: CareersPortal's college course filters.

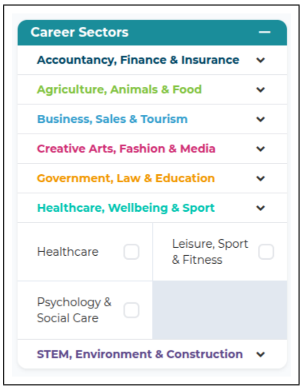


Figure 3.6: CareersPortal's career sector filter.

Figure 3.7: CareersPortal's career interests filter.

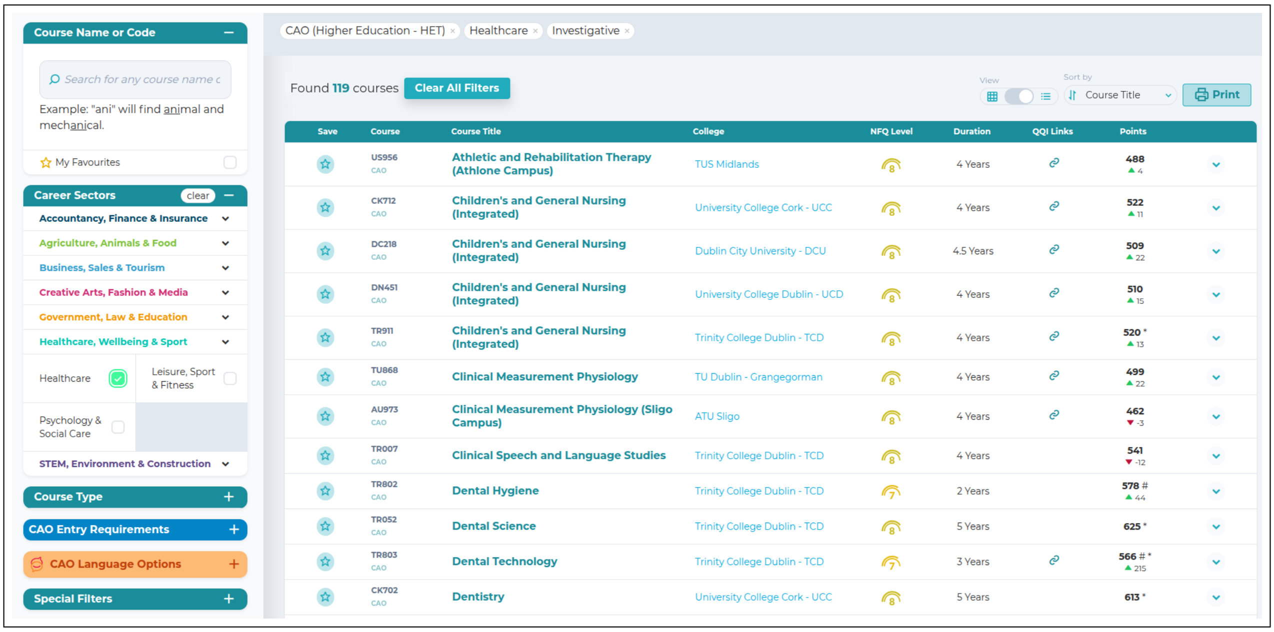


Figure 3.8: A sample result of what users would see after applying some filters to a course search in CareersPortal.

Qualifax operates similarly: they allow user to filter their college course database by numerous filters, including by college course category. After applying some of these filters, a user would be presented with a result very similar to the CareersPortal site as shown in figure 3.8.

Simple filters in college course search, such as the CareersPortal and Qualifax websites, are useful as they allow students to scan a large variety of courses. However, the strength of these filters are also its greatest weakness: the quantity of choice. Even after applying a handful of filters, for example in Figure 3.8, a user is still presented with 119 courses. This is a large quantity of courses, which may result in a phenomenon known as overchoice, coined in Toffler (1970). According to Toffler, overchoice is a paradoxical phenomenon where choosing between a large variety of options can be detrimental to decision-making processes. As a result, we find that simple filtering technologies present large quantities of college course options; however, this does not necessarily make the decision-making process easier. This calls for technologies to prune their results and present more tailored college courses to the student, setting the scene for our next subsection.

3.2.2 College Course Quizzes

In this subsection, there will be a discussion on online quizzes that provide users with a more tailored, personalised list of college course recommendations. As a side note, seeing as these are real-world products, we will not have an insight into the recommendation algorithms behind these quizzes; we will only have access to their front-facing user interface.

For this discussion, we will examine the college major quizzes offered by SuperProf[9], Anderson University[10] and Marquette University[11]. These quizzes ask the user close-ended questions, and generally the questions are tasked with discovering the student's interests in different areas of study. Examples of the kinds of questions that get asked are seen in Figures 3.9 and 3.10.

[9]`https://www.superprof.com/blog/college-major-quiz/`
[10]`https://anderson.edu/persona-quiz/`
[11]`https://www.marquette.edu/academics/majors/choose-your-major/`

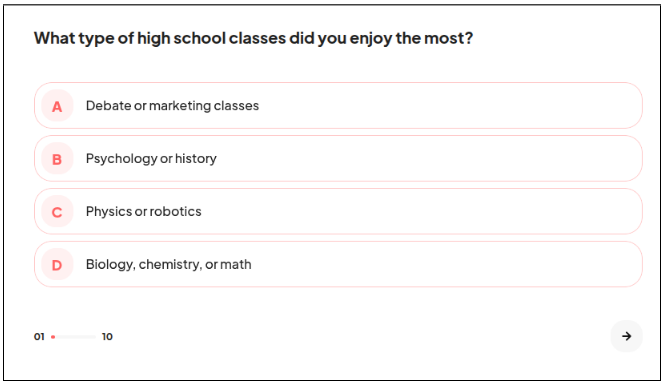


Figure 3.9: A sample question provided on SuperProf's college major quiz.

Figure 3.10: A sample question provided on Anderson University's college major quiz.

After answering all the questions, the quizzes provide the user with a suggested list of college majors based off the answers provided to the questions. Examples of results are given in Figure 3.11 and 3.12.

Figure 3.11: A sample results page after finishing the college major quiz on Anderson University.

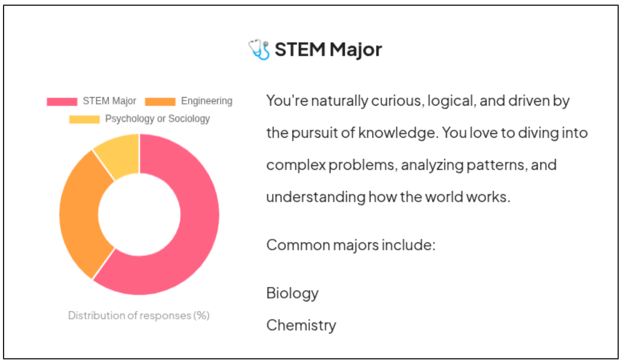


Figure 3.12: A sample results page after finishing the college major quiz on SuperProf.

In spite of the evident utility of these quizzes, they do pose some limitations. Firstly, considering the importance of the college course decision process, these quizzes are generally very short - the three quizzes provided had 10-25 questions each. If we take Marquette University's quiz as an example: 79 college majors are offered[12], and ten questions may not be considered sufficient to gauge a student's interest appropriately across the 79 college majors. In other words, the length of the quizzes imply they are trivialising and simplifying the solution to the problem.

In addition to this, generally the results of these quizzes may not refer to specific

[12]According to `https://www.marquette.edu/academics/majors/`

college majors, although they may refer to broader fields of study. For example as we see in Superprof's results in figure 3.12, STEM is being suggested to the user, however STEM encompasses a wide range of college majors including mathematics and engineering - similar, but still different fields of study. While this may guide the user toward narrowing their college major choices, there is still a lack of specificity in the college major recommendations.

However, if one wants to go beyond broader fields of study and have more specificity in their college course recommendations, the technology is required to have knowledge of specific college courses being offered at specific colleges. This introduces a new requirement of the technology having up-to-date, accurate college course information being offered at multiple different colleges. Moreover as mentioned in chapter 2, the college admissions system and entry requirements differ by country, and clearly the Irish college system differs to the American college system that these quizzes refer to. Seeing as this dissertation focuses on Irish college course recommendations, these requirements call for a technology with a working knowledge of the Irish college courses and college system, and one that could suggest specific, personalised college courses according to the Irish college system.

The technology described above already does exist: *FindMyCollegeCourse* (FMCC)[13]. To the best of our knowledge, this is the only online quiz that recommends Irish college courses. FMCC addresses many of the aforementioned limitations of other college major quizzes: the quiz has over 100 questions, demonstrating much more breadth in capturing student's interests and preferences. Additionally, FMCC recommends specific Irish college courses as opposed to broad fields of study, and the college course information is up-to-date and accurately maintained - a challenge given the volume of new courses being added each year. Naturally, the quiz is focused in the Irish college domain, which is also the focus of this dissertation. That being said, there are still some limitations and improvements which can be applied, although a more thorough discussion of this will be provided in section 3.3.

3.2.3 College Course Recommender Systems

In this subsection, we will turn our attention from real-world products, and examine the intersection of RSs and college course recommendations in the academic literature. In this discussion, we will particularly focus on the type of RSs being employed. A further

[13] `https://findmycollegecourse.ie/`

discussion of other features of these RSs will be provided in section 3.3. An understanding of the common RS types, as discussed in section 2.7, will be useful in understanding this discussion. A comparison of the different RS types is best demonstrated in the table 3.1:

Paper reference	**RS type**
Qamhieh et al. (2020)	Content-Based Filtering
Ghuge et al. (2023)	Hybrid: Content-Based Filtering & Collaborative Filtering
Lahoud et al. (2023)	Hybrid: Content-Based Filtering & Knowledge-Based Filtering
Su et al. (2021)	Collaborative Filtering

Table 3.1: A comparison of the RS types used across closely-related works in literature.

It is worth noting that in Lahoud et al. (2023), they tested a wide variety of RS types, and in their specific case, a hybrid RS with Content-Based Filtering and Knowledge-Based Filtering yielded the best college course recommendations. That being said, as is clear from table 3.1, there is no common consensus on the most optimal RS type for the problem of college course recommendations, suggesting that a solution to this problem may adopt many of the approaches above.

3.3 Research Gap

In this section, we will synthesise our background knowledge of college course choice fundamentals, gathered in section 3.1, with our knowledge of closely-related work as explored in section 3.2, in order to clearly define a research gap. To assist us with this, tables 3.2 and 3.3 define different features of the closely-related work referenced in section 3.2:

Table 3.2: Comparing features of all examined closely-related work (Part 1)

Name	**# Questions**	**# Results**	**Recommendation type**
Superprof	10	3	General area of study
Anderson Uni	25	50	College major
Marquette Uni	10	30	College major
FMCC	∼150	200	Irish college course
Qamhieh et al. (2020)	125	7	Engineering course
Ghuge et al. (2023)	6	5	College major
Lahoud et al. (2023)	6	5	Lebanese college course
Su et al. (2021)	90	5	Taiwanese college course

Explanation of columns:
 # Questions: the number of questions asked to the user prior to receiving their college course recommendations.
 # Results: The number of college course recommendations the user receives.
Recommendation type: the type of college course recommendation the user receives.

Table 3.3: Comparing features of all examined closely-related work (Part 2)

Name	**Can filter results by college?**	**Gathered user information**
Superprof	✗	Academic interests
Anderson Uni	✗[14]	Academic interests
Marquette Uni	✗[15]	Academic interests
FMCC	✗[16]	Academic interests
Qamhieh et al. (2020)	✗	Hobbies, grades, MBTI[17]
Ghuge et al. (2023)	✗	Hobbies, grades
Lahoud et al. (2023)	✓	Hobbies, gender, academic interests
Su et al. (2021)	✗[18]	RIASEC personality model

Explanation of columns:
Can filter results by college? Is it possible for the user to specifically receive college course recommendations from colleges of their choice?
Gathered user information: What information is the system gathering off the user in order to make personalised college course recommendations?

[14] The only college majors recommended are ones offered from Anderson University.
[15] The only college majors recommended are ones offered from Marquette University.
[16] You can filter your colleges by region, for example just the colleges in Dublin, Munster, etc.
[17] Myer-Briggs Type Indicator (MBTI); a personality test developed in Myers et al. (1962).
[18] The only college majors recommended are ones offered from Cheng Shiu University, Taiwan.

With a clearer overview of the differing features of all the examined closely-related work, we can now clearly define our research gap. By examining table 3.3 above, we can observe that no closely-related works have applied the combination of Holland's RIASEC model and academic interests. Given the ubiquity and robustness of Holland's RIASEC model, as described in subsection 3.1.4, and the statistically significant positive correlation between interests and job satisfaction, as described in subsection 3.1.5, we are presented with an opportunity to merge Holland's RIASEC model and the user's academic interests in providing personalised college course recommendations. It is worth noting that a more elaborate justification of related design decisions will be provided in section 4.3.

In addition to this, when we examine Table 3.2, few of the examined closely-related works recommend specific college courses in a particular country - with just one in Ireland (FMCC). The Irish college admissions system presents an interesting use-case for this problem for many reasons. Firstly, all college applications go through the Central Applications Office (CAO), which is a centralised system, as opposed to other countries, such as the USA, where students must apply to colleges directly. Secondly, the CAO requires the student to create a ranked list of their preferred college courses. Seeing as RSs provide their users with ranked lists of items, it is quite appropriate to apply an RS to the domain of Irish college course recommendations. As a result, the Irish CAO system presents an interesting case study for research to be conducted in the context of a college course RS problem.

As we discovered in subsection 3.1.2, the choice of college is an important factor in a student's college course decision-making process. In Ireland, there is no work that allows a user to be recommended college courses for specific colleges[19]. As a result, this research seeks to allow users to filter their college course recommendations by specific colleges, a research gap in the context of Irish college course RSs.

We can now succinctly and clearly define our research gap: a college course RS applied to the **Irish context,** utilising a user's **RIASEC model** and **academic interests**, allowing the user to filter their college course recommendations by **specific colleges.**

3.4 Chapter Summary

This chapter started with an investigation into the fundamentals of college course choice, which allowed us to answer the question posed by research objective 1: *"What college*

[19]As mentioned in table 3.3, FMCC allows users to filter their colleges by region, but not specific colleges.

course should people study?". Answering this question is relevant, as it provides us with the foundations to answer research objective 2: "Identify the relevant information an RS should question a user to recommend suitable college courses", which will be explored in the next chapter. In addition to this, we examined closely-related works and clearly defined our research gap. This provides us with more clarity and foundation to move towards designing our college course RS.

Chapter 4

Design

In this chapter, there will be an explanation into the high-level design of the college course RS, including justifications behind different design decisions. Before we dive into the design behind our RS, it would be important to revisit our research question: *"To what extent can a Recommender System assist an Irish student in exploring college courses they would consider studying?"* Recall that we hypothesised that an RS that provides relevant, diverse, and serendipitous recommendations would be best in answering this question. This describes the overall design philosophy behind our RS, and it shall be the underpinning behind almost every design decision.

4.1 Design Methodology

Prior to exploring the design of this dissertation, we shall describe the design methodology, which also describes the overarching structure of the chapter. To start, interviews were conducted with two professional Guidance Counsellors (GCs) in Ireland. This provided us with a deeper insight into the practice of guidance counselling, which will inspire the final design of our RS. With this information, we can more appropriately approach research objective 2: "Identify the relevant information an RS should question a user to recommend suitable college courses", which will be addressed in this chapter. With a clearer idea of the information our RS will gather from a user, we can discuss the dataset that will be utilised for this problem, including some dataset augmentations that will more appropriately answer this dissertation's research question. With our dataset designed, we can discuss the filtering algorithms used by the RS, which will be used to provide recommendations to the end user. Lastly, we can discuss the design of the quiz itself, which is an integral piece to the design of the RS as it will gather information from the user, allowing the RS to provide personalised college course recommendations. With our design methodology outlined, we can proceed with the rest of the chapter.

4.2 Interviews with Guidance Counsellors

As part of this research, interviews were conducted with two professional GCs in Ireland. The main purpose of these interviews was to gain a deeper understanding of the GC process. In particular, we were curious as to what information a GC asks a student in order to make college course recommendations, which would help achieve research objective 2: "Identify the relevant information an RS should question a user to recommend suitable college courses". Naturally, this exploration is relevant as we will be designing an RS that provides personalised college course recommendations. This section will feature some insights gleaned from both interviews, and they will inspire the final design of this dissertation's RS.

4.2.1 Academic Interests

As part of our defined research gap in section 3.3, we planned to utilise a user's academic interests to provide personalised college course recommendations. Both GCs were asked about the relevance of academic interests in making college course recommendations, and they were in agreement that academic interests are incredibly useful in this context. As part of their GC process, they generally ask their students about their favourite and least favourite subjects. While this information is evidently useful, we can recall the information about LC subjects in section 2.1: in student's last years of secondary school, they take at least six LC subjects. Arguably, if we only rely on a student's LC subjects as a source for information for academic interests, we may miss out on a student's academic interests outside of the LC subjects they take. For example, a student may demonstrate strong interest in history, but if they do not study LC history, we may miss out on this information.

To account for this when designing our RS, we should consider a set of academic interests with broad coverage across all areas of study. As a result, our RS can capture this information, even in the event where the student did not study the subject at school. It is worth mentioning that currently the term 'academic interests' is quite vague - a more specific definition of academic interests will be provided in subsection 4.4.2.

4.2.2 The RIASEC Model

As part of our defined research gap in section 3.3, we also planned to utilise a user's RIASEC model to provide personalised college course recommendations. In light of the ubiquity and robustness of the RIASEC model as established in section 3.1.4, we asked both GCs for their opinion on the RIASEC model. One of the GCs found the RIASEC model to be a limiting snapshot of a person, and they place little value on the RIASEC model in their GC process.

In fact, we are also in agreement that the RIASEC model provides a limited snapshot of the user. As mentioned in section 3.1.4, the RIASEC traits are nuanced. For example, a biologist and philosopher may separately demonstrate Investigative qualities; however they may not be interested in each other's field. Thus, we can agree that a user's RIASEC model is a limited snapshot of the user when considered in isolation. However, we would argue that a user's RIASEC model may be useful in conjunction with academic interests. For example, suppose a student demonstrates an interest in 'computers' as an academic interest. With the additional information of a user's RIASEC model, we can provide more informed recommendations. For example, if this user demonstrates a high Realistic trait, they may be suited for more hands-on college courses relating to computers, such as computer engineering. On the other hand, if this user demonstrates a high Enterprising trait, they may be more suited for a college course like computer science with business, as the Enterprising trait is common in business studies. This additional RIASEC model information may appear redundant; however, we would argue this provides more context to the RS, allowing the RS to potentially provide more serendipitous recommendations, which assists us in answering our research question.

4.2.3 Hobbies

Note: In this context, 'hobbies' is being used as an umbrella term for hobbies, extracurriculars and personal interests outside of academics.

Both GCs were asked about the relevance of hobbies in their GC process, and whether they consider the information as redundant or relevant to the college course decision-making process. Both GCs agreed that hobbies can be useful in suggesting potential college courses to their students. For example, a student that loves watching history documentaries may be suggested to study history.

While we agree that information regarding hobbies can be useful in suggesting relevant college courses to a student, we would argue that a lot of this information could be captured under an academic interests quiz with a broad range of coverage, as mentioned in the subsection 4.2.1. Continuing our previous example, a student that loves history documentaries would likely respond positively to questions relating to the 'humanities' academic interest, which could lead to a recommendation for history anyway. In other words, with an academic interest quiz with broad coverage, information about a user's hobbies may be redundant in the context of college course recommendations.

4.2.4 The Potential Role of an RS in Guidance Counselling

Both GCs were made aware of the research question of this dissertation, and were asked to provide their honest evaluation of how a technology, such as an RS, could fit into the GC process.

One of the GCs remarked that their process is very instinctive and intuitive, and their meetings often do not follow a rigid structure. In most cases, the process begins by asking the student about their LC subjects, and then the discussion continues from there. As is evident from our investigations in chapter 3, the college course decision-making process is incredibly nuanced and personally confronting for students. This gives credence to the human dimension to guidance counselling, a profession that requires empathy and understanding on a human level. This is incredibly challenging to replace with a technology; we and both GCs agreed that a technology, such as an RS, should not be designed to replace the crucial role of a GC, but rather to complement the GC's process, allowing the GC to shift their priorities towards the more human, nuanced parts of the college course decision-making process.

During one of the interviews, there was a discussion on the value of diversity in recommendations. Interestingly, one of the GCs remarked that diversity in recommendations is incredibly valuable, which is something they integrate into their GC process. The GC claimed that they like to challenge their students with different recommendations, as it helps develop the student's concepts about the college courses they would consider studying. These insights are incredibly valuable, as it supports the idea that diverse recommendations can help answer our research question, which is to assist Irish students in exploring college courses they would consider studying.

4.2.5 GC Interviews Summary

In summary, there was unanimous agreement in the value of gathering a student's academic interests for the purpose of college course recommendations. This is crucial, as our RS will involve gathering a student's academic interests. The RIASEC model did not receive the same levels of agreement; however, we hypothesise that the inclusion of a RIASEC model could lead to serendipitous recommendations. Moreover, the GCs both generally gather hobbies off their students for the purpose of college course recommendations; however, we argued that a broad and comprehensive academic interests quiz could capture this information instead. We then discussed the potential role an RS like this could play in the GC process. In particular, an RS should complement the GC process by assisting a student with their college course exploration, instead of trying to replace the human GC process completely.

After our examinations into the literature in chapter 3, and our interviews conducted in this section, we have gathered a large body of background knowledge that will allow us to address research objective 2: "Identify the relevant information an RS should question a user to recommend suitable college courses", which will be answered in the next section.

4.3 User Information to Gather

It would not be possible for a GC to provide a student with relevant college course recommendations if the GC knows nothing about the student. The GC must first ask the student some questions in order to gain some understanding, which would then fuel the GC's college course suggestions. This is true for humans, but the same applies for technologies such as an RS. This provides the context surrounding research objective 2: "Identify the relevant information an RS should question a user to recommend suitable college courses". This section will answer this research objective, and this will be instrumental in the design of our college course RS.

Let us turn our attention to section 3.1.6, in which we attempted to answer the question: *"What college course should people study?"* Our answer involved factoring in a student's:

- Academic interests.
- RIASEC model.

- Desire for salary or job prospects after graduation.
- Desire for meaningful work.

Moreover, in table 3.3 we examined what other information the closely-related works gather from users in order to provide personalised college course recommendations:

- Hobbies.
- School grades.
- Gender.
- Myers-Briggs Type Indicator (MBTI).

Clearly, we are presented with many different categories of information to gather from a user. Now, the task is to identity which categories are relevant to gather from a user in the context of a college course RS. In the next subsections we shall justify the inclusion or exclusion of different categories to gather from a user, allowing us to achieve research objective 2: "Identify the relevant information an RS should question a user to recommend suitable college courses".

4.3.1 Academic Interests and RIASEC Model

In sections 3.3 and 4.2, we justified gathering a user's RIASEC model and academic interests in order to provide personalised college course recommendations. This was also part of our research gap identified in section 3.3. As a result, in our college course RS, we will gather a user's academic interests and RIASEC model.

4.3.2 Hobbies and School Grades

Given our justifications in subsection 4.2.3, we can disregard hobbies as information to gather from a user. We may also disregard school grades because a user's LC subjects provides a limited snapshot of their interests and aptitudes, and we will also cover this information as part of a more broad academic interests quiz, as explained in subsection 4.2.1.

4.3.3 Salary and Job Prospects

As described in subsection 3.1.5, a user's desire for salary or job prospects after graduation would be relevant information in the context of college course recommendations. However to the best of our knowledge, data on the mean salary or employment rate of graduates from each college course in Ireland are unavailable. As a result, it would be very challenging to effectively integrate this information into our RS. It is worth mentioning this will be discussed as a piece of future work in subsection 7.2.2.

4.3.4 Meaning

As also described in subsection 3.1.5, a user's desire for meaningful work would be relevant information to gather from a user in the context of a college course RS. However, we believe this is challenging to work with, as this information is difficult to translate to something computational. For example, we could consider creating something akin to a 'meaning score' for each college course: where a value of 1.0 implies a high degree of meaning associated with the college course, and a value of 0.0 implies a low degree of meaning. However, this approach is limited by a lack of objectivity - for instance, how does one argue that one college course is more meaningful than another course? As a result, we can disregard a user's desire for meaning in the context of college course recommendations. It is worth mentioning this will be discussed as a piece of future work in subsection 7.2.2.

4.3.5 Gender, Race/Ethnicity and Parent's Educational Background

As is clearly evident from our investigations in subsection **??**, an individual's gender, race/ethnicity and parent's educational background are strong influences on an individual's college course choice. As a result, it could be reasonable to consider gathering these pieces of information from a user for the purpose of college course recommendations.

However, we would argue that these pieces of information should not be considered in the context of a college course RS. For example, by factoring in a user's gender, the RS would not only reinforce stereotypes surrounding gender, but it may also obscure the college course recommendation algorithm. To give an example, suppose that a male demonstrates interest in pre-school teacher education, which is 97% female-dominated according to Higher Education Authority (2024a). An RS algorithm may deem this course unfit for the user solely on the basis on gender, reinforcing the gender norms and stereotypes surrounding this particular course. As a result, we would argue that external influences such as gender, race/ethnicity and parent's educational background would not be relevant in the context of college course recommendations, due to their implications.

In spite of our above efforts to unbias the RS, it is worth remembering that a human, with all their biases and limitations, will interact with the RS, meaning that the element of bias is still unavoidably present in the RS. In this context, it is worth mentioning the work of Wang et al. (2022), in which they compared the results of biased and unbiased career recommendation algorithms. According to their findings, the users actually preferred the biased career recommendations over the unbiased recommendations. On the surface this may appear surprising; however when we consider our investigations in subsection **??**, we can recall that these external factors are large influences on an individual's college course choice, which explains these findings.

In addition to this, to the best of our knowledge, data about race/ethnicity and parent's educational background are not available for each college course in Ireland, posing a challenge for integrating this effectively into a college course RS. In spite of Wang et al.'s findings, we will disregard gender, race/ethnicity and parent's educational background in the context of college course recommendations, due to the reinforcement of stereotypes and limitations on accessible data.

4.3.6 Myers-Briggs Type Indicator

One of the closely-related works utilised the Myers-Briggs Type Indicator (MBTI), a personality model developed in Myers et al. (1962), to provide personalised college course recommendations. Seeing as we are already utilising Holland's RIASEC model, which is also a personality model but more specific to the context of work, we would consider the MBTI model to be unnecessary and redundant information to gather. Following this reasoning, in the context of college course recommendations we can disregard the MBTI personality model.

4.3.7 LC points, NFQ levels and Colleges

As explained in subsection 2.4.1, a user's LC points is an influential entry requirement to college courses in Ireland. In addition to this, Irish students may have preferences for particular types of NFQ levels in the context of college courses, as described in section

2.2. We can recall that relevance is an important quality of an effective RS, as described in section 1.3. For instance, it would not be relevant to suggest a course that is 500 LC points to a user that is expected to receive 300 LC points, or an NFQ level 8 course to a user who is only interested in NFQ level 6 courses. Thus, gathering a user's expected LC points and their NFQ level preferences would provide more relevant college course recommendations, which better answers our research question, which is to assist the Irish student in exploring their college course choices.

As is evident from our discussion in subsection 3.1.2, a student has a variety of justifications for choosing their college, which is a related, but separate decision to choosing their college course. In light of this, we should also gather the user's preferences towards colleges, in order to provide them with more relevant college course recommendations.

4.3.8 Section Summary

In order to provide personalised and relevant college course recommendations to a user, we must gather some information off the user. In this section, we examined the different information an RS should gather from a user in the context of college course recommendations. We justified the gathering of the following information from a user in the context of a college course RS, and we plan to integrate these into our RS design:

- RIASEC model
- Academic interests
- Expected LC points
- NFQ level preferences
- College preferences

In addition to this, we justified disregarding a user's hobbies, school grades, gender, race/ethnicity, parent's educational background, MBTI model, and their desire for salary, job prospects and meaningful work in the context of college course recommendations. This completes research objective 2: "Identify the relevant information an RS should question a user to recommend suitable college courses", providing us with a clearer picture of the design behind our college course RS.

4.4 Dataset

In order to create an RS, we need a dataset of Irish college courses. To the best of our knowledge, there is no publicly available dataset of all Irish college courses. As a result, we resorted to web-scraping the data from a publicly available source, solely for the sake of research and exploring this dissertation's research question. Seeing as CareersPortal (CP) is the leading careers website in Ireland, the public information on the site was web-scraped to form a dataset of all Irish college courses. Going forward, this will be known as the CP dataset. Each course in the CP dataset contains the course's title, college, NFQ level, LC points and other metadata, which are partially sufficient in satisfying our RS requirements as laid out in subsection 4.3.8.

However, the CP dataset poses a limitation: it does not have information regarding the RIASEC model or academic interests of the college courses. Naturally, this poses a challenge as our RS intends to utilise a user's RIASEC model and academic interests to make recommendations. In the next subsections, we will provide more details as to how these data were obtained.

4.4.1 RIASEC Model Metadata

Seeing as we are gathering a user's RIASEC model, we will need the RIASEC model associated with each Irish college course in order to make college course recommendations. The CP dataset already categorises each college course under eight working personality traits, described in figure 4.1 below:

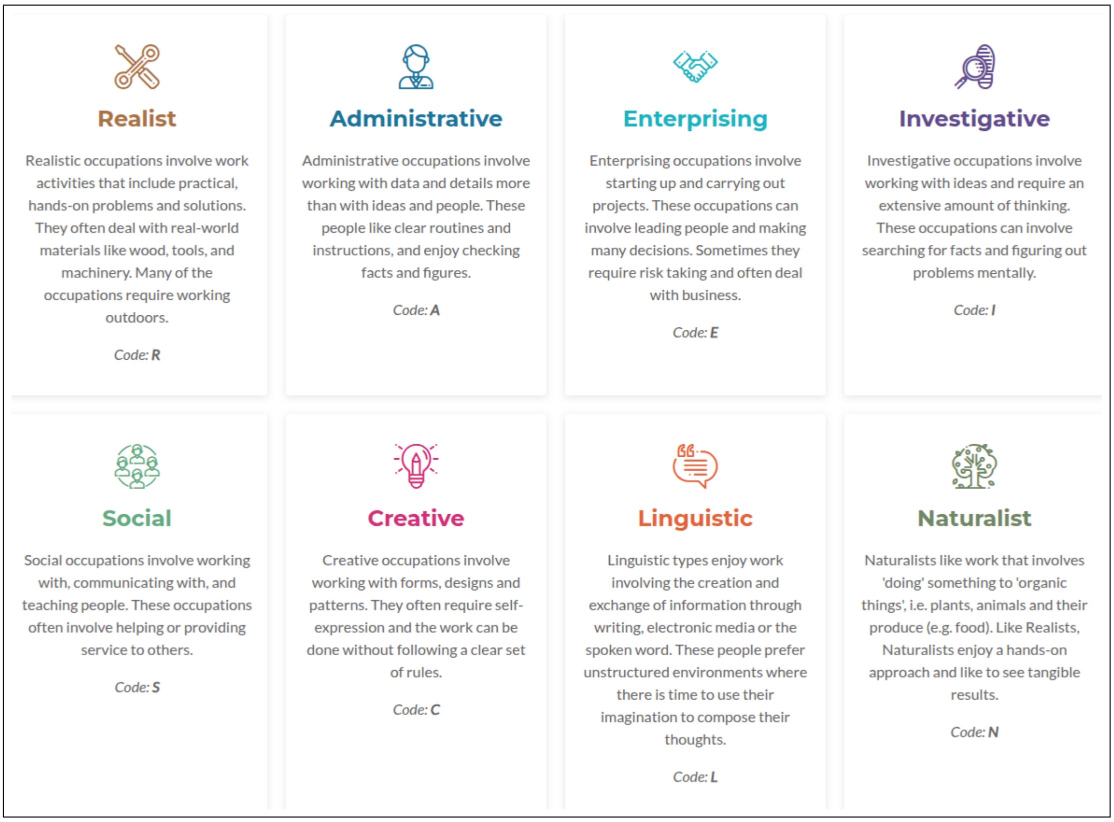


Figure 4.1: CP's eight-trait working personality model.

Seeing as this dissertation seeks to utilise Holland's six-trait RIASEC model, we wish to translate CP's eight-trait model to Holland's six-trait RIASEC model. We desire to translate these traits in a manner that avoids any information loss and preserves the original categorisation of CP's eight-trait model. Thus, we shall document our methodology clearly below.

As we can observe in figures 4.1 above, four of the eight CP traits are equivalent to four of the RIASEC traits: Realist, Investigative, Social and Enterprising. By also observing the definitions above, the 'Administrative' CP trait is equivalent to the Conventional RIASEC trait, and the 'Creative' CP trait is equivalent to the Artistic RIASEC trait. However, this leaves us with two leftover CP traits: 'Naturalist' and 'Linguistic'. According to CP's definition of Naturalists above, they enjoy hands-on work and seeing tangible results, much like Realists. As a result, we can map the 'Naturalist' trait under the Realistic RIASEC trait. Also according to CP's definition above, Linguists are interested in reading and writing, which is very creative, unstructured work. By following the definitions above, this work is very similar to the work exhibited by those with the Artistic RIASEC trait. As a result, we can map the 'Linguistic' trait under the Artistic RIASEC trait. It is worth mentioning that these mappings were performed on the entire dataset as a whole, and they were not performed on a course-by-course basis, allowing for more reproducibility. Our final mapping is best shown in table 4.1 below:

CareersPortal (CP) trait	**RIASEC trait**
Realist	Realistic
Naturalist	
Investigative	Investigative
Creative	Artistic
Linguistic	
Social	Social
Enterprising	Enterprising
Administrative	Conventional

Table 4.1: Mapping CP's eight-trait model to the six-trait RIASEC model.

This allows us to curate a dataset that is more generalisable to Holland's original six-trait RIASEC model, as opposed to working with CP's eight-trait model.

Manual revisions to the CP dataset

During the design of this dissertation's RS, a significant amount of time was invested into manually revising the RIASEC traits of some college courses in the CP dataset. Comparing the revised CP dataset to the original CP dataset, 279 (16%) of college courses in the CP dataset faced manual revisions for their RIASEC traits.

Some of the college courses were manually revised for a few reasons. Firstly, in some cases the RIASEC traits were inconsistent between identical college courses. For example according to the original CP dataset, general nursing in Maynooth University[1] is classified with the Social and Investigative RIASEC traits, whereas general nursing in UCD[2] is classified with the Social, Investigative and Realistic RIASEC traits. Seeing that these are both general nursing courses, it is confusing that they have different classifications of RIASEC traits.

Secondly, many of the RIASEC traits in the CP dataset were poorly representative of the college course. For example, the medicine college courses are only classified with the Investigative RIASEC trait[3]; however this RIASEC classification is ignorant of the

[1] `https://careersportal.ie/courses/coursedetail.php?course_id=26215`
[2] `https://careersportal.ie/courses/coursedetail.php?course_id=459`
[3] `https://careersportal.ie/courses/coursedetail.php?course_id=1079`

Realistic and Social RIASEC traits that the medical profession requires. Another example is the chemical science courses which are also only classified with the Investigative RIASEC trait[4]. This RIASEC is ignorant of the Realistic and Conventional RIASEC traits that a chemical profession requires.

In this context, it is worth mentioning O*NET[5], an online careers database that is regularly maintained and sponsored by the U.S. Department of Labour, Employment & Training. The O*NET database classifies different professions by their RIASEC code. An example from the O*NET database is medical physicians, which would match with the medicine college courses in Ireland, which is classified as the ISR Holland code[6]. As well as this, chemists, which would match the chemical science college courses, are classified as the IRC Holland code[7].

Using O*NET as a reference point, and with our domain knowledge, the RIASEC traits of some college courses (16%) were manually amended. Admittedly, this approach has limitations due to its lack of total reproducibility. Due to time constraints, we were unable to conduct a rigorous experiment to evaluate whether or not there was a performance increase from these revisions. That being said, the effectiveness of this solution with the curated CP dataset will be evaluated against a baseline in chapter 6.

4.4.2 Academic interests Metadata

Throughout this dissertation, there have been many mentions of academic interests, and in this subsection they will finally be described.

FindMyCollegeCourse (FMCC), the only other Irish college course RS, places college courses into 23 categories. With the permission of the team behind FMCC, the same 23 categories were used in this dissertation. However, these categories underwent some simplifications and renaming, and the final result used in this dissertation are the 21 academic interest categories below in table 4.2:

[4] https://careersportal.ie/courses/coursedetail.php?course_id=19282
[5] https://www.onetonline.org/
[6] https://www.onetonline.org/link/summary/29-1216.00
[7] https://www.onetonline.org/link/summary/19-2031.00

Table 4.2: Academic Interest Categories

Academic interests		
Agriculture	Architecture and Construction	Business
Chemical Science	Computers	Communications
Creative Arts	Education	Engineering
Environment	Healthcare	Hospitality
Humanities	Languages	Law
Life Science	Mathematics	Physical Science
Social Science	Sport	Welfare

The CP dataset does not feature any categorisation similar to the academic interests given in table 4.2 above. As a result, we need to additionally augment data regarding the academic interests associated with each Irish college course in the dataset. This task was initially performed by Gemini Model 3[8], however the model's categorisation underwent manual revisions to correct inaccuracies and inconsistencies. For example, initially medicine college courses were only categorised under the 'healthcare' academic interest; however, seeing as study of medicine involves the study of life sciences, the courses were re-categorised to include the 'life science' academic interest.

Similar to the RIASEC trait modifications in subsection 4.4.1, this approach is admittedly limited by a lack of complete reproducibility. That being said, augmenting academic interest metadata was necessary, as this type of metadata did not exist at all in the original CP dataset. In other words, to the best of our knowledge there was no other feasible method to augment this academic interest data completely objectively. As mentioned previously, the effectiveness of this solution will be evaluated in comparison to a baseline in chapter 6.

4.4.3 Final Dataset

After our additional data augmentations, we now have a dataset that can satisfy the requirements of our desired RS set out in section 4.3. An example of the data about a college course is given below in JSON format:

[8]`https://gemini.google.com/`

Listing 4.1: An example of a college course in the CP dataset after additional augmentation.

```
{
    "id": "TR052",
    "title": "Dental Science",
    "college": "Trinity College Dublin",
    "region": "Dublin",
    "duration": "5 Years",
    "nfqLevel": 8,
    "points": 625,
    "isAdditionalPortfolioTestInterviewRequired": false,
    "overview": "Dental Science is the study of ...",
    "riasecTraits": [
        "investigative",
        "social",
        "realist"
    ],
    "academicInterests": [
        "healthcare",
        "life science"
    ]
}
```

Note: as mentioned in section 2.4.3, an additional entry requirement to some Irish college courses are Portfolios, Tests or Interviews. This information was already present in the CP dataset, and that is what the boolean variable: `isAdditionalPortfolioTestInterviewRequired` refers to.

4.5 Recommender System Design

As described in section 2.7, there are many types of RSs, each with their own advantages and disadvantages. In this section, we will describe and justify the design to our RS.

It is worth mentioning that during this dissertation, a significant amount of time was dedicated towards exploring a Collaborative Filtering (CF) approach in designing our RS. The approach utilised some popular datasets others may use in solving similar problems to this dissertation, However, this experiment yielded unsatisfactory results, and as a result we had to explore other RS types outside of the CF approach. For the sake of brevity and relevance, the full content of this approach is outlined in the appendix in chapter A. This leaves us with the more traditional Content-Based Filtering (CBF) and Knowledge-Based

Filtering (KBF). In the next two subsections we will describe the integration of these approaches into our RS.

4.5.1 Knowledge-Based Filtering

As described in section 2.7, KBF requires the user to set explicit requirements for their recommendations. In subsection 4.3.7, we justified the gathering of the following information from a user:

- Expected LC points (e.g. 589)
- NFQ level preferences (i.e. NFQ level 6, 7 or 8)
- College preferences (e.g. Trinity College Dublin, etc.)

This type of information is more simply obtained by asking the user directly. As a result, a KBF approach is appropriately applied here. Figures 4.2, 4.3 and 4.4 below demonstrate the RS user interface asking these explicit questions to the user.

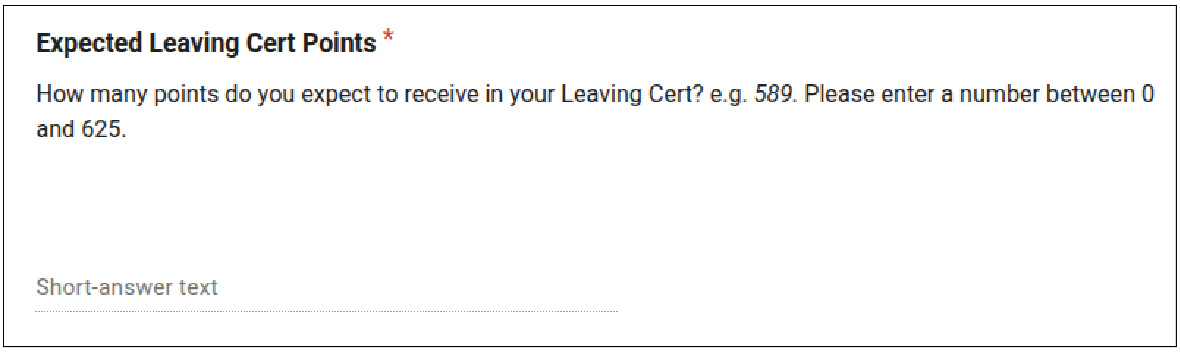


Figure 4.2: The RS user interface asking the user for their expected LC points.

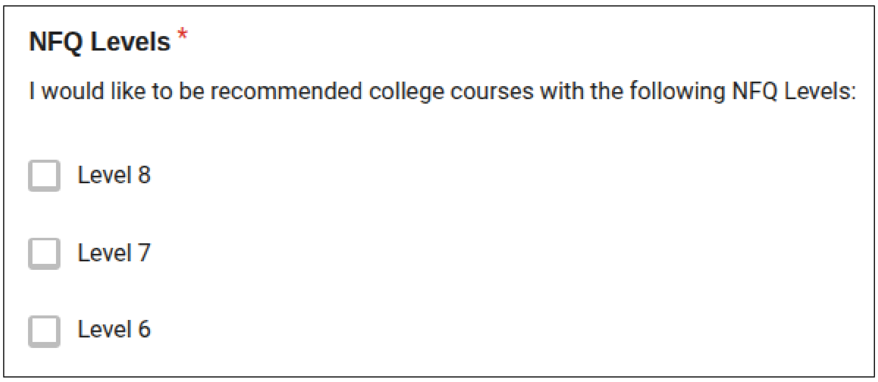


Figure 4.3: The RS user interface asking the user for their NFQ level preferences.

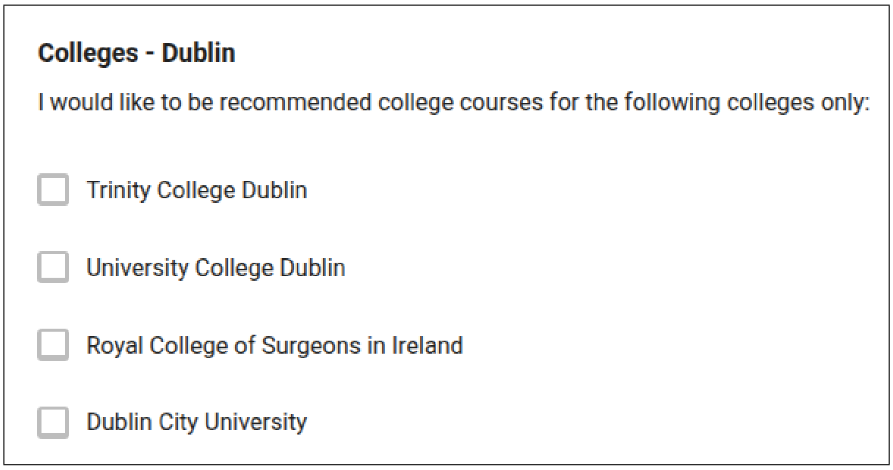


Figure 4.4: The RS user interface asking the user for their specific college preferences. The user can filter by all Irish colleges - this screenshot demonstrates a fraction of the choice.

Figure 4.5 below best describes this stage to our college course RS design: a user interacts with a KBF RS that utilises their expected LC points, NFQ level and college preferences to provide a narrower set of college courses. For example, a user that expects to receive 500 LC points, prefers NFQ level 8 college courses, and only wants to study at TCD, will filter the college course set down to college courses at TCD which are under 500 LC points and are NFQ level 8.

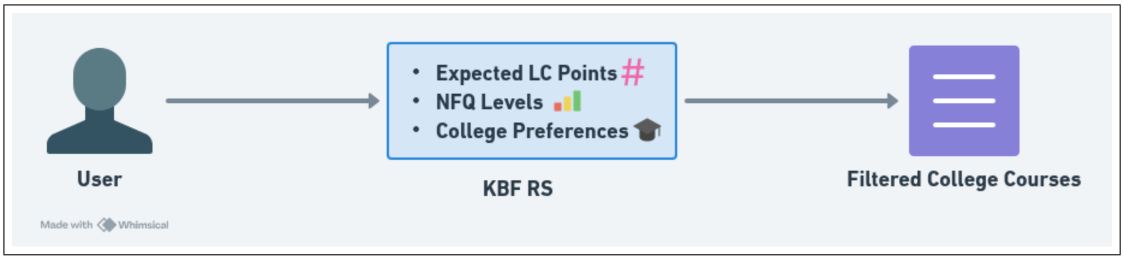


Figure 4.5: A diagram of the Knowledge-Based Filtering RS.

After interacting with the KBF RS, the user will be provided with a filtered set of college course recommendations, not a personalised set, simply because the above information is not sufficient to provide personalised college course recommendations to the user. This filtered set of college course recommendations will be instrumental in the next stage of our college course RS.

4.5.2 Content-Based Filtering

After the user interacts with the KBF RS, the candidates for their college course recommendations are refined through their explicitly set preferences. However, we must now provide a more personalised list of college course recommendations to the user. We will utilise the the user's RIASEC model, academic interests and expected LC points to bridge this gap. Theoretically, we could utilise a KBF approach to provide more personalised recommendations. For instance, the RS could ask the user to explicitly filter college courses by academic interests. For example, the user could explicitly filter college course recommendations by biology and mathematics. However, this approach is not significantly different to the more traditional college course search filters, such as CareersPortal as described in subsection 3.2.1.

Thus, for the sake of novelty we desire a more sophisticated type of RS, which leads us to the Content-Based Filtering (CBF) approach. CBF directly utilises the information of the user - their RIASEC model, academic interests and expected LC points - to provide personalised college course recommendations. For example, a user demonstrating an interest in the 'chemical science' academic interest, the Investigative RIASEC trait and expects 550 LC points may be recommended chemistry in TCD with 542 points[9]. A detailed explanation of the CBF RSs implementation is provided in section 5.3. The complete design of our college course RS best described in figure 4.6 below:

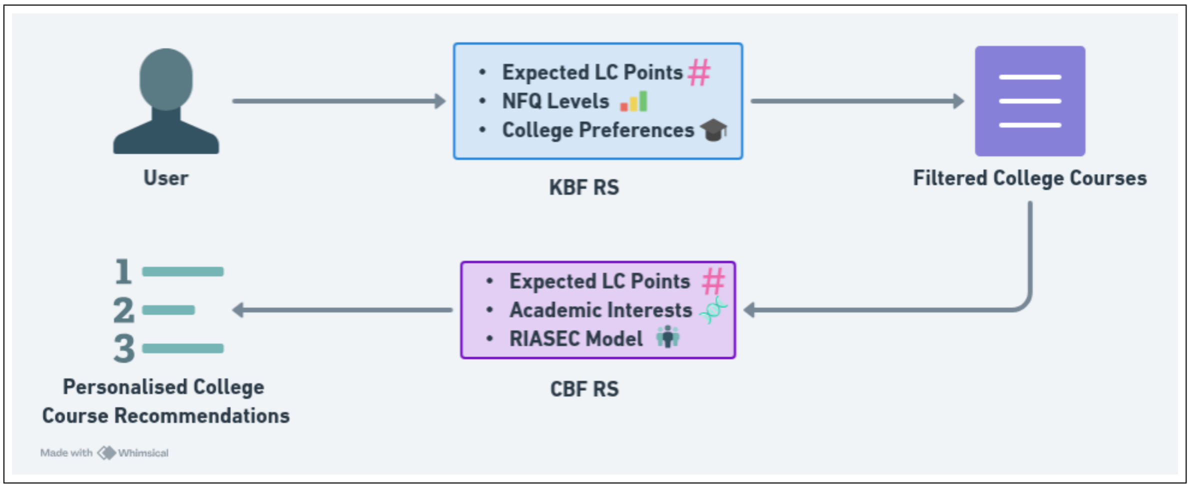


Figure 4.6: A diagram of our hybrid RS with KBF and CBF approaches.

[9]https://www.tcd.ie/courses/undergraduate/courses/chemistry-chemical-sciences/

4.5.3 Section Summary

In summary, this section described the high-level design of our RS. A KBF RS will filter college course recommendations utilising the user's expected LC points, NFQ level and college preferences, and a CBF RS will provide college course recommendations utilising the user's RIASEC model, academic interests and expected LC points. In other words, our college course RS is a hybrid RS with Knowledge-Based Filtering and Content-Based Filtering.

4.6 Quiz Design

This section will describe the quiz behind the RS and the design of the quiz questions.

In order to gather a user's academic interests and RIASEC model, we must design a quiz to gather their preferences. The quiz must have sufficient coverage across all 21 academic interests and the six RIASEC traits. We can then design the quiz with six questions being asked per academic interest, resulting in a total of 126 questions. Given the established importance of the college course decision-making process, and the requirement for broad coverage across all academic interests, we believe a quiz of this length is justified. The user's response to the question must be quantifiable with a discrete measurement, in order to make their response more easily represented numerically. In light of this, we opted for a 5-point Likert scale, which prompts users to rate each question on a scale of 1 to 5, where 1=Strongly Disagree, 2=Disagree, 3=Neutral, 4=Agree and 5=Strongly Agree.

4.6.1 Question Design

Six questions are being asked for each of the 21 academic interests, and we must now discuss what questions we ask the user. The questions were designed to reflect the typical work done in each academic interest, to give the student a clearer idea of their preferences towards each academic interest. For example, a student who has never studied biology may find it difficult to answer a more general question such as: *On a scale of 1 to 5, how interested are you in biology?*, and they would likely find it easier to answer a more specific question such as: *On a scale of 1 to 5, how interesting would you find investigating how bacteria react to medicine?* This enhanced specificity helps the student explore their college course preferences, which assists us in answering our research question.

Each of the six questions are not only designed to capture typical work performed in each academic interest, but also to gather a student's RIASEC traits. For example, the activity: "Investigate how bacteria react to medicine" is a 'life science' question, but it also targets the Investigative RIASEC trait, as the work requires critical analysis. On the other hand, the activity: "Dissect biological specimens in a lab" is also a 'life science' question, but it targets the Realistic RIASEC trait, as it involves working with your hands. In this sense, the six academic interest questions target different RIASEC traits as well as typical work in that field.

However, among the six academic interest questions, the RIASEC traits are not targeted evenly. For example, among the six 'life science' questions, four questions target the Investigative RIASEC trait, and two questions target the Realistic RIASEC trait. This was done because generally, each academic interest is clustered around specific RIASEC traits; some traits are more relevant than others depending on the context. This idea is also seen in the O*NET database as referenced in subsection 4.4.1. To assist in discovering the relevant RIASEC traits for each academic interest, the RIASEC trait distribution for each academic interest is plotted in appendix B.2. These data allow us to appropriately target the relevant RIASEC traits in each of the six academic interests questions.

With this context and knowledge, we can design the 126 questions in our quiz to target different types of typical work and relevant RIASEC traits across the 21 academic interests. The 126 questions were initially formed by Gemini model 3; however, the questions underwent significant manual revisions by the author in order to more accurately reflect the academic interest and RIASEC traits. The full list of 126 questions can be found in appendix B.3.

4.6.2 Question Phrasing

With our questions designed, we must now discover how to phrase the question. At first, the questions were phrased as:

On a scale of 1 to 5, how much would you agree with the following statement?

I would like to do this activity: Investigate different philosophical ideas.

However, early user testing exposed a flaw: due to the phrasing of the question, users were being recommended college courses in areas they were recreationally interested in. As a result, they would not realistically consider studying the college course for the sake of job prospects or salary. For example, a student may be interested in exploring philosophy as a hobby, but not for full-time work. Thus, the question required some re-phrasing to reflect this, and we settled on: *I would like to do this activity* ***for a living.*** The 'for a living' pushes the user to consider the activity in the context of full-time work, as opposed to a hobby. This is a subtle but vital addition to the question, resulting in more relevant college course recommendations that the user would realistically consider studying, assisting us in answering our research question.

4.6.3 Examples of Quiz Questions

Two examples of questions, as shown in the RS user interface, are given below in figure 4.7:

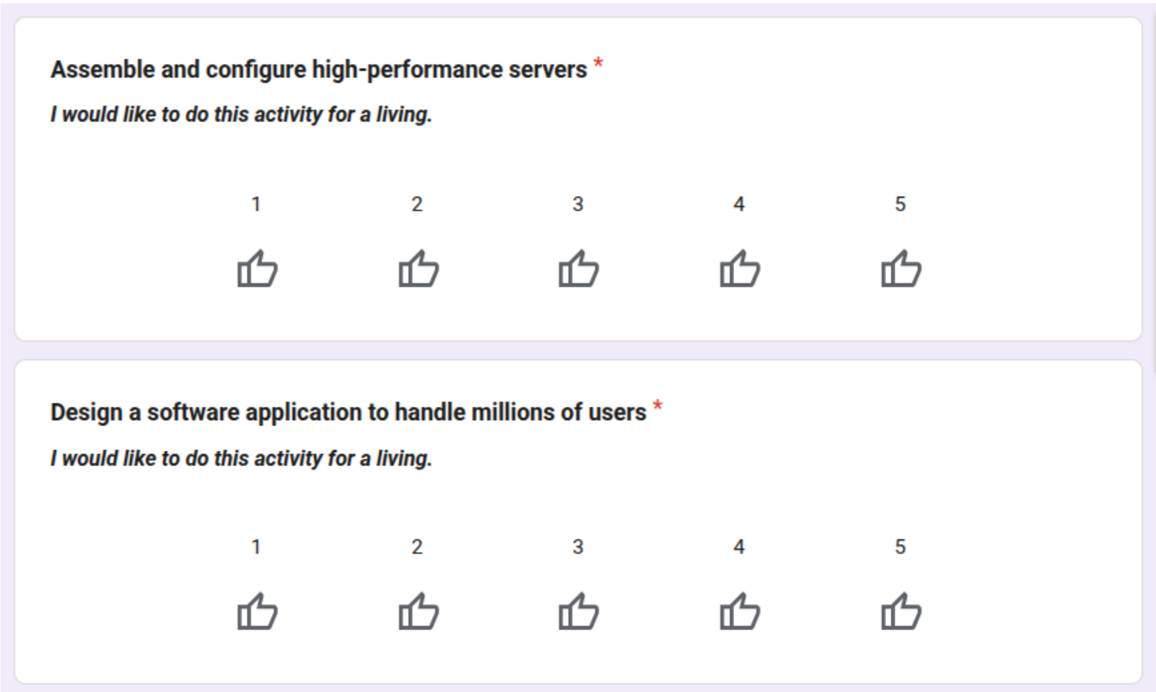


Figure 4.7: Two examples of questions being asked to the user in order to gather their academic interests and RIASEC Model.

4.7 The Recommendations

We have described the quiz which gathers information from a user, and the design behind our RS. The last step is to provide the final college course recommendations to the user, which will be described in this section.

4.7.1 Recommendation Set Size

Generally speaking, RSs recommend a finite number of recommendations to a user, and in this subsection we will justify the number of college course recommendations the college course RS will present to the user.

In table 3.3, we examined the number of college course recommendations given by other closely-related works. We observed that some of the closely-related works provided large numbers of recommendations, even as high as 200 college course recommendations. In light of this, we can recall two findings in the review of our literature in chapter 3: Firstly, the findings of Long et al. (2025), in which smaller recommendation sets are equally as effective as larger recommendation sets. Secondly, the phenomenon of overchoice as coined by Toffler (1970), which remarks that choosing between a large variety of options can be detrimental to the decision-making process. As a result of these two findings, we are motivated to provide a smaller list of college course recommendations to the user.

As described in section 2.3, a student can put up to 20 college courses on their CAO application - ten NFQ level 8 courses, and ten NFQ level 6/7 courses. Inspired by this, our college course RS will return 20 college course recommendations to the user, a relatively small number which is in line with our aforementioned findings.

4.7.2 Recommendation Diversity and Serendipity

As mentioned in section 1.3, diversity is a quality of an effective RS. Suppose a user demonstrates strong interest in computers, and as a result they are recommended five different computer science courses in Dublin. While these recommendations are relevant to the user's interests, the recommendations are neither diverse nor serendipitous. As a result, many of these computer science recommendations are redundant; after the first computer science recommendation, the other computer science recommendations do not provide significantly new information to the user. Seeing as our research question involves assisting the user in exploring college courses, this is detrimental. Instead, a more different, but relevant college course could take its place - such as *Immersive Software Engineering*[10]. Not only does this benefit the diversity of the recommendations, but it also gives the RS a better chance to make a serendipitous recommendation - the gold standard of RSs. This design decision will better answer our research question.

Following from this reasoning, only college courses with unique titles are recommended.

[10] `https://software-engineering.ie/`

For example, if computer science in TCD has been recommended to the user, then computer science in UCD will not be recommended to the user, and another recommendation will take its place. The implementation for this is given in subsection 5.4.1.

In addition to this, for the sake of diversity and serendipity we do not want one academic interest to dominate all 20 recommendations. Following this reasoning, the RS should strive to recommend college courses across many of the user's preferred academic interests. From a high level, suppose a user demonstrates interest towards computers, life science and sport, then across the 20 recommendations, college courses will be recommended according to their interests. For example, ten recommendations can go towards computers, six recommendations towards life science and four recommendations towards sport. The implementation details for this is given in subsection 5.4.2.

4.7.3 Final Result

In figure 4.8, an example of the college course RS output is shown.

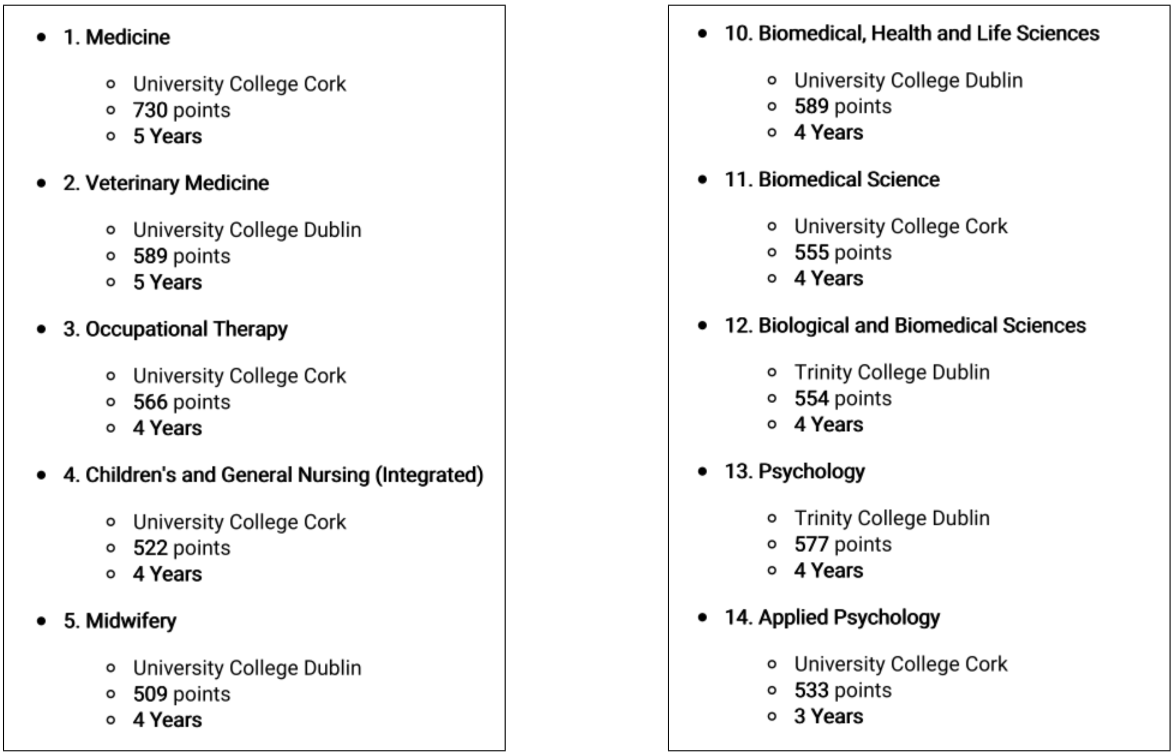


Figure 4.8: Examples of the final recommendations presented to the user.

In the above example, the user's strongest academic interest was in 'healthcare', which describes their first five recommendations. The user also demonstrated academic interests in the 'life science' and 'social science' categories, describing their 10-14th recommendations, showcasing the RS's ability to recommend college courses across a user's academic interests, as opposed to focusing on just one academic interest, for the sake of diversity. It can also be observed that no identically-named college courses are recommended, enhancing the diversity and potential serendipity of the recommendations.

4.8 Chapter Summary

This chapter opened with insights gathered from interviews with two GCs. These insights, along with our knowledge in chapter 3, allowed us to achieve research objective 2: "Identify the relevant information an RS should question a user to recommend suitable college courses". In particular, we justified the gathering of a user's academic interests, RIASEC model, expected LC points, NFQ levels and college preferences in order to provide personalised college course recommendations. Additionally, the high-level design of our RS was described: a hybrid RS with KBF and CBF, along with explanations of the quiz design and different design decisions to enhance the diversity and serendipity of the final college course recommendations.

In the next chapter, we will delve deeper into the implementation behind the high-level design ideas presented in this chapter.

Chapter 5

Implementation

In this chapter, we will provide the implementation details behind the design ideas exposed in chapter 4. Firstly, we will examine the technologies utilised in the implementation of the college course RS. Seeing as our RS is a Hybrid RS with KBF and CBF, we will describe their implementations in detail.

5.1 Technologies

In this section, we will describe the technologies powering the college course RS. The user interface for the RS, as seen in chapter 4, was a quiz hosted on Google Forms[1]. After interacting with the RS, the user's quiz results were stored in a `.csv` file. The RS is designed in Python, and processes the user's quiz results from this `.csv` file, and then prints the college course recommendations to a Markdown `.pdf` file which is viewable to the user. All data is stored and retained on private storage on OneDrive[2]. This technology stack was intentionally lightweight because its primary purpose was to serve as a prototype for evaluation.

5.2 Knowledge-Based Filtering Implementation

Recall from figure 4.5, that the purpose of the KBF RS is to provide a filtered list of college course recommendations. The KBF RS utilises a user's expected LC points, NFQ

[1] `https://forms.google.com/`
[2] `http://onedrive.live.com/`

levels and college preferences to perform this filtering, the code for this is given below in listing 5.1:

```
def kbf_rs(user_college_course_preferences):
    filtered_college_courses = []

    for course in college_courses:
        if is_course_filtered(course, user_college_course_preferences):
            filtered_college_courses.append(course)

    return filtered_college_courses

def is_course_filtered(course, user_college_course_preferences):
    return isNfqLevelSatisfied(course, user_college_course_preferences)
        and isCollegeSatisfied(course, user_college_course_preferences)
        and isPointsSatisfied(course, user_college_course_preferences)

def isNfqLevelSatisfied(course, user_college_course_preferences):
    return course["nfqLevel"] in
        user_college_course_preferences["nfqLevels"]

def isCollegeSatisfied(course, user_college_course_preferences):
    return course["college"] in
        user_college_course_preferences["colleges"]

def isPointsSatisfied(course, user_college_course_preferences):
    return course["points"] <=
        user_college_course_preferences["expectedPoints"]
        or course['isAdditionalPortfolioTestInterviewRequired']
```

Listing 5.1: Code for the KBF RS

As described in section 2.4.3, despite 625 being the maximum achievable points in the LC, a college course may require a combined LC points of 700, for example. This occurs when a course has additional entry requirements, such as a portfolio, test or interview, which are added on to the course's total LC points. As a result of this edge case, if a college course requires an additional entry requirement, represented by the boolean variable `isAdditionalPortfolioTestInterviewRequired`, the course is still included in the filtered list, even if the user's expected LC points are below the college course's LC points. For example, if a user is expected to receive 550 LC points, they may still be recommended medicine at 732 LC points. This limitation is further discussed in section 5.5.

This KBF RS now returns a list of filtered college courses that satisfy a user's expected

LC points, NFQ level and college preferences. This will be instrumental in the next stage of our college course RS: the CBF RS.

5.3 Content-Based Filtering Implementation

The CBF RS will provide a more personalised set of college course recommendations from the KBF RS's filtered list, utilising a user's academic interests, RIASEC model and expected LC points, as described visually in figure 4.6. From a high level, this type of RS will generate similarity scores between a user and all the college courses, and the top-ranked results will be returned. In order to calculate similarity scores, we must numerically represent both users and college courses, both of which will be described in detail in the coming discussions.

5.3.1 Numerical Representation of a User

This subsection will feature an in-depth description of how a user is numerically represented. Ultimately, we wish to numerically represent a user's RIASEC model, academic interests and expected LC points as float values between 0.0 and 1.0: six values represent the user's RIASEC model, one value represents the user's expected LC points, and 21 values represent the user's academic interests. A value of 1.0 implies strong interest or preference, and a value of 0.0 implies weak interest or preference. For example, a Realistic RIASEC trait score of 1.0 implies a user's strong interest towards this trait, whereas a score of 0.0 implies a user's weak interest towards this trait.

Numerically Translating Quiz Results

With the end-goal established, we must first numerically translate a user's quiz results. The user's 5-point Likert scale response, x, to each question is converted to a numerical value using the non-linear mapping function, $f_{\text{non-linear}}$, in figure 5.1b:

$$f_{\text{linear}}(x) = \begin{cases} 1.0 & \text{if } x = 5, \\ 0.75 & \text{if } x = 4, \\ 0.5 & \text{if } x = 3, \\ 0.25 & \text{if } x = 2, \\ 0.0 & \text{if } x = 1 \end{cases}$$

(a) Linear mapping function.

$$f_{\text{non-linear}}(x) = \begin{cases} 1.0 & \text{if } x = 5, \\ 0.2 & \text{if } x = 4, \\ 0.04 & \text{if } x = 3, \\ 0.0 & \text{if } x = 2, \\ 0.0 & \text{if } x = 1 \end{cases}$$

(b) Non-linear mapping function.

Figure 5.1: Comparison of mapping functions, where x is a 5-point Likert scale response to a question.

Initially, a linear mapping function, f_{linear}, as shown in figure 5.1a, was utilised. However, a linear function would be inappropriately applied to this problem. For example, suppose a user gives two ‘life science’ questions a ‘3’ rating, and one ‘computers’ question a ‘5’ rating. Clearly, a user demonstrates strong interest in ‘computers’, and a moderate interest towards ‘life science’. However, when using the linear mapping function, f_{linear}, the totals of both academic interests equal each other, as shown below:

$$2 \cdot f_{\text{linear}}(3) = 2 \cdot 0.5 = 1.0 = f_{\text{linear}}(5)$$

As a result, a linear mapping function is limited. In the context of this problem, we wish to strongly emphasise ‘5’ ratings, as they are the best indicators of a user’s interest, whereas the other ratings (‘4’, ‘3’, etc.) do not provide as much information. Following this line of reasoning, we decided to use a non-linear mapping function, $f_{\text{non-linear}}$, as shown in figure 5.1b. This function places a strong weighting (1.0) towards a ‘5’ rating, and exponentially decreasing weights towards the other ratings. Seeing as there are five different ratings, each subsequent rating decreases by 80%, which describes $f_{\text{non-linear}}(4) = 0.2$ and $f_{\text{non-linear}}(3) = 0.04$. Weights of 0.0 were assigned to ‘1’ and ‘2’ ratings, as they indicate dislike towards an interest, which describes $f_{\text{non-linear}}(2) = 0.0$ and $f_{\text{non-linear}}(1) = 0.0$. From this point onward, $f := f_{\text{non-linear}}$.

Recall in section 4.6 that each quiz question targets a specific academic interest and RIASEC trait. For example: the question *Investigate how bacteria react to medicine* targets the ‘life science’ academic interest and the Investigative RIASEC trait. From a high level, if a user responds positively to that question, we wish the user to score points towards the ‘life science’ academic interest and the Investigative RIASEC trait, and if they respond negatively, no points will be scored towards those categories.

These numerical representations are then added to a total score for each academic interest and RIASEC trait. For example, if a user gives four 'life science' questions a '5' rating, and two 'life science' questions a '4' rating, then the 'life science' total will be:

$$4 \cdot f(5) + 2 \cdot f(4) = 4 \cdot 1.0 + 2 \cdot 0.2 = 4.4$$

Similarly, if a user gives ten Realistic questions a '3' rating, and five Realistic questions a '2' rating, the total for the Realistic RIASEC trait will be:

$$10 \cdot f(3) + 5 \cdot f(2) = 10 \cdot 0.04 + 5 \cdot 0 = 0.4$$

Converting total scores

After answering all 126 questions in the quiz, the user will have total scores for each of the 21 academic interests and six RIASEC traits. We wish to convert these total scores to float values between 0.0 and 1.0, where 0.0 implies weak interest and 1.0 implies strong interest for that particular academic interest or RIASEC trait. Inspired by the sigmoid function, $\frac{1}{1+e^{-x}}$, we created a function, g, which maps total scores, x, to a float value between 0.0 and 1.0:

$$g(x) = 1 - e^{-x}$$

Continuing our previous example, suppose a user has a 'life science' total score of 4.4, their numerical interest towards the 'life science' academic interest will be:

$$g(4.4) = 0.99$$

This implies a user's strong interest towards this academic interest. Seeing as the user gave four 'life science' questions a '5' rating and two 'life science' questions a '4' rating, a value of 0.99 closely reflects the user's interest in this case. On the other hand, suppose the same user has a Realistic total score of 0.4, their numerical interest towards the Realistic RIASEC trait will be:

$$g(0.4) = 0.33$$

This implies a user's moderate interest towards this RIASEC trait. Seeing as the user gave ten Realistic questions a '3' rating, and five Realistic questions a '2' rating, a value of 0.33 closely reflect the user's interest in this case too.

This total conversion function, $g(x)$, is applied to each total score of the user's 21 academic interests and six RIAESC traits. Each of the 21 academic interests and six RIASEC traits now have scores between 0.0 and 1.0, where 0.0 implies weak interest, and 1.0 implies a strong interest towards that particular academic interest or RIASEC trait.

Numerically Representing LC points

We have now numerically represented a user's RIASEC traits and academic interests as scores between 0.0 and 1.0. Now, we must numerically represent a user's expected LC points as a float between 0.0 and 1.0. The maximum achievable points in the LC is 625, so the numerical representation of a user's LC points is simply divided by 625. For example, if a user is expected to receive 300 LC points, this will be numerically represented as:

$$\frac{\text{Expected LC points}}{\text{Maximum Achievable LC points}} = \frac{300}{625} = 0.48$$

Final Numerical Representation of a user

The CBF RS will gather a user's RIASEC model, academic interests and expected LC points, and we have demonstrated how this information is numerically represented as floats between 0.0 and 1.0. We can combine all of the above information together to numerically represent a user as a 28-dimensional vector - one vector dimension for expected LC points, six dimensions for RIASEC trait scores and 21 dimensions for academic interests scores. This is best shown in table 5.1 below:

RIASEC traits							Academic interests					
Realistic	**Investigative**	**Artistic**	**Social**	**Enterprising**	**Conventional**	**Expected LC points**	**Agriculture**	...	**Life Science**	...	**Sport**	**Welfare**
0.33	0.95	0.1	0.65	0.3	0.4	0.48	0.05	···	0.99	···	0.1	0.1

Table 5.1: An example of the numerical representation of a user as a vector

By following the example of the user's numerical representation in table 5.1 above, the user demonstrates strong interest in the Investigative RIASEC trait, as is demonstrated by a score of 0.95. The user demonstrates almost no interest in the Artistic RIASEC trait, demonstrated by a score of 0.1. Their expected LC points is encoded as 0.48, which is translated to 300 LC points. Moreover, the user demonstrates almost no interest in the 'agriculture' academic interest, as they have a score of 0.05, whereas the user demonstrates very strong interest in the 'life science' academic interest, with a score of 0.99.

5.3.2 Numerical Representation of a College Course

As described above, a user's RIASEC traits, LC points and academic interest scores are represented as 28-dimensional vectors. Each college course in our dataset will also be numerically represented as 28-dimensional vectors in the same manner. Take the following college course as an example:

```
{
    "id": "TR052",
    "title": "Dental Science",
    "college": "Trinity College Dublin",
    "region": "Dublin",
    "duration": "5 Years",
    "nfqLevel": 8,
    "points": 625,
    "isAdditionalPortfolioTestInterviewRequired": false,
    "overview": "Dental Science is the study of ...",
    "riasecTraits": [
        "investigative",
```

```
        "social",
        "realist"
    ],
    "academicInterests": [
        "healthcare",
        "life science"
    ]
}
```

Listing 5.2: An example of a college course in the dataset

Following the above example, the vector dimension associated with the Realistic, Investigative and Social RIASEC traits are set to 1.0, because they are associated with the dental science college course, whereas all other RIASEC vector dimensions are set to 0.0. Similarly, the vector dimensions associated with the 'healthcare' and 'life science' academic interests are set to 1.0, whereas all other academic interest vector dimensions are set to 0.0. The dental science course's LC points are numerically represented as 1.0, as a result of following the same LC points conversion rules as mentioned in subsection 5.3.1:

$$\frac{\text{College Course LC points}}{\text{Maximum Achievable LC points}} = \frac{625}{625} = 1.0$$

Following these rules, the dental science college course is then numerically represented as a 28-dimensional vector as shown below in table 5.2

RIASEC traits							**Academic interests**						
Realistic	**Investigative**	**Artistic**	**Social**	**Enterprising**	**Conventional**	**LC Points**	**Agriculture**	...	**Healthcare**	**Life Science**	...	**Sport**	**Welfare**
1.0	1.0	0.0	1.0	0.0	0.0	1.0	0.0	...	1.0	1.0	...	0.0	0.0

Table 5.2: Example college course vector representation

5.3.3 Similarity Score

As mentioned earlier, from a high level, an RS calculates similarity scores between a user and multiple college courses, and the top-ranked results are returned. Seeing as we have numerically represented users and college courses as 28-dimensional vectors, we need some method to calculate the similarity between a user and a college course.

There are many different measures of similarities between vectors to choose from. For example: the dot product, Euclidean distance or cosine similarity. Some of these similarity measures factor a vector's magnitude and direction, and some similarity measures factor only a vector's direction. In our specific case, we prefer a similarity measure that only factors a vector's direction. This is because college courses with many academic interests or RIASEC traits will have higher vector magnitudes compared to college courses with fewer academic interests or RIASEC traits. This is a result of how they are encoded, as outlined in subsection 5.3.2.

For example, a 625-point college course with Realistic and Investigative RIASEC traits, and 'healthcare' and 'life science' academic interests, has a vector magnitude of $\sqrt{5}$. On the other hand, a 625-point college course with an Investigate RIASEC trait and 'computers' academic interest would have a vector magnitude of $\sqrt{3}$. As a result, if our similarity measure factors in a vector's magnitude, our algorithm will be unfairly biased towards college courses with more RIASEC traits and academic interests. Thus, we desire a similarity measure that only factors in a vector's direction, in the interest of algorithmic fairness.

As a result, the appropriate method of choice is cosine similarity, which measures the angle between two vectors. A score of 1.0 means that the vectors point in the same direction, whereas a score of 0.0 means that the vectors are perpendicular to each other. Cosine similarity is defined mathematically as:

$$\frac{A \cdot B}{||A||\,||B||}$$

Suppose a student has a high Investigative RIASEC trait, high 'computers' academic interest and 589 expected LC points. From a high level, we desire the RS to recommend college courses that generally fit this signature. Cosine similarity is an appropriate measure of similarity in this case, because college courses with the Investigative RIASEC trait, 'computers' academic interest and high LC points will result in a high similarity score with the user, whereas college courses without those features will have lower similarity scores with the user.

With our similarity measure defined, we can now move on to the final step of our RS implementation.

5.4 The Recommendations

In this section, we will build off our previous explanations, and describe the final steps in generating the college course recommendations. In particular, we will expand upon the high-level design ideas exposed in subsection 4.7.2.

5.4.1 Filtering Equivalent College Course Names

As described in subsection 4.7.2, college courses with equivalent names are not recommended for the sake of diversity and serendipity. Performing a simple string match between college course titles would not be effective because the naming can differ slightly. Table 5.3 defines some college courses that are effectively equivalent, but traditional string matching would fail to recognise that the college course titles are equivalent.

College Course Title A	**College Course Title B**
Dental Science	Dentistry
Computer Science and Business	Business with computer science
Mathematics and Irish with Education	Teaching with Mathematics and Irish

Table 5.3: Examples of equivalent college course titles that traditional string matching would fail to recognise.

To solve this problem, the college course titles can be preprocessed into a simpler form using a custom preprocessing script, which addresses many edge cases in the domain of college courses. Further implementation details regarding this script is given in section B.1. These preprocessed college course titles are then used for string comparison, which allows us to avoid recommending college courses with essentially equivalent titles, enhancing the diversity and serendipity of the recommendations. The above example is continued in table 5.4 below:

College Course Title A	**College Course Title B**	**Preprocessed Titles**
Dental Science	Dentistry	*dentistri*
Computer Science and Business	Business with computer science	*busi comp*
Mathematics with Education	Teaching and Mathematics	*educ mathemat*

Table 5.4: The preprocessing script successfully capturing essentially equivalent college course titles.

5.4.2 Recommending Across Multiple Academic Interests

In section 4.7.2, we justified recommending college courses across different academic interests to a user, rather than just one academic interest. This was done in the interest of relevance, diversity and serendipity. In this subsection, we shall describe how college courses are being recommended across different academic interests.

In this college course RS, a user's top-5 ranked academic interests are considered in the final 20 college course recommendations. In an early implementation of this RS, four college courses were recommended for each academic interest, bringing the total up to the desired 20 college course recommendations. However, a limitation with this static approach is that it does not factor in the user's different scores towards the academic interests. For example, suppose a user's top ranked academic interest is 'computers' with a score of 1.0, and their fifth-ranked academic interest is 'sport' with a score of 0.2. Following this reasoning, it would be more intuitive to recommend college courses that are proportional to the scores of their interests. This will be made clearer through an example: suppose a user's top-5 ranked academic interests are given below in table 5.5:

Ranking, i	**Academic Interest**	**academicInterestScore$_i$**
1	Computers	1.0
2	Engineering	0.8
3	Life science	0.6
4	Welfare	0.4
5	Sport	0.2

Table 5.5: An example of a user's top-5 ranked academic interest scores.

With these scores, we can calculate the proportional number of recommendations for academic interest i using the following formula:

$$\#\ \text{Recommendations}_i = n \cdot \frac{\text{academicInterestScore}_i}{\sum_{j=1}^{5} \text{academicInterestScore}_j}$$

where $n = 20$, the total number of college course recommendations. In our above example, the number of recommendations for the 'computers' academic interest at ranking $i = 1$ is:

$$\#\ \text{Recommendations}_1 = 20 \cdot \frac{1.0}{1.0 + 0.8 + 0.6 + 0.4 + 0.2} = 20 \cdot \frac{1.0}{3.0} \approx 7\ \text{Recommendations}$$

Naturally, we need the # Recommendations to be integers, and not float values. As a result, we can round the numbers to the closest integer. Continuing our example, we can augment table 5.5 above by adding a column for the number of proportional recommendations per academic interest as shown in table 5.6 below, to enhance our understanding of the given example:

Ranking, i	**Academic interest**	**academicInterestScore$_i$**	**# Recommendations$_i$**
1	Computers	1.0	7
2	Engineering	0.8	5
3	Life Science	0.6	4
4	Welfare	0.4	3
5	Sport	0.2	1

Table 5.6: An example of a user's top-5 ranked academic interest scores.

This results in the RS providing recommendations that are more proportional to the user's preferences towards different academic interests. The motivation behind this implementation decision is to enhance the diversity, relevance and potentially serendipity of the recommendations, all of which are qualities of an effective RS which assists us in answering our research question.

5.5 Limitations

During the design and implementation of this dissertation's RS, some limitations were exposed. We do not believe these limitations are significant; however, they are worth addressing in this section.

As mentioned in section 2.4, some colleges require a third language to be permitted into the college, for example Royal College of Surgeons in Ireland (RCSI). Thus, it would not be relevant to recommend an RCSI course to a user who is not studying a third language for their LC. As well as this, some college courses have specific grade requirements for subjects. For example, computer science in TCD requires a H4 in LC mathematics, so it would be irrelevant to recommend this course to someone who is studying ordinary level mathematics. A limitation of our RS is that we do not take a student's subjects or grades into account, and we hypothesise this would result in less relevant recommendations to the user. This limitation could be resolved by attaining more specific entry requirements for each college and college course, and then augmenting this metadata to college courses in the dataset.

As mentioned in section 5.2, some college courses require additional entry requirements in the form of portfolios, tests or interviews, and this information is simply encoded as a boolean variable in the dataset. The student's grades in these additional entry requirements are added to their base LC points, which is why some courses in the dataset are over the max 625 LC points, e.g. medicine at 732 LC points. A disadvantage is that this number is unfairly compared to other courses without additional entry requirements. This limitation could be resolved by calculating the base LC points for each course with additional entry requirements. However, given that there are about 100 college courses in the dataset with additional entry requirements, and the fact that grading is done differently for each unique portfolio, test or interview, this would be a long and arduous task, constrained by the time of this dissertation. Due to these constraints, we made the design decision to recommend these types of college course to the user, even if their LC points are not satisfied. For example, a student that is expected to achieve 400 LC points could still be recommended medicine. Naturally, this would not be a relevant recommendation, although it could be amended with more accurate LC points metadata.

5.6 Chapter Summary

In this chapter, we provided the implementation details behind this dissertation's RS. In particular, we described the implementations behind the KBF and CBF RSs, and our efforts in enhancing the relevance, diversity and serendipity of the final 20 recommendations. We also described a handful of known limitations regarding our approach. With the implementation of our RS designed, we can move on to evaluating its performance in the next chapter.

Chapter 6

Evaluation

As part of this research, an experiment was conducted on human participants, in which they interacted with the college course RS. The results of this experiment are evaluated in this chapter. The full methodology of the experiment will be described, as well as an analysis of the experiment results.

6.1 Experiment Methodology

This section will describe the complete methodology behind the experiment conducted in order to evaluate this dissertation's RS. We will describe the ethics behind the research, the sample of the experiment, and the experiment itself.

6.1.1 Research Ethics

The proposed experiment was approved by the TCD research ethics committee. Prior to beginning the experiment, participants read a participant information leaflet and an informed consent form. If participants were willing to take part in the experiment, they signed the informed consent form. All data collected during this experiment was anonymised with no traceable link back to the participant.

6.1.2 Experiment Sample

Seeing as our RS is focused on college course recommendations in the Irish context, we sought participants that were over the ages of 18, and are currently enrolled in a college course in Ireland. According to Higher Education Authority (2024b), the experiment population size is about 250,000. No direct benefits or payments were provided to participants. As part of the experiment, participants provided their original CAO college course list that they applied to prior to their first year of study. An example of this CAO list is given in figure 6.1 below:

CAO Application No. : [redacted]

1. TR033 Computer Science
2. DN201 Computer Science (Common Entry)
3. TR031 Mathematics
4. DN200 Science (options)
5. TU856 Computer Science
6. DN230 Actuarial and Financial Studies
7. TR035 Theoretical Physics
8. TR034 Management Science and Information Systems Studies
9. DC121 Computer Applications
10. TR032 Engineering

Figure 6.1: An example of an NFQ level 8 CAO college course list.

Participants that fit the above requirements were sought opportunistically, and a sample size of $n = 31$ was obtained for this experiment.

6.1.3 The Experiment

After the participant voluntarily accepted to participate in the study, experiments were either conducted in-person, or over online video conferencing software. The PI was present constantly, and the participant was made aware that they can ask questions at any time. From start to finish, the experiments took no longer than 60 minutes.

The first part of the experiment was for the user to interact with the college course RSs user interface, as given in this Google Form. This involved the user filling out their expected LC points, NFQ level and college preferences, as shown in subsection 4.5.1. As well as this, the user answered the questions regarding their academic interests and RIASEC traits, as shown in subsection 4.6.3. It is worth mentioning that these questions were randomly shuffled to obscure users from noticing patterns in the academic interest categories.

Upon completion of the quiz, the PI ran the recommendation algorithm on the user's quiz results. This produced a pdf file with the user's 20 college course recommendations, which was then shared with the user. As will be described in section 6.3, a baseline RS designed as part of this experiment. As a result, the pdf file contains 20 college course recommendations generated from this dissertation's RS, and 20 college course recommendations generated by the baseline RS. The participants were told that one set of recommendations was generated by this dissertation's RS, and the other set was generated by an anonymous RS, meaning the participants were unaware which recommendations were the baseline or the actual RS; the recommendation sets are simply denoted as 'set 1' and 'set 2', as show in figure 6.2 below:

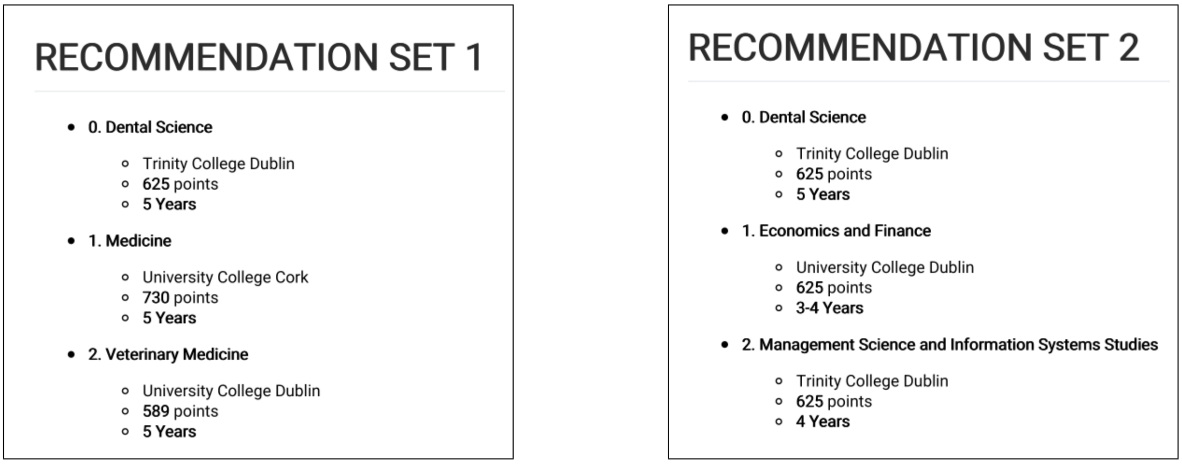
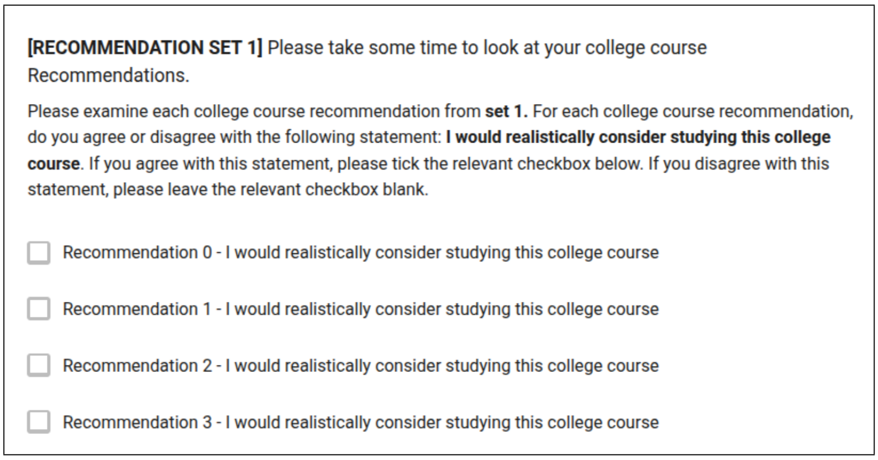


Figure 6.2: Examples of the two different recommendation sets being provided to the user in the same pdf file.

Participants were given about five minutes to familiarise themselves with the college course recommendations across set one and two. They were then asked if they needed more time to familiarise themselves with the recommendations. If they said no, they were asked to fill out a second quiz which evaluates the performance of the RS.

This quiz asks the user for their original, NFQ level 8 CAO college course list. Afterwards, the quiz asks the user to evaluate each of the college course recommendations across both set two and set two. For each college course recommendation, the user is asked: "Do you agree or disagree with the following statement: *I would realistically consider studying this college course*", as shown below in figure 6.3:



[RECOMMENDATION SET 1] Please take some time to look at your college course Recommendations.

Please examine each college course recommendation from **set 1.** For each college course recommendation, do you agree or disagree with the following statement: **I would realistically consider studying this college course**. If you agree with this statement, please tick the relevant checkbox below. If you disagree with this statement, please leave the relevant checkbox blank.

- [ ] Recommendation 0 - I would realistically consider studying this college course
- [ ] Recommendation 1 - I would realistically consider studying this college course
- [ ] Recommendation 2 - I would realistically consider studying this college course
- [ ] Recommendation 3 - I would realistically consider studying this college course

Figure 6.3: The evaluation quiz asking the user to evaluate each specific college course recommendation. Users are asked to evaluate 20 recommendations from set 1, and 20 from set 2. The screenshot shows a snippet of this.

As well as this, the user is asked to subjectively evaluate different aspects to the RS. More details of the evaluation questions are given in section 6.2

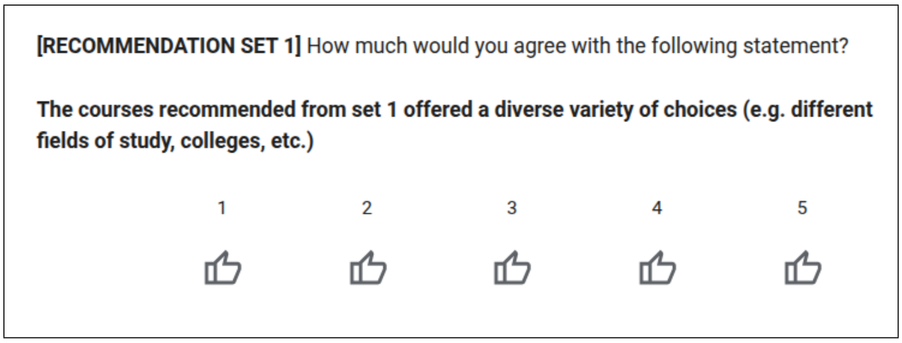
6.2 Evaluation Metrics

Recall that this dissertation's research question is: *"To what extent can a Recommender System assist an Irish student in exploring college courses they would consider studying?"*, and we hypothesised that an RS that provides relevant, diverse, and serendipitous recommendations would answer this research question. In order to measure the effectiveness of our answer to this research question, we need measurable evaluation metrics, and this section will describe the evaluation metrics for this dissertation. It is worth mentioning that evaluation metrics are gathered for both the baseline and actual RS.

Note: in this chapter, when we say that a user 'likes' a college course recommendation, this is equivalent to saying that a user 'would realistically consider studying' the college course recommendation - this is done for brevity.

6.2.1 User-perceived Diversity

Diversity is one of the qualities of an effective RS. Objectively measuring diversity in the context of college course recommendations is challenging and may differ according to the end user's interpretation. As a result, we decided to opt for user-perceived, or subjective, diversity. This was evaluated by asking the user the question shown in figure 6.4 below:


[RECOMMENDATION SET 1] How much would you agree with the following statement?

The courses recommended from set 1 offered a diverse variety of choices (e.g. different fields of study, colleges, etc.)

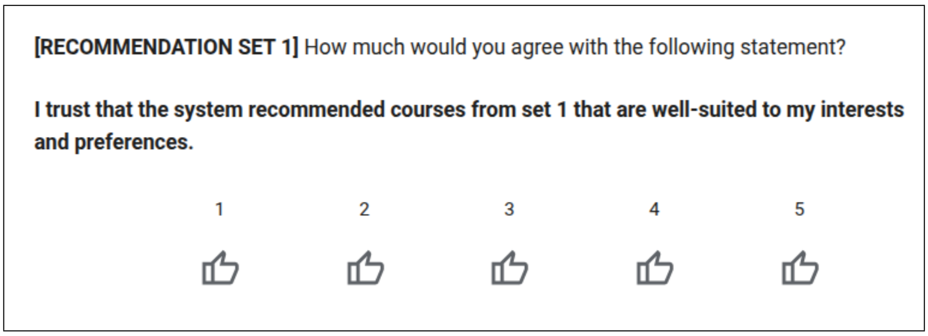
1 2 3 4 5

Figure 6.4: Asking the user for their user-perceived diversity of recommendation as part of the experiment.

The user-perceived diversity metric is an integer between 1 and 5, where 1 implies low diversity and 5 implies high diversity.

6.2.2 User-perceived Trust

In Aggarwal et al. (2016), an additional quality of an effective RS is a user's trust in the recommendations and the RS behind it. In light of this, recall research objective 4: "In comparison to a baseline college course RS, design an RS with statistically significant improvements for the user-perceived trust metric". Naturally, measuring trust objectively is not possible as it is user-dependant. Following this reasoning, we will measure the user's trust subjectively in a similar manner to user-perceived diversity. We evaluated the user's perceived trust in the RS by asking the user the question shown in figure 6.5 below:


[RECOMMENDATION SET 1] How much would you agree with the following statement?

I trust that the system recommended courses from set 1 that are well-suited to my interests and preferences.

1 2 3 4 5

Figure 6.5: Asking the user for their user-perceived trust of recommendation as part of the experiment.

The user-perceived trust metric is a integer between 1 and 5, where 1 implies low trust and 5 implies high trust in the RS and the recommendations behind it.

6.2.3 Precision

Precision is a common metric in RS problems. Recall that users evaluated whether or not they would realistically consider studying each recommended college course in figure 6.3. In the context of college course recommendations, precision can be verbally defined as: *"What proportion of the 20 college course recommendations did the user like?"*. In the context of our problem, precision is expressed mathematically as:

$$\text{Precision} = \frac{\#\text{ Liked college course recommendations}}{20}$$

6.2.4 Recall

Recall is also a common metric in RS problems. In the context of our problem, recall can be defined as: *"Of all the college courses that the user likes, what proportion was recommended by the RS?"*. Naturally, obtaining a truly representative set of 'all the college courses that the user likes' would require the user to go through all ~1740 college courses in the dataset, and asking the user if they like the college course or not. However, this would be a very time-consuming and mentally draining task. As a result, we would not consider this approach feasible in the context of an experiment.

The original CAO list the user provides would not be considered a set of 'all the college courses that the user likes', because this constrains the list to ten college courses, when a user could theoretically like more than ten college courses. Due to these limitations, we have decided not to include recall as a metric in the context of college course recommendations.

6.2.5 Serendipity

Serendipity is a recommendation that is both useful and unexpected to the user. In the Irish context, a user's CAO list is a representation of the user's known, relevant college courses. Following this reasoning, a college course that is recommended to the user that is not present on their CAO list can be seen as serendipitous. To be more specific to the context of this problem, serendipity can be defined as: *"What proportion of the 20 college* *course recommendations did the user like AND was not on their CAO list?"*. Serendipity can also be mathematically expressed as:

$$\text{Serendipity} = \frac{\#\text{ (Liked college course recommendations NOT in user CAO List)}}{20}$$

6.2.6 Evaluation Metrics Summary

From the above investigations, we have now gathered and defined four measurable evaluation metrics. As described by Aggarwal et al. (2016), the three qualities to an effective RS are: relevance, diversity and serendipity. Aggarwal et al. also described trust as an additional quality of an effective RS, meaning we have four qualities of consideration when it comes to an effective RS. We can measure each of these four qualities with our established four metrics, as is more clearly defined in table 6.1 below:

Effective RS Quality	**Evaluation Metric**
Relevance	Precision
Diversity	User-perceived diversity
Serendipity	Serendipity
Trust	User-perceived trust

Table 6.1: Effective RS qualities and their associated evaluation metrics

With this knowledge, we can continue with our RS evaluation.

6.3 Baseline RS

In isolation, providing this dissertation's RS performance on the established evaluation metrics does not provide significant information. Generally in RS problems, the performance of the actual RS is compared to a baseline RS. Recall research objective 3: "Design an objective, easily reproducible baseline RS appropriate to the Irish college course context". This section will describe the implementation of this baseline RS. This allows others to easily implement the baseline RS, should they approach a similar problem to this dissertation.

As mentioned in section 2.4.1, the LC points associated with a course are a reflection of the demand of the course. As a result, an appropriate baseline in the context of Irish college course recommendations would simply be an RS that provides a list of college course sorted descending by LC points. So, the baseline RS would return college courses from 625 LC points and below.

However, this baseline RS may be incredibly irrelevant to some users. For example, it would be unfair for the baseline to recommend 625-point courses to users that expect to receive 300 LC points. As well as this, it would be unfair for the baseline to recommend college courses in TCD to a user that is only interested in studying in University of Limerick, or NFQ level 8 college courses to a user that is only interested in studying NFQ level 6 courses.

As a result, in the interest of a fairness, the baseline RS will filter college courses by the user's specified college preferences, expected LC points and NFQ levels. After this filtering, the baseline RS will return the top 20 college courses sorted descending by LC points. If there is a tie in LC points, the tied courses should be sorted alphabetically.

With the baseline RS established, we have achieved research objective 3, and we can move forward to observing the results of the experiment.

6.4 Results

In this section, we will provide the results of the experiment, and compare the performance of the actual RS against the baseline RS. There will also be an analysis on the statistical significance of the results. The complete data for the experiment are provided in section B.4.

6.4.1 Visualising Evaluation Metrics

In this subsection we will visualise and compare the evaluation metrics obtained by the actual and baseline RS.

User-perceived Metrics

Recall in subsection 6.2 that two of the four evaluation metrics are user-perceived metrics, as opposed to objective ones. In particular, we are referring to: user-perceived diversity and trust. The actual performance of the user-perceived diversity and trust is compared to the baseline's performance in figure 6.6 below:

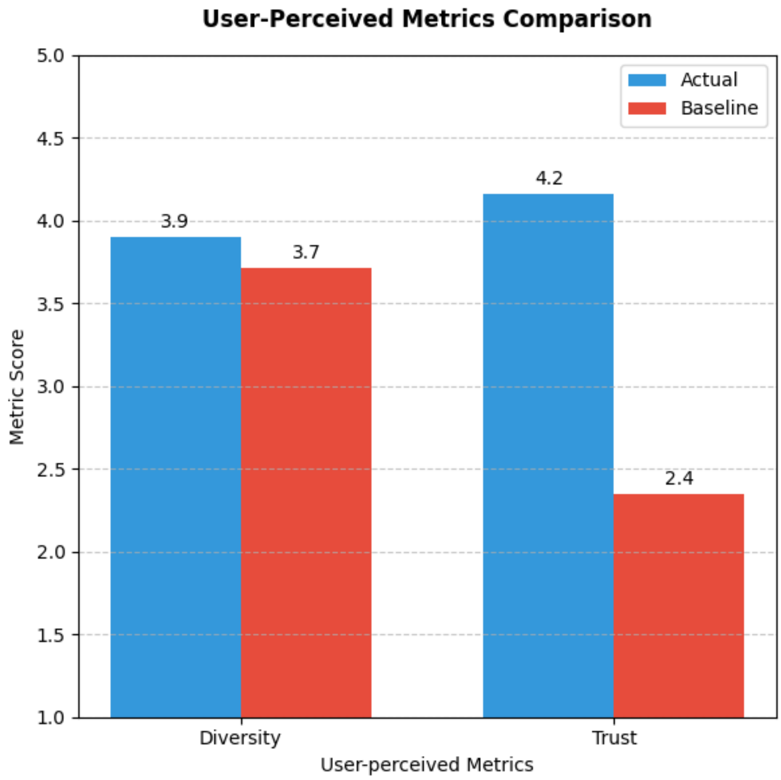


Figure 6.6: Comparing the performance of the actual against the baseline for user-perceived diversity and trust.

When observing figure 6.6 above, we can see that the user-perceived diversity of the actual RS (3.9/5) slightly outperforms the baseline (3.7/5) by a difference of 0.2. Intuitively speaking, it makes sense that the baseline RS has high diversity because the algorithm is completely unaware of the type of college course being recommended. The baseline simply returns high-point courses, meaning that courses will be recommended across healthcare, law, engineering, etc., resulting in high diversity. Considering that the actual RS still outperforms the baseline is a promising result in regards to the diversity of the actual RS.

We can also observe the user-perceived trust of the actual RS (4.2/5) to strongly outperform the baseline (2.4/5), with an improvement of 1.6, indicating users have much higher perceived trust in the actual RSs recommendations. Seeing as a '4' trust score indicates moderately strong trust in the RS and its recommendations, an actual RS trust score of 4.2 is promising. On the other hand, the baseline's 2.4 trust score is relatively close a trust score of '2', indicating the user's moderate dissatisfaction and distrust in the baseline RS. Intuitively speaking, this is sensible as baseline simply returns high-LC point courses with no other algorithmic basis.

Objective Metrics

The other two of our four evaluation metrics are objective metrics. To be specific, we are referring to precision and serendipity. The performance of the actual RS compared to the baseline RS in terms of precision and serendipity is shown below in figure 6.7.

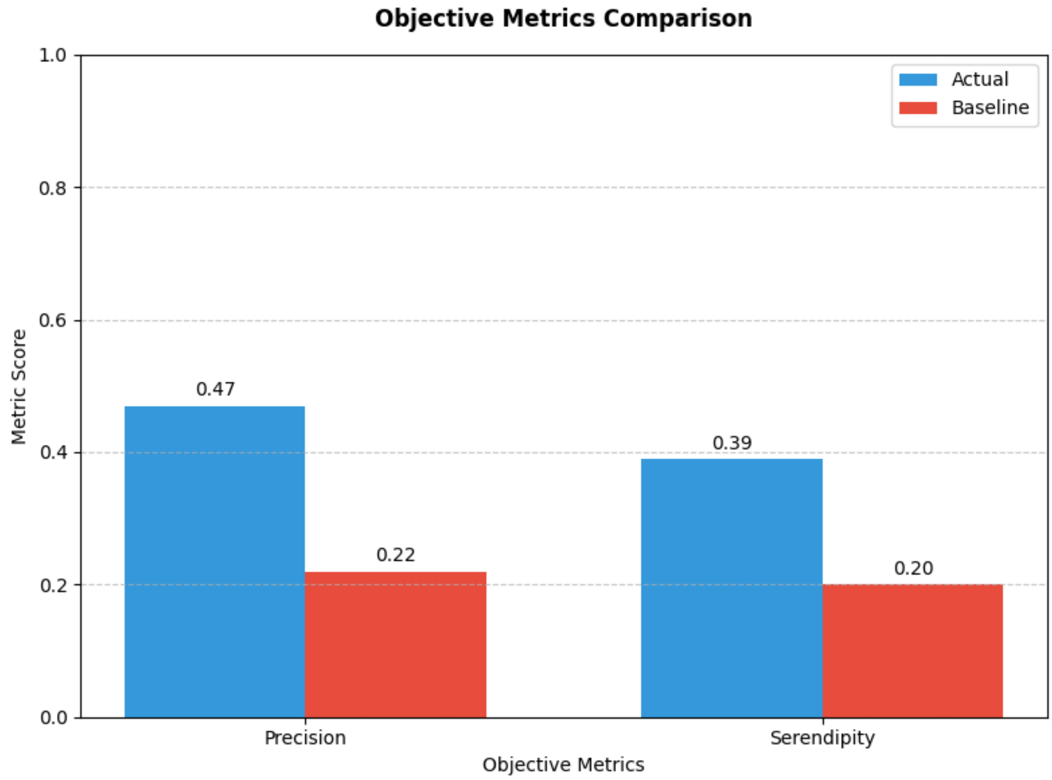


Figure 6.7: Comparing the performance of the actual against the baseline for precision and serendipity.

When observing figure 6.7 above, we can see the actual RS (0.47) strongly outperforming the baseline (0.22) when considering the precision metric. The baseline's mean precision score is 0.22, meaning that on average, users will like about four out of 20 recommendations given by the baseline RS. In comparison, the actual RS achieves a mean precision of 0.47, meaning that users will like about nine out of 20 recommendations given by the actual RS, indicating a significant improvement from the actual RS to the baseline.

We can observe a similar performance increase when considering the serendipity metric. The baseline achieves a mean serendipity score of 0.2, meaning that on average, users will like four recommendations out of 20, all of which are not on their original CAO list. On the other hand, the actual RS provides about eight serendipitous recommendations out of 20, indicating performance that is almost twice as effective when comparing the actual and baseline RSs.

We can gather more insights when we compare the precision and serendipity performances of the actual RS directly. A precision score of 0.47 means that on average, users like nine out of 20 recommendations given by the actual RS, and a serendipity score of 0.42 means that on average, eight out of 20 recommendations are liked by the user and these courses are not on their original CAO list. Comparing these metrics directly, this implies that eight out of nine, or $\frac{0.42}{0.49} \approx 89\%$, of college course recommendations liked by the user are serendipitous. Considering that the gold standard of RSs is serendipity, this high ratio is a very promising result and implies satisfactory RS performance.

6.4.2 Statistical Significance of Results

In the previous subsection, we observed the actual RS improving upon the baseline's performance across all four evaluation metrics. However, we have yet to analyse whether or not these results are statistically significant, which is the objective of this subsection. We will work with a standard 95% level of statistical confidence in this analysis.

Measuring the Normality of Evaluation Metrics

First, we must determine if our four gathered evaluation metrics come from a normal distribution. We can use the Shapiro-Wilk test for normality, with a null hypothesis that the data follow a normal distribution. If $p \leq 0.05$, we reject the null hypothesis in favour of the alternative hypothesis, which claims the data no not follow a normal distribution. Seeing as our experiment involves the same user being evaluated by the actual and baseline RS, for each metric we will run the Shapiro-Wilk test on the differences between the actual and baseline, with the results of our test shown in table 6.2 below:

Evaluation Metric	Shapiro-Wilk p-value	Is Normally Distributed?
Precision	0.60	✓
Diversity	0.23	✓
Serendipity	0.93	✓
Trust	0.023	✗

Table 6.2: Results of the Shapiro-Wilk test for normality on our four evaluation metrics.

From observing the results of table 6.2, all of our metrics follow a normal distribution except for the trust metric.

Investigating Significant Statistical Differences between Samples

If the examined metric follows a normal distribution, we can use a paired t-test (PTT) to determine a statistically significant difference between the sample means. Otherwise, we can use a Wilcoxon signed-rank test (WSRT), which evaluates the median differences between the samples. If $p \leq 0.05$, we have sufficient evidence to reject the null hypothesis, that means/medians of both samples are equal, in favour of the alternative hypothesis, that the means/medians of two samples differ significantly. The results of this test is shown in table 6.3 below:

Evaluation metric	Test type	p-value	Significant difference between means/medians?
Precision	PTT	<0.001	✓
Diversity	PTT	0.58	✗
Serendipity	PTT	<0.001	✓
Trust	WSRT	<0.001	✓

Table 6.3: Results of the PTT or WSRT on our four evaluation metrics.

From observing the results of table 6.3 above, we can conclude that the means/medians of the precision, serendipity and trust metrics are statistically significant when comparing the actual RS to the baseline. On the other hand, we have insufficient evidence to claim that the means of the diversity metric differ significantly between the actual RS and the baseline.

Measuring the Magnitude of Difference between Sample Means

Seeing we have evidence that three out of four of our evaluation metrics differ significantly from the actual compared to the baseline, it is also useful to see how much the means of the samples differ. For normally distributed samples, we can use the Cohen effect size, d, a measure of difference between two sample means. According to Cohen (2013), $d \geq 0.8$ is considered a 'large' difference between sample means, $d \geq 0.5$ is a 'medium' difference, and $d \geq 0.2$ is a 'small' difference. We can visually display our Cohen effect size, d, for our three normally-distributed evaluation metrics in figure 6.8 below:

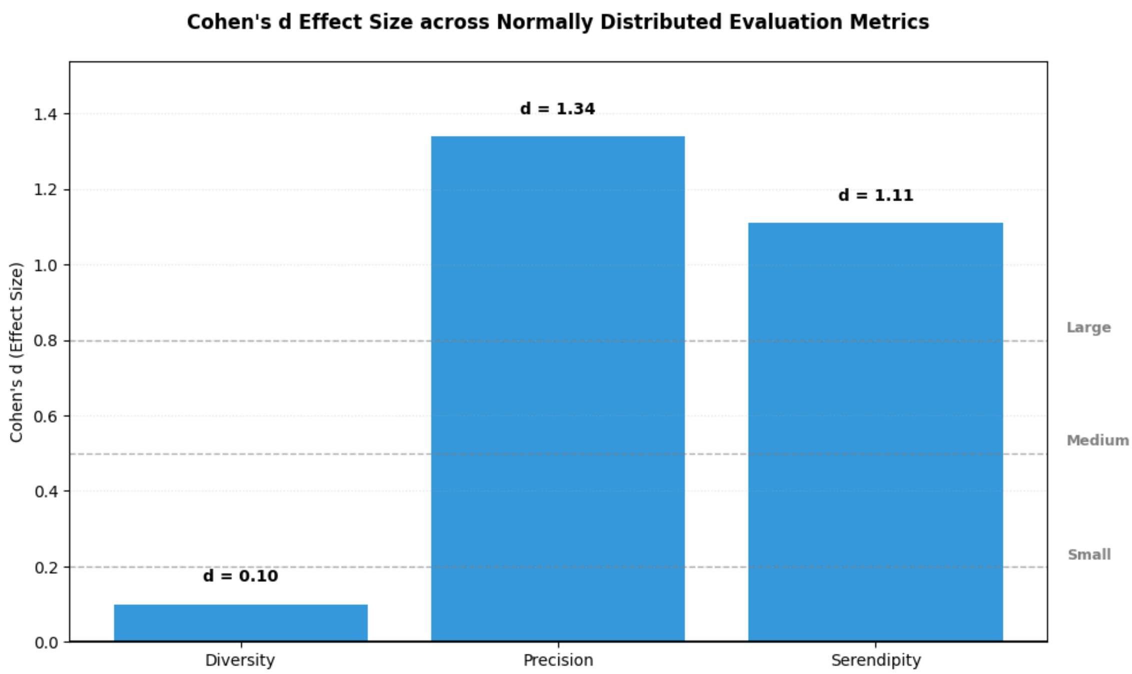


Figure 6.8: The Cohen effect size, d, of our three normally-distributed evaluation metrics.

From observing figure 6.8 above, the Cohen effect size for precision and serendipity all lie well above the 'large' threshold of 0.8, at 1.34 and 1.11 respectively, indicating a very large magnitude of difference between the sample means of the actual and the baseline. On the other hand when considering diversity, a value of $d = 0.1$ indicates a negligible increase from the actual compared to the baseline.

Seeing as the trust metric is not normally distributed and we used the WSRT, we can use the matched-pairs rank biserial correlation (MPRSC) to calculate the effect size r. According to Cohen (2013), $r \geq 0.5$ is considered a 'large' difference, $r \geq 0.3$ is a 'medium' difference, and $r \geq 0.1$ is a 'small' difference. Using the MPRSC, the effect size for the trust metric is calculated as $r = 0.96$. This value of r lies well above the 0.5 threshold, indicating a very large magnitude of difference between the actual and the baseline medians for the trust metric. Thus, we can consider research objective 4 achieved: "In comparison to a baseline college course RS, design an RS with statistically significant improvements for the user-perceived trust metric.".

6.4.3 Results Summary

This section demonstrated large, statistically significant ($p \leq 0.05$) improvements for the actual RS compared to the baseline for the three metrics: trust, precision and serendipity, which is a satisfactory result. That being said, no statistically significant improvement was found for the diversity metric. As mentioned earlier, it is a very difficult task for an actual RS to outperform the baseline in terms of diversity, as the baseline RS only recommends courses sorted by LC points, and thus the baseline's recommendations will be very diverse by nature, as is evident by its mean diversity score of 3.7/5. Seeing that the actual RS slightly outperformed the baseline in terms of diversity (3.7/5 vs. 3.9/5), we would see this as a success. In addition to this, users liked nine out of 20 college course recommendations provided by the actual RS on average, where eight of these recommendations were serendipitous. In comparison, users liked four out of 20 recommendations provided by the baseline RS, four of which were serendipitous, indicating the actual RS was twice as effective as the baseline in providing serendipitous recommendations. Considering all the above factors, we find the actual RS to demonstrate satisfactory performance in the context of our research question.

6.5 Experiment Findings and Analysis

From our analysis in the previous section, the aggregated results of our experiment tell quite a straightforward story. That being said, there were numerous findings gathered from the the experiments as they were being conducted, either from the PI's observations, or participant's verbal feedback. Recall that this describes research objective 5: "Analyse the qualitative insights and findings from a conducted experiment". In some cases, these insights were gathered from just one participant's interactions with the RS. This section will discuss these findings.

6.5.1 Quiz Comprehension and Cognitive Overload

After completing the 126-question quiz, all participants were asked if they could understand the questions and tasks at hand. There was unanimous agreement that the questions were straightforward and easily understandable. This is a positive insight as an RS like this should be designed without requiring the participant to have specific domain knowledge.

Additionally, participants were asked if they found the 126-question quiz to be too long, or if they felt mentally exhausted after answering the quiz. There was also unanimous agreement that the quiz was not cognitively exhausting or demanding, demonstrating an appropriate level of quiz length and complexity. As well as this, participants felt the quiz was an appropriate length given that there are many different fields of study - in our case, that is 21 different academic interests and six RIASEC traits. That being said, a handful of participants expressed that some of the questions felt repetitive, particularly the questions around the natural sciences which includes: chemical science, life science, physical science, engineering and healthcare, suggesting that some of the questions around these academic interests could be refined or removed to avoid repetition or redundancy.

6.5.2 The Role of Subjectivity in Answering the Quiz

Despite the question being phrased as: *"I would like to do this activity for a living"*, one participant approached the quiz differently. The participant was well-aware that the intention of the quiz was to provide college course recommendations, and they also highly valued salary in work, so they only highly rated questions when the job behind the activity was high-paying, even if they did not find the activity particularly interesting. After answering the quiz in this way, the participant received recommendations in high-paying fields which aligned with their working values, resulting in satisfactory end-user performance.

In section 3.1.1, we discovered that users have many different values and justifications for choosing their college course. In a way, the user can construct their own way of answering the quiz and tune the recommendations to align with their values. For example, a user that values passion in their college course can only highly rate questions when the activity involves something they love, like art or music. Or a user that values social impact can only highly rate questions when the activity involves something meaningful, like social policy or the environment. In this manner, regardless of how the question is phrased (*"I would like to do this activity for a living"*), users can still answer the quiz in a way that aligns with their values, which is unique to every individual.

In our opinion, there is nothing negative about the prospect of this. If anything, this is a promising and positive insight, as it implies that users can answer the quiz in a way that reflects their working values. It is the user's own college course choice, and the RS should be designed to complement this process.

6.5.3 Limitations of Academic Interests Labels

As seen in table 4.2, 'languages' is one of the academic interests. Some users in our experiments demonstrated interests in languages, and in one case the user was recommended 'Law and French' from TCD. This user did not know any French, and instead was more interested in a language like Irish or German. As a result, the recommendation was irrelevant. While this is a specific example, it makes the greater point that the 'languages' academic interest poses an interesting edge case, and in the context of college course recommendations, the label is too broad on its own. If the RS has further knowledge on the language the user likes (e.g. Irish or German), it may be able to recommend a more relevant recommendation to the user's interest, such as 'Law and German'.

Another limitation of the academic interest labels can be given with a few specific examples. For instance, economics, psychology and sociology are all classed under 'social science'. In one case, a participant responded positively to an economics-related question, and then they received recommendations for psychology and sociology, which they found irrelevant and unsatisfactory. In this case, the 'social science' category could be broken into more granular categories to improve the relevance of the recommendations. Another example is with the 'creative arts' academic interest, which includes college courses like creative writing, drawing, interior design, theatre and film studies. One participant expressed interest in creative writing and film studies, however they received recommendations for courses that involved drawing, pottery or interior design, as they were also classed under the same 'creative arts' category. While these are particular examples, it demonstrates the greater point that some of the academic interests labels were simply too broad, and at times resulted in irrelevant recommendations.

6.5.4 The Prestige of Leaving Certificate Points

As mentioned in subsection 2.4.1, the LC points associated with a college course is commonly misunderstood as reflecting the difficulty of the course, rather it is more accurate to say the LC points reflect the demand of the course. While that is the technical definition, the LC points associated with a course are commonly associated as a status symbol. In Kelly (2026), the former president of Maynooth university accused Irish higher education institutions of deliberating driving up the LC points of their college courses to create an illusion of quality, which supports the idea that LC points are a symbol of prestige for a college course.

In light of this, recall that our baseline RS simply recommends the highest LC point-courses that satisfy the college, NFQ level and LC points requirements. In one particular case, a participant explicitly expressed weak interest in law and chemical science through their quiz answers. However, when the participant received recommendations from the baseline RS for law and pharmacy courses, they grew interested in the idea of studying those courses. The PI was struck by this, and asked why the participant was interested in studying law and pharmacy, when only minutes ago the participant poorly rated law- and pharmacy-related questions in the quiz. The participant remarked that the high LC points associated with the courses attracted them to it. While this is one particular example, it supports a larger point and corresponds with our prior findings that the LC points of a college course can be seen as status symbols, even in cases where the user previously expresses disinterest in the category.

That being said, having a user's preferences change during the interaction with an RS is not unusual. In the context of RSs, Chen et al. (2013) describes the concept of *Preference Construction*: a phenomenon where humans do not have a clear idea of their preferences prior to a decision-making process, and when they are confronted with making a decision, they construct their preferences in that moment. Clearly, these preferences are dynamic, and they can change upon learning new information. For example, discovering that a college course one previously expressed no interest in, has high LC points, shifting one's perception towards the college course.

Naturally, the inconsistency and changeability of human preferences complicates the RS process. However, it is worth remarking this insight as it is evidently relevant to the context of college courses recommendations.

6.5.5 Limitations of Question Phrasing

Recall that in subsection 4.6.2, early prototypes of the RS phrased questions originally as: *"I would like to do this activity: ..."*. However, this approach was turned down as users were positively rating activities they were recreationally interested in - like philosophy, art, etc. - although they would not realistically consider studying related college courses due to job prospects or salary. In light of this, the question was then rephrased as: *"I would like to do this activity* ***for a living:*** *..."*, to turn the user's focus to a college course they would realistically study, which better answers our research question.

As the experiments were being conducted, participants received some college course recommendations in areas they found interesting, such as linguistics or mathematics education. In many cases, these recommendations were a direct reflection of their quiz results. For example, they would receive teaching recommendations due to their positive response to teaching-related questions. However, when it came down to realistically considering to study the college course, a handful of participants turned the recommendation down, due to perceived low salaries or job prospects from these college courses.

These findings demonstrate a weakness in relevance for the college course RS. Evidently, rephrasing the question to include 'for a living' was unsatisfactory in resolving this problem completely. Arguably, this problem could better be addressed by an additional rephrasing of the question. We hypothesise that a more explicit phrasing of the question could resolve the problem, such as: *"I would like to do this activity* ***as a job:*** *..."*, or *"I would like to do this activity* ***as part of a college course:*** *..."*.

6.5.6 Limitations of Unique College Course Recommendations

As described in subsection 4.7.2, college courses with identical titles are not recommended. For example, if computer science from TCD is recommended, then computer science at UCD cannot be recommended. Admittedly, there is a potential loss in relevance due to this design decision. However, it was justified as a trade-off as it could result in more diverse, and serendipitous recommendations.

This trade-off was experienced firsthand in the experiments. In a handful of cases, participants expressed interest in a particular college course, although they remarked they would prefer to study the same course at a different college. In one specific case, a participant was recommended general nursing in University College Cork (UCC), although they would not realistically consider studying the course due to location constraints, and they would consider studying the course in a more convenient location. The PI inquired the participant about this decision, as the participant expressed an interest in studying at UCC in their quiz. The participant remarked that UCC was a lower-priority college, and they would have a preference for studying elsewhere.

While the given example was specific, it makes the greater point that users do not have equal preferences towards colleges. Evidently, filtering recommendations by the user's explicit college filters, as shown in figure 4.4, did not resolve this problem completely. To account for this limitation, we suggest that a college course recommendation should present multiple locations, as shown in figure 6.9 below:

Figure 6.9: A proposed solution, where multiple colleges are offered for a particular college course recommendation.

6.5.7 Experiment Findings Summary

Recall research objective 5: "Analyse the qualitative insights and findings from a conducted experiment", which was explored in this section. This section featured a stronger focus on qualitative insights that cannot be directly inferred from the raw data shown in section 6.4. A handful of these insights exposed limitations in the RS design, such as the broad academic interest labels. Moreover, some insights were positive, for example the finding that users can tune their recommendations to their unique working values. Lastly, some of the insights highlighted the complexity and nuance of human preferences, such as the prestige attached to a course's LC points. Overall, we can consider research objective 5 achieved due to the explorations completed in this section.

6.6 Chapter Summary

This chapter first described the experiment conducted to evaluate this dissertation's RS. The results of the actual RS against a baseline were evaluated on four established metrics, with significant improvements observed for the trust, precision and serendipity metrics. Negligible improvements were observed for the diversity metric; however we argued this result is still considered satisfactory, seeing as the baseline RS is very diverse by nature.

As well as this, we examined the qualitative findings gathered from the conducted experiments, providing additional insights beyond the narrative of the raw experiment data. With our evaluation complete, we can move toward the final chapter.

Chapter 7

Conclusions & Future Work

This chapter will conclude the dissertation, and discuss future work in the domain of college course recommendations.

7.1 Conclusions

Choosing a suitable college course is a difficult, multi-faceted decision, with 31% of recent Irish college graduates reporting they were not likely to study their college course again, according to Higher Education Authority (2018). In the Irish context, GCs are turning their priorities from career counselling towards mental health counselling, signalling the applicability of a technology in this domain. In light of this, we proposed an Irish college course RS to assist students in their college course exploration, by providing them with diverse, relevant and serendipitous recommendations: all qualities of an effective RS.

Designing an effective college course RS requires a strong grasp on relevant domain knowledge. This prompted a broad and comprehensive review of the fundamentals of college course choice, in which we provided an answer to the question: *"What college course should people study?"*. Additionally, during our examination of closely-related works, we identified a potential research gap by unifying Holland's RIASEC model with academic interests, as well as providing users with a smaller, more manageable recommendation set of specific college courses, as opposed to generalised areas of study.

Our review on the fundamentals of college course choice, along with interviews conducted with two professional GCs, were instrumental in our RS design. This provided us with more clarity on the role of an RS in Irish guidance counselling, desiring an RS that complements the human process of guidance counselling, as opposed to replacing this process entirely. This review also allowed us to identify the relevant information an RS should gather off a user in order to provide personalised college course recommendations. This inspired the final design of this dissertation's RS: a hybrid RS with Knowledge-Based Filtering to filter college courses by a user's NFQ levels, expected LC points and colleges, and a Content-Based Filtering RS utilising academic interests, RIASEC model and expected LC points to provide 20 personalised college course recommendations. Courses were proportionately recommended across the user's top-5 academic interests, and courses with identical titles were not recommended. All of which were efforts to enhance the diversity and serendipity of the recommendations.

This dissertation's RS was evaluated by conducting an experiment with $n = 31$ participants. An objective baseline RS was designed to return courses sorted descending by LC points, satisfying the user's explicit filters (expected LC points, NFQ levels and college preferences). Compared to the baseline, the actual RS demonstrated significant improvements across the trust, serendipity and precision metrics, with competitive performance on the diversity metric. On average, the user liked nine out of 20 college course recommendations provided by the actual RS, eight of which were serendipitous recommendations. This is compared to the baseline's four out of 20 serendipitous recommendations, indicating satisfactory performance of this dissertation's RS in the context of the research question.

7.1.1 Revisiting the Research Question

In section 1.3 we introduced this dissertation's research question: *"To what extent can a Recommender System assist an Irish student in exploring college courses they would consider studying?"*. Recall that we hypothesised that an RS that provides relevant, diverse, and serendipitous recommendations would answer this research question. Therefore, to measure the extent this dissertation achieved in context to the research question, we shall examine the results for the relevance, diversity and serendipity metrics.

In chapter 6, we compared the performance of an actual RS to a baseline. During our evaluation, we observed the actual RS (3.9/5) slightly outperforming the baseline (3.7/5) in terms of user-perceived diversity, implying a marginal improvement for an Irish student's college course exploration. On average for the baseline, users only considered studying four out of 20 recommendations, all four of which were serendipitous recommendations. Alternatively for the actual RS, users considered studying nine out of 20 college course recommendations, eight of which are serendipitous recommendations. This indicates the actual RS was twice as effective in providing relevant and serendipitous recommendations compared to the baseline, a substantial improvement for an Irish student's college course exploration. Ultimately, this demonstrates the extent to which this dissertation has answered this research question, which is to assist Irish students in exploring college courses they would consider studying.

7.1.2 Revisiting Research Objectives

In this subsection, we will revisit each research objective and examine whether the objective was achieved or not.

1. *Provide an answer to the question: "What college course should people study?"*
   An answer to this question was provided in subsection 3.1.6, which largely inspired the design of this dissertation's RS. As a result, we can consider this research objective achieved.

2. *Identify the relevant information an RS should question a user to recommend suitable college courses.*
   The relevant information was identified in subsection 4.3.8, directly informing the design of this dissertation's RS. Thus, we can also consider this research objective achieved.

3. *Design an objective, easily reproducible baseline RS appropriate to the Irish college course context.*
   A baseline RS that fits this criteria was explicitly described in section 6.3. We may consider this research objective achieved.

4. *In comparison to a baseline college course RS, design an RS with statistically significant improvements for the user-perceived trust metric.*
   As observed in section 6.4.2, statistically significant improvements were found for the trust metric, when comparing the actual RS to the baseline. As a result, we can consider this research objective achieved.

5. *Analyse the qualitative insights and findings from a conducted experiment.*
   These insights were described in section 6.5, providing necessary qualitative analysis that goes beyond raw metrics. Therefore, we can consider this research objective achieved.

With our evaluation of research objectives complete, we can now move on to discussing this dissertation's contribution.

7.1.3 Contribution

To demonstrate this dissertation's contribution, we will augment tables 3.2 and 3.3 introduced in section 3.3 to include this dissertation's RS, which will be referred to as *Irish College Course Recommender System* (ICCRS).

Table 7.1: Comparing features of all examined closely-related work (Part 1)

Name	**# Questions**	**# Results**	**Recommendation type**
Superprof	10	3	General area of study
Anderson Uni	25	50	College major
Marquette Uni	10	30	College major
Qamhieh et al. (2020)	125	7	Engineering course
Ghuge et al. (2023)	6	5	College major
Lahoud et al. (2023)	6	5	Lebanese college course
Su et al. (2021)	90	5	Taiwanese college course
FMCC	∼150	200	Irish college course
ICCRS	**126**	**20**	**Irish college course**

Explanation of columns:
 # Questions: the number of questions asked to the user prior to receiving their college course recommendations.
 # Results: The number of college course recommendations the user receives.
Recommendation type: the type of college course recommendation the user receives.

In observing table 7.1 above, ICCRS features a quiz with significant length, demonstrating its breadth at 126 questions. Additionally, inspired by the findings of Toffler (1970) and Long et al. (2025), in which smaller recommendation sets yield more performant results, 20 college course recommendations are provided to the user. To the best of our knowledge when considering the Irish context, providing the user with a more refined set of college course recommendations (20) is a novel contribution.

Table 7.2: Comparing features of all examined closely-related work (Part 2)

Name	**Can filter results by college?**	**Gathered user information**
Superprof	✗	Academic interests
Anderson Uni	✗[1]	Academic interests
Marquette Uni	✗[2]	Academic interests
Qamhieh et al. (2020)	✗	Hobbies, grades, MBTI[3]
Ghuge et al. (2023)	✗	Hobbies, grades
Lahoud et al. (2023)	✓	Hobbies, gender, academic interests
Su et al. (2021)	✗[4]	RIASEC personality model
FMCC	✗[5]	Academic interests
ICCRS	✓	**Academic interests, RIASEC personality Model**

Explanation of columns:
Can filter results by college? Is it possible for the user to specifically receive college course recommendations from colleges of their choice?
Gathered user information: What information is the system gathering from the user in order to make personalised college course recommendations?

In observing table 7.2 above, ICCRS is one of the few works that allows users to filter college courses by particular colleges, at least in the Irish context, ICCRS is the only RS capable of this. To the best of our knowledge, utilising a user's RIASEC model and academic interests is a novel contribution to the domain surrounding college and career RSs.

7.2 Future Work

In this section, we will discuss a list of future works in the context of college course RSs. Some of these future works will be relevant to the Irish context specifically, and others will be relevant beyond the Irish context.

[1] The only college majors recommended are ones offered from Anderson University.
[2] The only college majors recommended are ones offered from Marquette University.
[3] Myer-Briggs Type Indicator (MBTI); a personality test developed in Myers et al. (1962).
[4] The only college majors recommended are ones offered from Cheng Shiu University, Taiwan.
[5] You can filter your colleges by region, for example just the colleges in Dublin, Munster, etc.

7.2.1 Collaborative Filtering

This dissertation's RS utilises Knowledge-Based Filtering (KBF) and Content-Based Filtering (CBF) in generating its recommendations. According to Aggarwal et al. (2016), CBF works by recommending items solely based on the user's gathered preferences. A disadvantage of CBF is known as the 'filter bubble': users are only being recommended items with similar attributes to items they have already interacted with. As a result, users find themselves in an 'echo chamber', with recommendations lacking diversity and most importantly, serendipity. Another disadvantage of CBF is that the RS design requires significant metadata and labelling, requiring a strong grasp on domain knowledge. For instance in this dissertation's RS implementation, we required significant metadata regarding the academic interest labels and RIASEC model associated with each college course.

On the other hand, Collaborative Filtering (CF) works by recommending items that other users with similar preferences also liked. For example, recommending biology to a user because other users with similar preferences chose to study biology. Due to the nature of its implementation, an advantage of CF lies in its ability to capture complex human preferences that manually-labeled CBF would be unable to achieve. For example, a CF RS may recommend a historical epic film to a user with past interest in sci-fi films, as other sci-fi lovers enjoy long films on a grand scale - not only is this recommendation diverse, it may also be incredibly serendipitous. This highly-nuanced information is incredibly difficult to encode with a dataset label, posing a disadvantage for CBF.

Due to the unsupervised nature of CF, we hypothesise that a CF RS would be more optimal for a college course RS. Naturally, a limitation of the CF approach is the access to data regarding individual's preferences and their subsequent college course choice. In chapter A, we explored a CF approach with a popular RIASEC dataset, although the results were unsatisfactory due to a lack of coverage of quiz questions in the dataset. To the best of our knowledge in the Irish context, this kind of data is unavailable, posing a limitation to implementing CF in the Irish college course RS context.

Another limitation posed to CF data gathering is that college course choice does not imply a successful or correct decision. For example, just because a student chooses to study biology, does not imply that the student will like the biology course. Through the duration of the student's degree, they may grow to dislike the course, or discover years later that the decision was one they regret. This limitation should be kept in mind regarding a CF implementation.

According to table 3.1, two pieces of works in the literature have employed CF for college course/career recommendations. To the best of our knowledge in the Irish context, the CF approach has not been explored, suggesting a piece of future work.

7.2.2 Additional Information to Gather from a User

In section 3.1.6, we provided an answer to the question posed by research objective 1: "*What college course should people study?*". The purpose of this was to provide insights into achieving research objective 2: "Identify the relevant information an RS should question a user to recommend suitable college courses". As part of our answer to research objective 1, we mentioned that the following two pieces of information would be relevant to gather from a user in the context of college course recommendations:

- The user's desire for salary or employment after graduation.
- The user's desire for meaningful or socially impactful work.

In subsection 4.3.3, we discussed the viability of integrating salary or employment information into our college course RS. Given the lack of Irish course-specific data regarding salary or employment rates, this approach was not deemed possible. That being said, as a piece of future work in the context of college course recommendations, we hypothesise that gathering a user's desire for salary or employment would result in more relevant recommendations. In terms of effectively implementing this information, we hypothesise that a CF RS would be optimal for solving this problem, due to the CF RS's ability to implicitly capture nuanced human preferences without explicit metadata. For instance, if a user highly values salary and employment, with a CF RS they may be recommended college courses that other students who also highly valued salary and employment chose to study, such as medicine or law.

Moreover, in subsection 4.3.4, we discussed the viability of integrating a user's desire for meaning or social impact into our college course RS. For instance, if an RS has knowledge that a student highly values social impact, the RS may be able to provide more tailored recommendations in light of this. For example, the RS may recommend courses in social policy, nursing, or other fields with conventionally higher means for social impact. That being said, this approach was turned down due to the difficulty in numerically encoding this information. However, in terms of possible implementation ideas, we hypothesise that a CF RS may resolve this problem. This is due to the CF RS's ability to implicitly capture nuanced human interaction that traditional metadata/labelling fails to capture. For example, a CF RS may recommend a college course in social policy to a student due to their interest in meaningful work, as other students who also valued meaningful work went into similar college courses.

To the best of our knowledge in table 3.3, no pieces of work in the literature have utilised a user's desire for meaning, salary or employment rates for the purposes of college course/career recommendations. This suggests an area for future implementation in the context of college course recommendations.

7.2.3 Large Language Models

The usage of Large Language Model (LLM) chatbots have become ubiquitous in today's society, with OpenAI reporting 900 million[6] weekly active users on their LLM platform *ChatGPT*[7]. According to Yankouskaya et al. (2026), over 40% of adults in the United Kingdom are happy to use ChatGPT as a mental health counsellor, indicating a significant amount of trust in LLM's towards important life decisions. Given the widespread use and trust in LLMs, it is worth discussing the applicability of LLMs, such as ChatGPT, for college course recommendations.

Firstly, we shall discuss the potential concerns of LLMs for college course recommendations. As described in Alansari and Luqman (2025), LLMs have the capability of hallucinating: presenting false or fabricated information as facts. This presents the possibility of an LLM providing inaccurate or false information regarding college courses. For example, the LLM may suggest a course that does not exist, or provide inaccurate entry requirements.

Secondly, as we discovered in section 3.1.6, answering the question: *"What college course should I study?"* is multi-faceted and nuanced. Without sufficient domain knowledge, users may be unaware of what relevant context to provide to the LLM for college course recommendations. Moreover, LLMs such as ChatGPT are intentionally designed to have simplistic interfaces, where the user selectively provides the context to the LLM. The LLM can only provide college course recommendations based off the context the user provided, which may be limited and result in ineffective college course recommendations.

Despite these concerns, LLMs pose many advantages in the context of college course recommendations. For instance, a design challenge encountered during this dissertation was translating user preferences - such as academic interests - into something numerical,

[6]According to `https://openai.com/index/scaling-ai-for-everyone/`
[7]https://chatgpt.com/

i.e. a float value between 0.0 and 1.0. A consequence of this design decision was that the quiz was constrained by close-ended questions, without any option for the user to provide responses in the form of natural language. For example, a user may give a '5' rating to a mathematics question for one reason (e.g. interest), and a '5' rating to a law question for another reason (e.g. status). Reducing the user's responses to a discrete numerical representation causes a loss of nuance, resulting in the RS having a simplistic understanding of the user. As a result of this, the RS's recommendations may not be as relevant or effective as they could be.

A key feature of LLMs are their ability to comprehend natural language, which addresses the previously mentioned concerns. Another advantage of this is that users can express their more nuanced values and preferences that may not be addressed in a close-ended quiz, such as their desire for work/life balance or meaning. For these reasons, we believe that LLMs could pose a great benefit for college course recommendations, and this suggests a piece of future research in this domain.

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Appendix A

Initial Prototypes

This chapter explores a failed approach to utilise Collaborative Filtering (CF) in attempting to solve this dissertation's research question. As part of the approach, popular datasets[1] were utilised that others may use in solving similar problems to this dissertation. A machine learning model classifier was trained on this dataset, and the results were unsatisfactory due to a lack of coverage by the questions in the dataset.

This chapter will describe the motivations behind the chosen approach, the implementation details, and then evaluate the results and describe why the experiment yielded unsatisfactory results.

A.1 Design Motivation

In this section, we will describe the motivation behind this approach. If we recall the research question as outlined in section 1.3, we are seeking to design an RS that assists a user's exploration into college courses by providing with them relevant, diverse, and serendipitous recommendations. Among the three types of RSs described in section 2.7, Collaborative Filtering (CF) is more likely to provide serendipitous recommendations, according to Aggarwal et al. (2016). This is because CF does not solely rely on what the user likes, rather it leverages the data of what other similar users also liked, increasing the likelihood of more surprising, but relevant recommendations, i.e. serendipity. As also described in section 3.3, serendipity is the gold standard of RSs, and it would be incredibly useful in answering our research question by assisting the user in their exploration of college courses. As a result, this chapter will explore a CF approach as it potentially

[1] `https://openpsychometrics.org/_rawdata/`

provides more serendipitous recommendations.

It is worth mentioning that for the sake of brevity, this chapter will not feature a vivid description of the different strategies attempted to improve model performance. Instead, we will focus on our best attempt in solving this problem through the CF approach.

A.2 Datasets and Initial Challenges

In section 3.3, we identified a research gap by leveraging a user's RIASEC model and academic interests to provide personalised college course recommendations. However, in order to design a CF RS in this context, we need to find datasets with other users' college choice, and either their academic interests or RIASEC model. To the best of our knowledge, this kind of data is not available in the Irish context. As a result, we explored similar datasets outside of the Irish context.

A promising dataset in this context was published by OpenPsychometrics[2] with data on over 140,000 individuals. In this dataset, which will be referred to as the 'RIASEC dataset', feature the results of a 48-question quiz which determined the user's RIASEC model. This quiz asked the user how much they enjoyed certain activities on a 5-point Likert scale, i.e. 1 implied they dislike the activity, and a 5 implied they would enjoy the activity. An example of a question is given in figure A.1. As part of the quiz, the user was asked to provide their college major, which is crucial information in our context. After some initial data cleaning, we were left with over 70,000 user's RIASEC quiz results and their college majors. This kind of data would be essential in designing a CF RS.

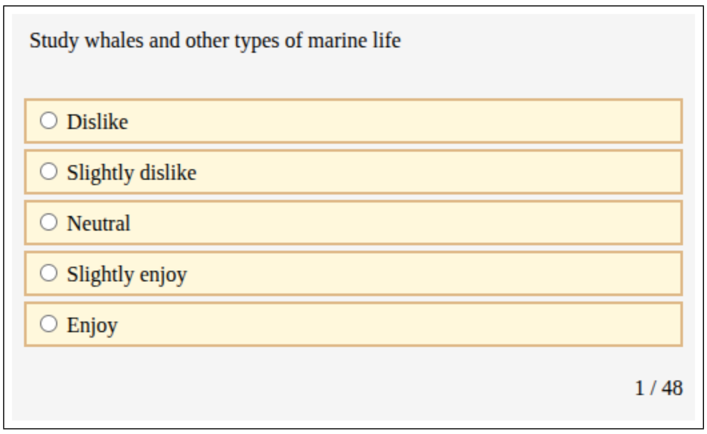


Figure A.1: An example of a question given on the OpenPsychometrics RIASEC quiz.

[2] https://openpsychometrics.org/_rawdata/

However, as described in subsection 3.3, this dissertation seeks to design an RS that recommends specific Irish college courses, as opposed to general areas of study. The RIASEC dataset has been answered by users from all over the world, particularly from the USA. As a result, there is a challenge in mapping the college majors that users provided in the RIASEC dataset, to specific Irish college courses. Moreover, new Irish college courses are being created each year, compounding the difficulty of this problem.

A.3 Solution to Introduced Problems

As a solution to the problem of specificity provided above, we can create more general categories of related college courses. For example, we can group college majors/courses such as medicine, Nursing and Occupational Therapy under a "healthcare" category. The college majors in the RIASEC dataset can be placed into these categories, and the college courses in the Irish college course dataset can also be placed into these categories. Admittedly, a disadvantage of this solution is a loss of nuance that would occur in the mapping. In other words, specific traits of particular college majors will get lost as they are combined into a more general category with other college majors.

We now need to find a dataset with college majors and associated college major categories. A popular dataset on college majors in the USA[3] was most appropriately applied to this problem. This dataset, which will be referred to as the 'college major dataset', initially featured 174 college majors and categorised each college major under 15 categories. However, in its initial form, this college major dataset failed to capture many of the college majors provided in the RIASEC dataset. In addition to this, many of the college major categories were too broad, and were then split up into different college major categories. As a result, this college major dataset underwent significant manual revisions, and the final form of this dataset is provided in table A.1. After these significant manual revisions, the college major dataset featured over 200 college majors with 16 categories.

[3] `https://github.com/fivethirtyeight/data/tree/master/college-majors`

Major	Category	Major	Category	Major	Category
acting	arts	english education	education	culinary arts	hospitality
animation	arts	health education	education	event management	hospitality
art and design	arts	history education	education	hospitality management	hospitality
commercial art	arts	language education	education	hotel management	hospitality
dance	arts	library science	education	tourism	hospitality
design	arts	mathematics education	education	tourism management	hospitality
drama	arts	music education	education	anthropology	humanities
fashion	arts	physical education	education	art history	humanities
fashion design	arts	psychology education	education	civilization studies	humanities
fashion merchandising	arts	science education	education	classical studies	humanities
film studies	arts	secondary education	education	comparative literature	humanities
fine arts	arts	social science education	education	english	humanities
graphic design	arts	special education	education	english literature	humanities
interior design	arts	aeronautical engineering	engineering	history	humanities
music	arts	aerospace engineering	engineering	intercultural studies	humanities
performing arts	arts	agricultural engineering	engineering	international studies	humanities
photography	arts	architectural engineering	engineering	liberal arts	humanities
product design	arts	architecture	engineering	literature	humanities
studio arts	arts	aviation studies	engineering	philosophy	humanities
theater	arts	biological engineering	engineering	religious studies	humanities
visual arts	arts	biomedical engineering	engineering	rhetoric	humanities
writing	arts	biotechnology	engineering	theology	humanities
child development	beh. health*	chemical engineering	engineering	criminal justice	law
clinical psychology	beh. health*	civil engineering	engineering	criminology	law
comm. disorders	beh. health*	computer engineering	engineering	government	law
counseling	beh. health*	construction studies	engineering	intl. relations	law
family science	beh. health*	electrical engineering	engineering	legal studies	law
human development	beh. health*	engineering mgmt.	engineering	paralegal studies	law
human services	beh. health*	engineering tech.	engineering	political science	law
mental health	beh. health*	environmental eng.	engineering	public admin.	law
occupational therapy	beh. health*	geological engineering	engineering	public policy	law
psychology	beh. health*	geophysical engineering	engineering	agricultural science	life science
rehabilitation	beh. health*	industrial engineering	engineering	biochemistry	life science
social work	beh. health*	industrial management	engineering	biology	life science
speech pathology	beh. health*	manufacturing eng.	engineering	biomedical science	life science
accounting	business	marine engineering	engineering	biopsychology	life science
accounting & finance	business	materials engineering	engineering	botany	life science
actuarial science	business	materials science	engineering	cognitive science	life science
administration	business	mechanical engineering	engineering	ecology	life science
business admin.	business	metallurgical eng.	engineering	environmental science	life science
business mgmt.	business	mining engineering	engineering	food science	life science
e-commerce	business	nuclear engineering	engineering	genetics	life science
economics	business	petroleum engineering	engineering	health science	life science
entrepreneurship	business	chinese	foreign languages	horticulture	life science
finance	business	english language	foreign languages	marine biology	life science
human resources	business	french	foreign languages	microbiology	life science
HR management	business	german	foreign languages	molecular biology	life science
intl. business	business	greek	foreign languages	neuroscience	life science
logistics	business	italian	foreign languages	pharmacology	life science
management	business	japanese	foreign languages	physiology	life science
mgmt. info. systems	business	korean	foreign languages	zoology	life science
marketing	business	latin	foreign languages	astronomy	physical science
operations mgmt.	business	linguistics	foreign languages	astrophysics	physical science
advertising	comms.	portuguese	foreign languages	atmospheric science	phys. science
journalism	comms.	russian	foreign languages	chemistry	physical science
mass media	comms.	spanish	foreign languages	earth science	physical science
public relations	comms.	translation	foreign languages	forensic science	phys. science
computer info. sys.	comp. & math	community health	healthcare	geology	physical science
computer networking	comp. & math	dental hygiene	healthcare	geoscience	physical science
computer science	comp. & math	dental technology	healthcare	industrial tech.	phys. science
computer systems	comp. & math	dentistry	healthcare	meteorology	physical science
cyber security	comp. & math	health admin.	healthcare	physics	physical science
information science	comp. & math	health services	healthcare	radiology	physical science
information systems	comp. & math	medic	healthcare	behavioral science	social science
info. technology	comp. & math	medical admin.	healthcare	geography	social science
mathematics	comp. & math	medical assisting	healthcare	sociology	social science
programming	comp. & math	medicine	healthcare	athletic training	sport
software eng.	comp. & math	medicine technology	healthcare	exercise & sport	sport
statistics	comp. & math	midwifery	healthcare	exercise physiology	sport
telecommunications	comp. & math	nursing	healthcare	exercise science	sport
adult education	education	nutrition science	healthcare	kinesiology	sport
art education	education	pharmacy	healthcare	physical therapy	sport
early childhood	education	public health	healthcare	sports management	sport
educational admin.	education	veterinarian	healthcare	sports medicine	sport
elementary education	education	vet. technology	healthcare		

**beh. health = behavioural health and human services*

Table A.1: Complete list of College Majors and Categories.

We now have a college major dataset with over 200 college majors and 16 associated college major categories, and a RIASEC dataset with over 70,000 user's college majors and the results of their RIASEC quiz. Using a pre-processing script as given in section B.1, we can categorise each of the ~70,000 users in the RIASEC dataset into one of 16 college major categories as provided in the college major dataset. This augmented dataset will be instrumental in training a Machine Learning model to predict a user's college major category from the results of this RIASEC quiz, which will be explored in the next section.

A.4 Machine Learning Model Selection

From the work in the last section, we now have a dataset with over 70,000 user's RIASEC quiz results, and their college major category. This can be reconceptualised in the context of a Machine Learning (ML) problem: we have 48 input features, and one output variable. In this case, each input feature is a rating from 1–5, indicating the user's enjoyment towards a particular task, and the output is one of 16 college major categories.

Many different ML models were trained on this data, and they were tested on unseen data to gather evaluation metrics, in particular: top-1 and top-3 accuracy. The ML model with the best top-1 and top-3 accuracy was the Histogram-based Gradient Boosting Classifier (HGBC), with its implementation details given in listing A.1 below:

```
hist_gradient_boosting_machines_model = HistGradientBoostingClassifier(
        class_weight='balanced',
        l2_regularization=1.0
    )
```

Listing A.1: Histogram Gradient Boosting Classifier Implementation

This implementation of a HGBC achieved a top-1 accuracy of 0.319, and a top-3 accuracy of 0.606. In other words for the unseen test set, this HGBC implementation predicted the user's college major category correctly in 31.9% of cases (top-1 accuracy), and in 60.6% of cases, the model predicted the user's college major category correctly in its first 3 predictions (top-3 accuracy). Seeing as this implementation of the HGBC model had the best performance metrics, we can evaluate its performance further in the next section.

A.5 Model Evaluation

In this section we will take a closer look in evaluating the chosen HGBC model's performance. In figure A.2 below, we can observe the Confusion Matrix of the chosen model, showcasing the recall scores.

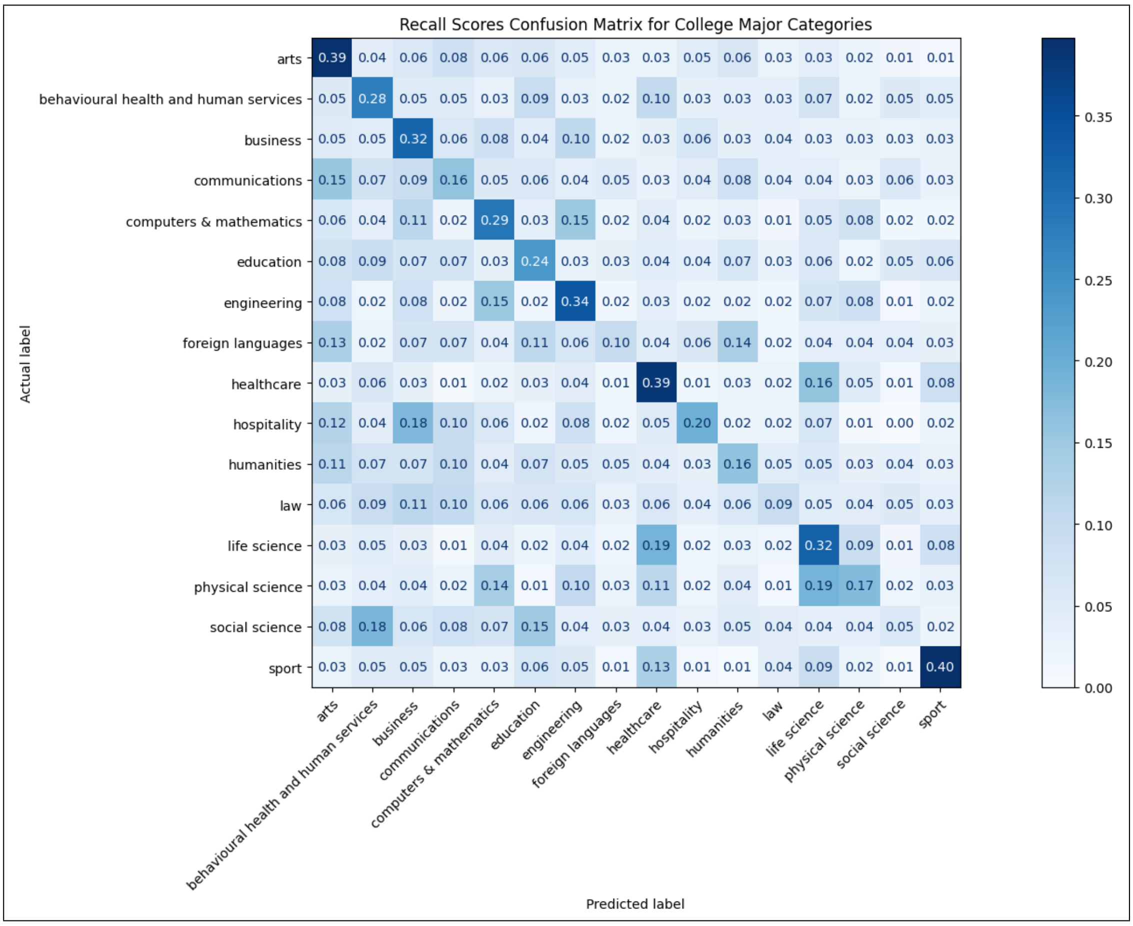


Figure A.2: A Confusion Matrix for Recall scores on the HGBC model.

To take one example, we can observe that the model achieved a recall score of 0.4 for the “sports” college major category, meaning that of all the users with “sports” as their college major category, the model correctly predicted “sports” for these users 40% of the time. Relatively speaking, we can see that this recall score of 0.4 is the highest performing recall score out of all the other college major categories. In particular, we can see the model struggling with some college major categories, such as “foreign languages” with a recall score of 0.1, and “social science” with a score of 0.05, meaning that the model correctly predicts the user’s college major category 10% and 5% of the time respectively.

Evidently speaking, the performance of this model was disappointing as demonstrated through the low recall scores referenced above. Many different ML models with differing configurations were applied to the problem, failing to provide any improvements to this HGBC implementation. As well as this, the college major dataset underwent significant manual revisions - adding new college majors and splitting up different college major categories, all failed attempts to improve the model’s performance.

A.6 Limitations of the Approach

In the previous section, we established the poor performance of this model in predicting the user's college major category given the results of a 48-question RIASEC quiz. In this section, we will provide more insights as to why the model with these popular datasets performed poorly.

In the context of predicting a user's college major, the primary limitation of the chosen RIASEC dataset is the lack of coverage by the questions. In other words, the 48 questions do not adequately cover many areas of study, which we believe is the primary reason behind the poor performance behind the model. For example, if we revisit figure A.2, we can observe very poor model performance on the "foreign languages" category, with a recall score of 0.1. This is simply because there are no questions in the RIASEC dataset that are explicitly related to languages or linguistics. To provide another example, we can observe very poor recall scores for the "law" category, most likely because there are no questions in the quiz that are explicitly related to law or government. Without this data, the model finds it very difficult to discern if a user is studying in the "foreign languages" or "law" categories, explaining the model's poor performance on these categories.

Due to this limitation, we found this dataset and subsequent approach to be inadequate in solving our problem, due to the poor recall scores as reported in figure A.2. In spite of many efforts to improve the model's performance through significant manual revisions to the college major dataset, and testing multiple different ML models and configurations, the performance of the model was still deemed unsatisfactory.

A.7 Chapter Summary

In this chapter, we explored an approach to solving this dissertation's research question by utilising a very popular RIASEC dataset. There was a description of the motivation in using this dataset, followed by an explanation of the various data engineering performed on both datasets in order to make it more suitable for a traditional ML problem. We evaluated the performance of the best performing model, and found the results to be unsatisfactory. Ultimately, we believe the unsatisfactory results to be a result of the poor coverage of the 48 questions on many different fields of study. As a result, we believe that a dataset such as this may not be particularly useful in solving a problem similar to the one addressed in this dissertation. That being said, we felt it was worth writing about our inadequate solutions in this chapter, firstly because of the popularity of this dataset, and secondly to add a useful contribution to the literature.

Appendix B

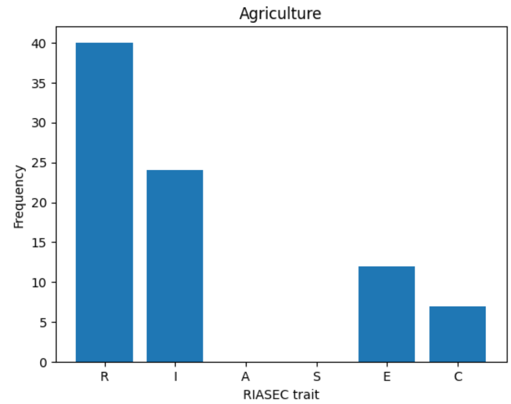
B.1 Preprocessing script for college course/major classification

The following Python script was used to preprocess the names of college course titles and college majors.

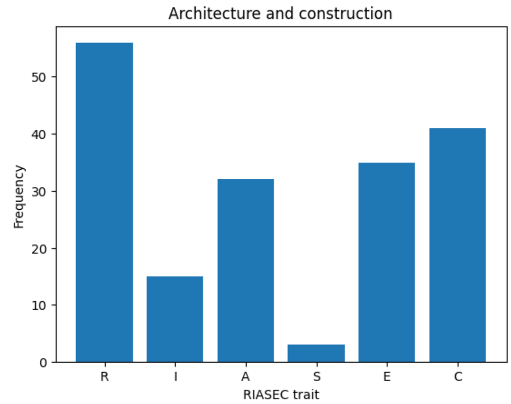
```
stemmer = SnowballStemmer("english")

def preprocess_college_title(text):
    text = text.lower()
    text = re.sub(r'[\.\?=!#\'`\*]', '', text) # remove certain symbols
    text = re.sub(r'\d+', '', text) # remove numbers

    for abbreviation, expanded_college_major in ABBREVIATIONS_MAP.items():

        pattern = r'\b' + re.escape(abbreviation) + r'\b'
        text = re.sub(pattern, expanded_college_major, text)

    # substitute certain symbols for spaces
    text = re.sub(r'[\(\)\{\}\[\]&/+,;:\\|\-]', ' ', text)

    text = text.strip()

    tokens = word_tokenize(text)
    cleaned_tokens = []
    for word in tokens:
        if word not in stop_words:
            stemmed_word = stemmer.stem(word)

            if stemmed_word not in cleaned_tokens:
```

```
                cleaned_tokens.append(stemmed_word)

            if len(cleaned_tokens) > 1 and "scienc" in cleaned_tokens:
                cleaned_tokens.remove("scienc")

    cleaned_tokens.sort()
    return ' '.join(cleaned_tokens)

ABBREVIATIONS_MAP = {
    'maths': 'mathematics',
    'teacher': 'teaching',
    'teaching': 'education',
    'pharmaceutical': 'pharmacy',
    'technician': 'technology',
    'admin': 'administration',
    'bio': 'biological',
    'phy': 'physics',
    'chem': 'chemistry',
    'chemistry': 'chemical',
    'dentist': 'dentistry',
    'dental': 'dentistry',
    'agri': 'agriculture',
    'ag': 'agriculture',
    'hrm': 'human resource management',
    'it': 'information technology',
    'ict': 'information and communication technology',
    'tech': 'technology',
    'llb': 'law',
    'bcl': 'civil law',
    'mgt': 'management',
    'gaeilge': 'irish'
}
```

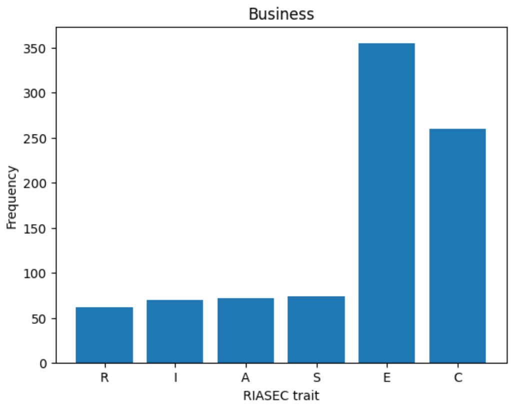
B.2 RIASEC Trait distribution per Academic Interest



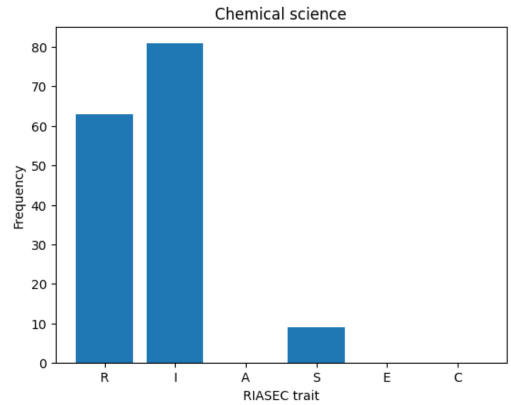
(a) Agriculture



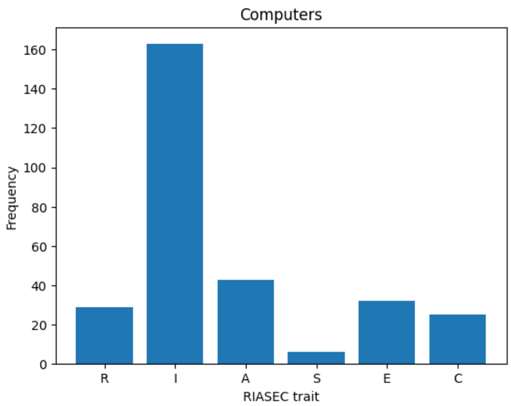
(b) Architecture & Construction



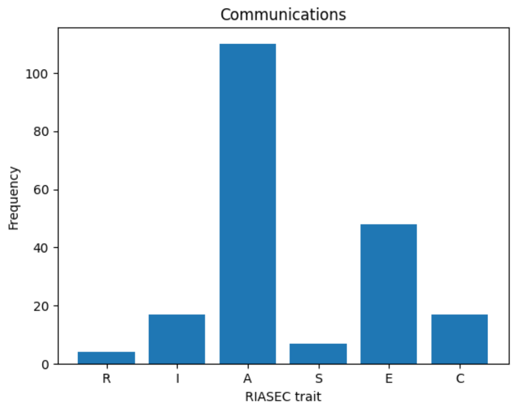
(c) Business



(d) Chemical Science



(e) Computers



(f) Communications

Figure B.1: RIASEC trait distribution across academic interests.

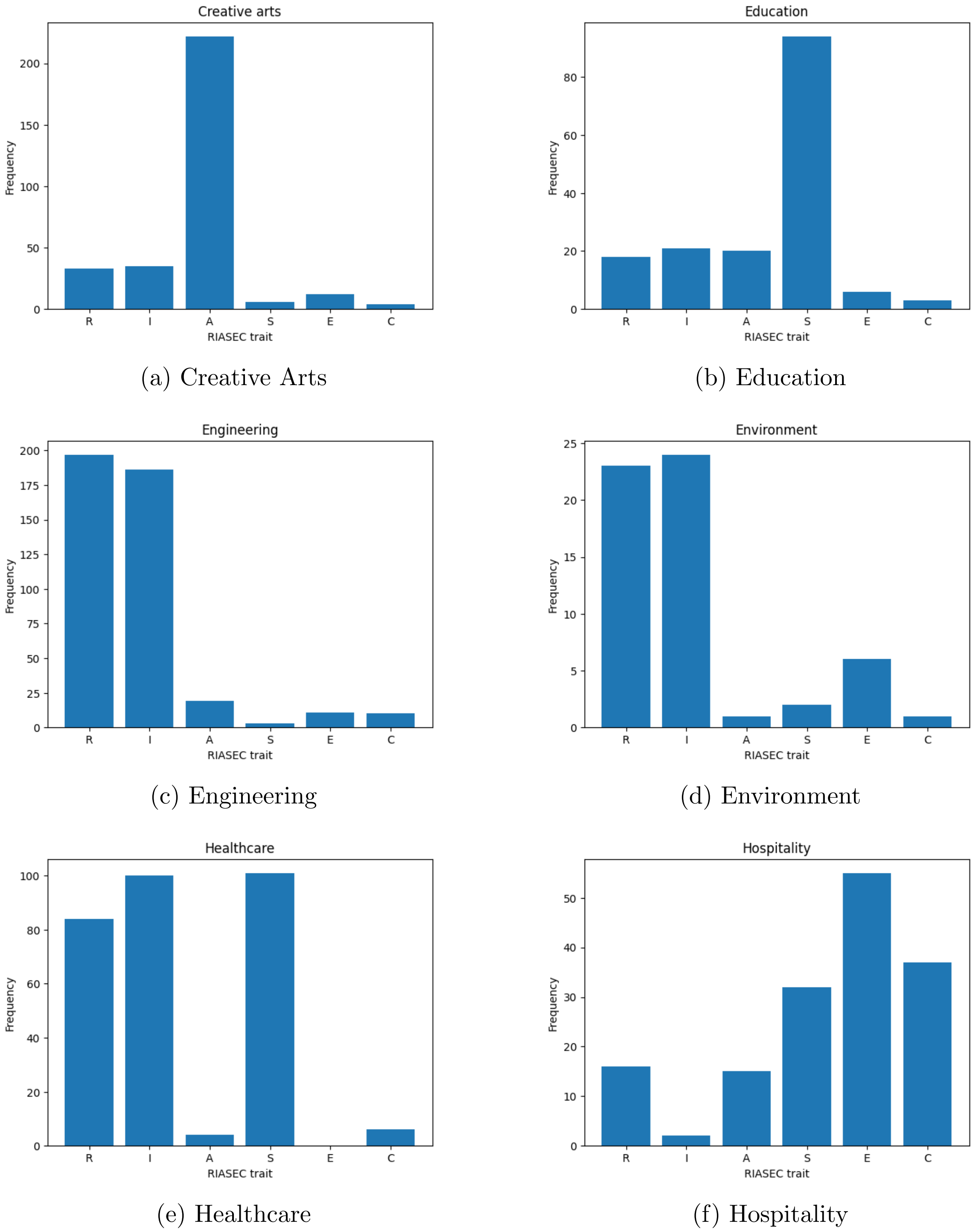
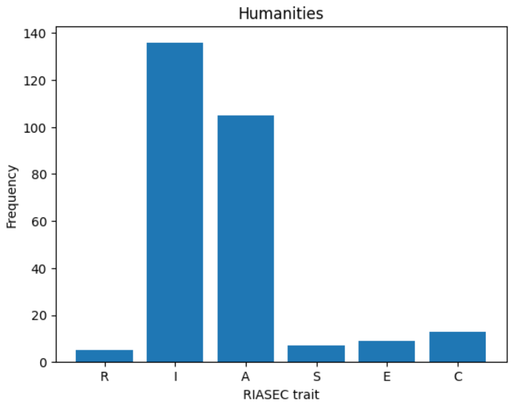
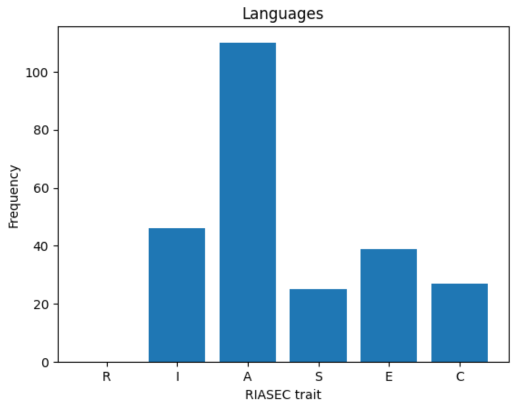


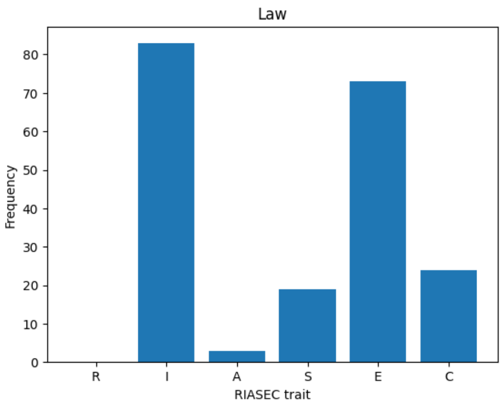
Figure B.2: RIASEC trait distribution across academic interests.



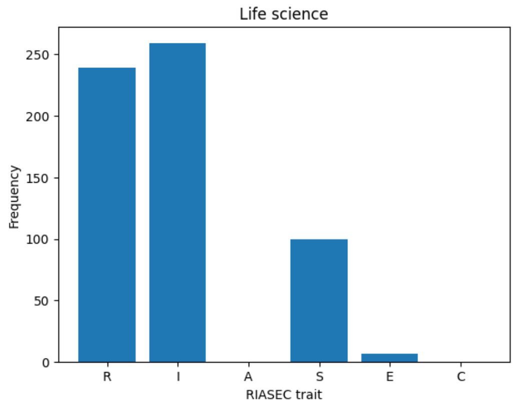
(a) Humanities



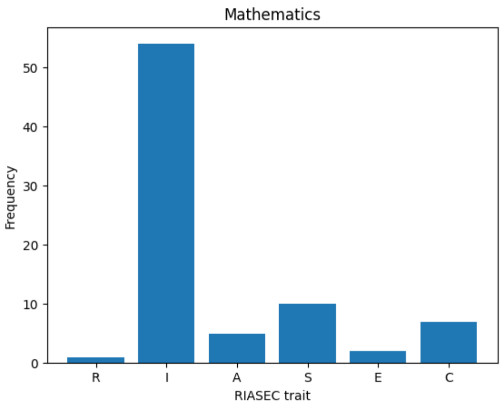
(b) Languages



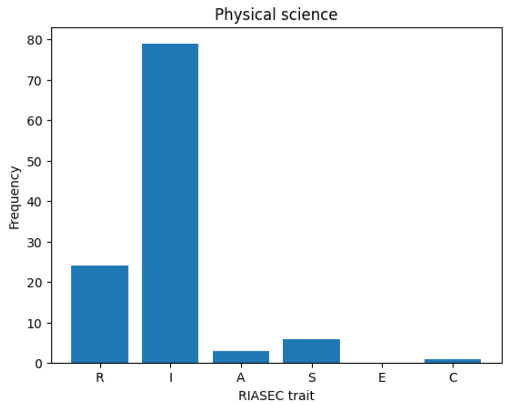
(c) Law



(d) Life Science

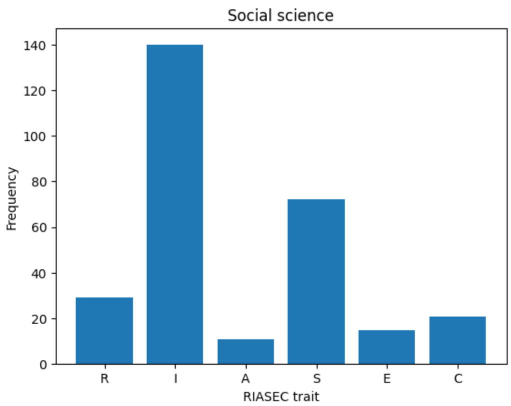


(e) Mathematics

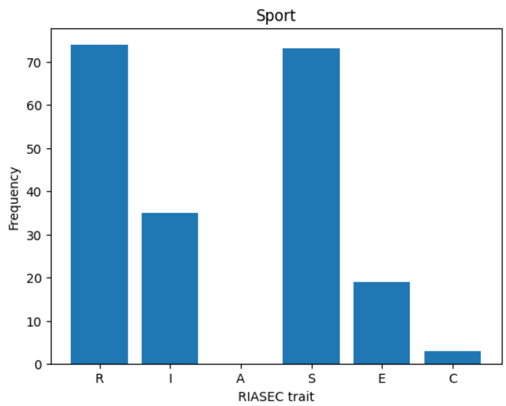


(f) Physical Science

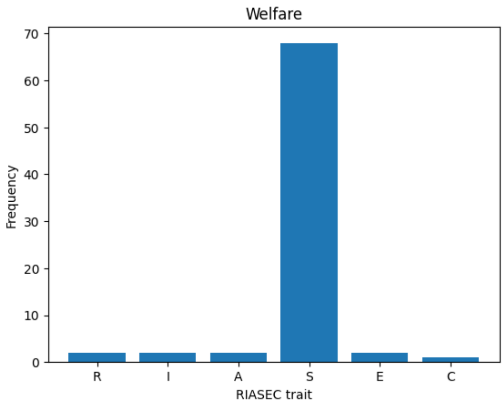
Figure B.3: RIASEC trait distribution across academic interests.



(a) Social Science



(b) Sport



(c) Welfare

Figure B.4: RIASEC trait distribution across academic interests.

B.3 All 126 Academic Interest Questions

Agriculture	
Activity	**RIASEC Trait**
Handle and care for livestock on a farm	realistic
Repair and maintain high-tech farm machinery	
Research plant genetics for better crop yields	investigative
Analyse soil chemistry to improve land health	
Sell agricultural products at a market to farmers	enterprising
Organise and track the financial accounts of a farm	conventional

Architecture & Construction	
Activity	**RIASEC Trait**
Draft technical drawings of building joints	realistic
Build scale models to test structural strength	
Analyse how new building materials affect temperature	investigative
Design the aesthetic look of a new building	artistic
Lead site meetings to keep projects on track	enterprising
Check that building plans follow all safety laws	conventional

Business	
Activity	**RIASEC Trait**
Negotiate with suppliers to lower business costs	enterprising
Manage a sales team to exceed revenue targets	
Pitch innovative business ideas to investors	
Organise company budgets and financial reports	conventional
Maintain accurate business records and data	
Coordinate complex logistics and supply chains	

Chemical Science	
Activity	**RIASEC Trait**
Research how proteins change to fight specific diseases	investigative
Discover how new drugs interact with human cells	
Research how catalysts speed up chemical reactions	
Analyse unknown substances to find their chemical makeup	
Mix raw ingredients to create skincare products	realistic
Mix chemical compounds to create new products	

Computers	
Activity	**RIASEC Trait**
Assemble and configure high-performance servers	realistic
Design a software application to handle millions of users	investigative
Fix hardware problems found in error logs	realistic
Organise and manage large databases of information	conventional
Optimise a computer program to run more efficiently	investigative
Develop the code for a search engine	

Communications	
Activity	**RIASEC Trait**
Build a viral social media campaign for a brand	enterprising
Pitch a creative vision to launch a new product	
Design the logo and visual identity for a music festival	artistic
Write an opinion piece for a respected newspaper	
Direct a short film for a company's marketing campaign	
Investigate a news article to ensure 100% truth	investigative

Creative Arts	
Activity	**RIASEC Trait**
Create original paintings or drawings for exhibition	artistic
Choreograph and perform expressive dance routines	
Compose and perform an original piece of music	
Perform lead roles in professional theatre or film	
Write original stories or poetry	
Hand-craft original ceramic or pottery artworks	realistic

Education	
Activity	**RIASEC Trait**
Create original lesson plans and activities for a class	artistic
Teach new concepts to a class of students	social
Tutor students who need specialised support	
Manage classroom behavior and group dynamics	
Mentor young people to reach their potential	
Explain complex theories in very simple ways	

Engineering	
Activity	**RIASEC Trait**
Analyse thermal data to improve engine efficiency	investigative
Analyse brain signals to control robotic limbs	
Analyse airflow patterns to reduce aircraft drag	
Build a mini airplane with electrical motors	realistic
Fix technical faults in industrial machinery	
Build and calibrate electronic medical devices	

Environment	
Activity	**RIASEC Trait**
Restore natural habitats by planting trees	realistic
Build solar-powered water filtration systems	
Research new laws to reward companies for lowering carbon	investigative
Analyse water samples to track the spread of microplastics	
Encourage firms to use sustainable packaging	enterprising
Lead a campaign to convince a city to go carbon neutral	

Healthcare	
Activity	**RIASEC Trait**
Evaluate a patient's symptoms to diagnose illnesses	investigative
Research the biological mechanisms of the human body	
Provide direct care to sick patients	social
Assist patients with physical rehabilitation	
Perform clinical procedures or surgeries	realistic
Administer emergency first aid treatments	

Hospitality	
Activity	**RIASEC Trait**
Manage and coordinate large-scale events	enterprising
Lead the daily operations of a hotel	
Serve and help guests in a luxury setting	social
Cook and prepare food in a professional kitchen	realistic
Organise detailed travel plans for clients	conventional
Direct and lead staff in a busy restaurant	enterprising

Humanities	
Activity	**RIASEC Trait**
Write creative essays on culture or history	artistic
Design museum exhibits to tell history stories	
Critique and review modern works of art	
Investigate different philosophical ideas	investigative
Analyse themes in classic literature and art	
Research how ancient civilisations lived	

Languages	
Activity	**RIASEC Trait**
Analyse the grammatical structure of languages	investigative
Speak fluently in a foreign language	social
Translate for people in professional settings	
Teach a language to non-native speakers	
Engage with foreign literature in their original language	artistic
Create a cultural guide based on local traditions	

Law	
Activity	**RIASEC Trait**
Research past legal cases and government acts	investigative
Analyse evidence to build a legal argument	
Argue a case in court to defend a client's rights	enterprising
Organise a campaign to advocate for human rights	
Help refugees apply for legal protection and asylum	social
Check that legal documents follow official regulations	conventional

Life Science	
Activity	**RIASEC Trait**
Investigate how bacteria react to medicine	investigative
Analyse genetic sequences and DNA samples	
Study how different nutrients affect human metabolism	
Research the decline of coral reefs	
Dissect biological specimens in a lab	realistic
Conduct field research on local wildlife	

Mathematics	
Activity	**RIASEC Trait**
Prove a mathematical theorem using formal logic	investigative
Research the hidden patterns within prime numbers	
Design algorithms to recommend movies to users	
Predict sports results using statistics	
Analyse crash data to estimate car insurance prices	
Clean large datasets to improve model accuracy	conventional

Physical Science	
Activity	**RIASEC Trait**
Research the physics behind planetary motion	investigative
Research new ways to create clean energy	
Calculate force and motion in physics systems	
Analyse seismic waves to map the Earth's core	
Run laboratory tests on matter and energy	realistic
Use technical tools to collect rock samples	

Social Science	
Activity	**RIASEC Trait**
Analyse how rising prices change spending habits	investigative
Study the development of the human mind	
Research societal trends in demographics	
Analyse patterns in human behavior and culture	
Analyse the personality traits of criminals	
Counsel people to help overcome challenges	social

Sport	
Activity	**RIASEC Trait**
Teach healthy nutrition habits to athletes	social
Coach a sports team to improve performance	
Analyse the biomechanics of body movement	investigative
Lead the operations of a sports club	enterprising
Help athletes recover from physical injuries	realistic
Demonstrate and teach new exercise routines	

Welfare	
Activity	**RIASEC Trait**
Help vulnerable people access vital support	social
Organise outreach programs for the community	
Advocate for social justice and equality	
Provide emotional support to people in crisis	
Assess if families need extra social support	
Plan creative activities to help kids develop	

B.4 Complete Data of Experiment

	Diversity		**Trust**		**Precision**		**Serendipity**	
ID	Actual	Baseline	Actual	Baseline	Actual	Baseline	Actual	Baseline
1	3	4	4	3	0.65	0.35	0.50	0.35
2	5	2	4	2	0.30	0.10	0.25	0.10
3	4	5	5	3	0.25	0.20	0.20	0.15
4	4	4	4	1	0.40	0.00	0.35	0.00
5	3	5	4	2	0.75	0.35	0.70	0.35
6	3	5	5	1	0.50	0.05	0.35	0.00
7	3	5	5	1	0.50	0.05	0.45	0.05
8	4	2	4	1	0.20	0.00	0.15	0.00
9	2	5	4	3	0.60	0.35	0.50	0.35
10	4	1	4	1	0.25	0.05	0.10	0.00
11	3	4	4	4	0.60	0.55	0.55	0.55
12	4	2	4	2	0.45	0.20	0.25	0.20
13	4	3	5	2	0.50	0.15	0.35	0.05
14	3	3	3	4	0.35	0.35	0.15	0.25
15	3	5	4	5	0.40	0.35	0.30	0.25
16	3	5	5	2	0.90	0.30	0.85	0.30
17	3	4	4	3	0.10	0.30	0.00	0.20
18	5	5	4	4	0.30	0.35	0.30	0.35
19	4	3	4	2	0.40	0.15	0.30	0.15
20	4	3	4	3	0.45	0.25	0.45	0.25
21	5	4	4	1	0.60	0.20	0.45	0.20
22	5	5	5	1	0.80	0.35	0.65	0.35
23	5	5	4	4	0.65	0.60	0.60	0.50
24	4	2	2	2	0.20	0.05	0.15	0.05
25	5	4	2	2	0.55	0.30	0.50	0.30
26	5	1	5	1	0.50	0.00	0.50	0.00
27	5	1	4	2	0.55	0.20	0.55	0.20
28	5	5	5	5	0.15	0.10	0.15	0.05
29	2	5	5	2	0.50	0.20	0.40	0.20
30	4	5	5	2	0.55	0.05	0.45	0.05
31	5	3	4	2	0.65	0.30	0.55	0.30

Table B.1: Raw evaluation metric data for all participants ($n = 31$), comparing the actual RS and the baseline.